Application of Generative AI for Scholarly work

October 2026

Alexander Botzki, Bruna Piereck and Jolan Heyse



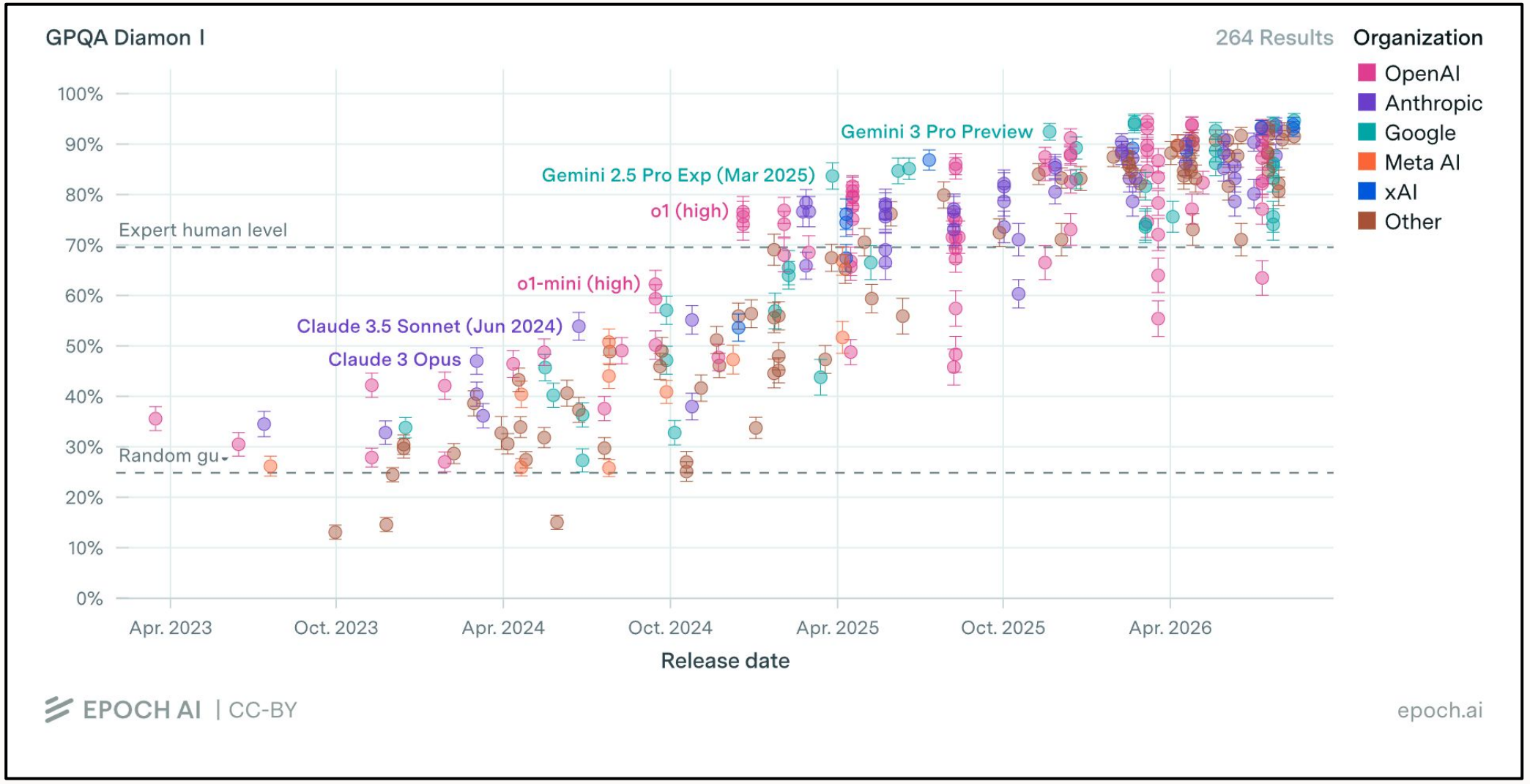


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AI in fast improvement

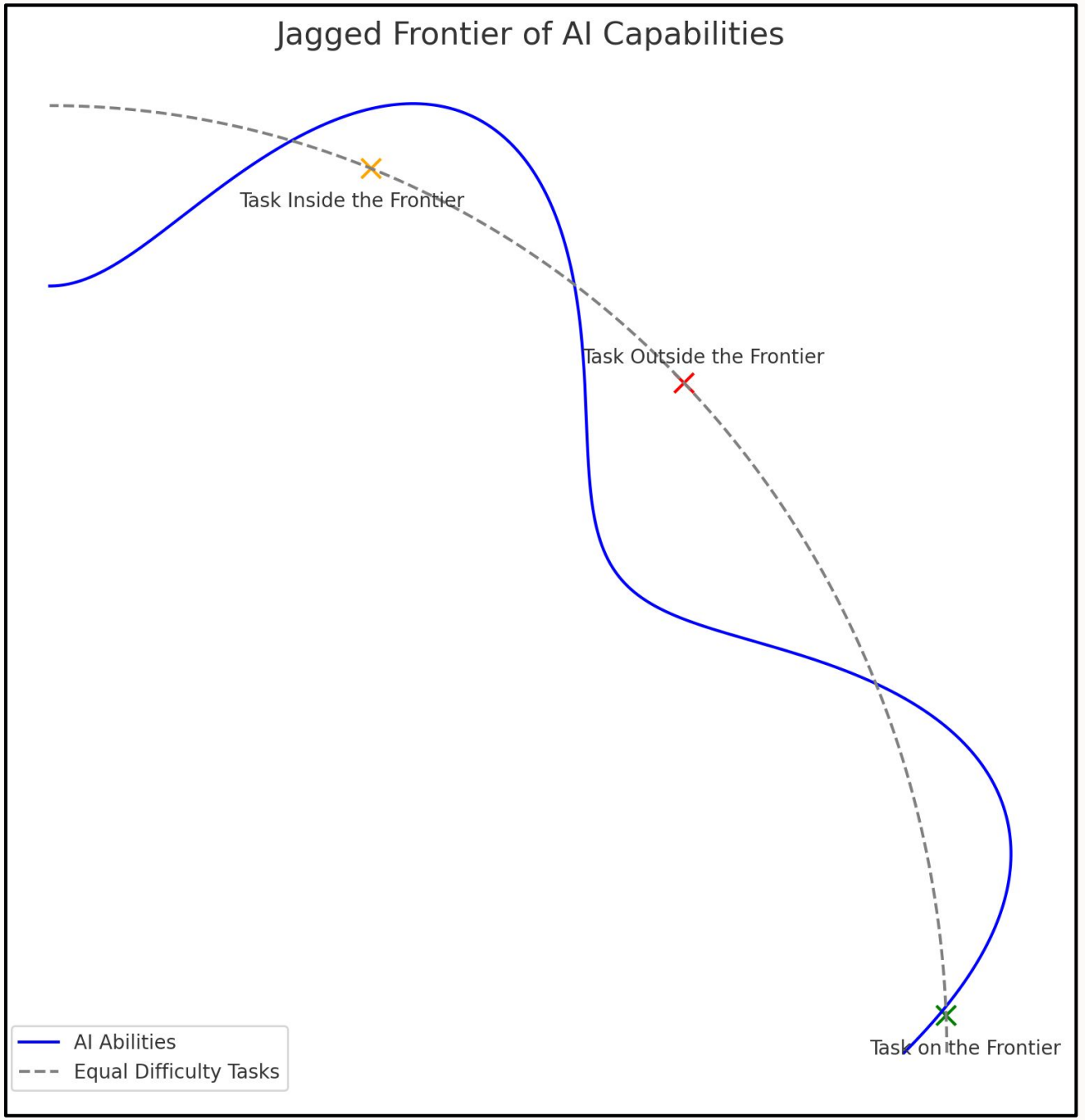
How it is going

On PhD-level, models are now above expert level



Progress is hugh,
but we see a jagged frontier

- Real-world challenges require
  - nuanced understanding
  - adaptable solutions



VIB

AI is going to change how we do research

Same research, done faster

- Faster coding.
  - Stata, R and Python
  - Drafts, debugging error logs,
  - Robustness checks, replication packages.

- Better writing.
  - Structured critique,
  - edits in your own voice,
  - pre-submission referee reports.

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New outputs, new projects

- New ways to communicate a paper.
  - An interactive website,
  - Policy brief,
  - Dashboard or an audio summary,

- New kinds of projects.
  - Mass automation (Project APE),
  - Automated replication and extension (Schwartz et al. 2026),
  - AI agents as simulated humans (Expected Parrot)

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01

introduction

Copyrights
When to be concerned

Patents
AI and the risks

Biases
How it applies in research

Hallucinations
General and specifics for research

What do you already know?



https://www.mentimeter.com/app/presentation/alzz5i5eov4umtatv9ctmj1ywx37wwsc/edit?question=ubeiyegirwu6

Copyrights
When to be concerned

Ownership

- You own
- You are accountable



Image recolored and cropped, by rawpixel.com on freepik.com

Generative AI and Copyright in the EU

Training, Creation, Regulation (Study 09-07-2025, by Nicola LUCCHI, University Pompeu Fabra, Barcelona)



- Outputs from iterative human + algorithmic tools
  - Member States (EU) differ in interpretation
  - Legal uncertainty



- EU copyright law is human-centric
  - It requires human creativity and authorship
  - (most) AI-generated outputs fall outside the scope of protection



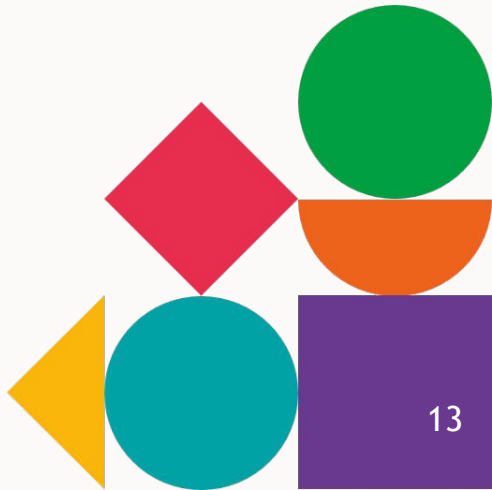
- Fully machine-generated content
  - No benefit from copyright protection
  - Remain in public domain

Generative AI and Copyright

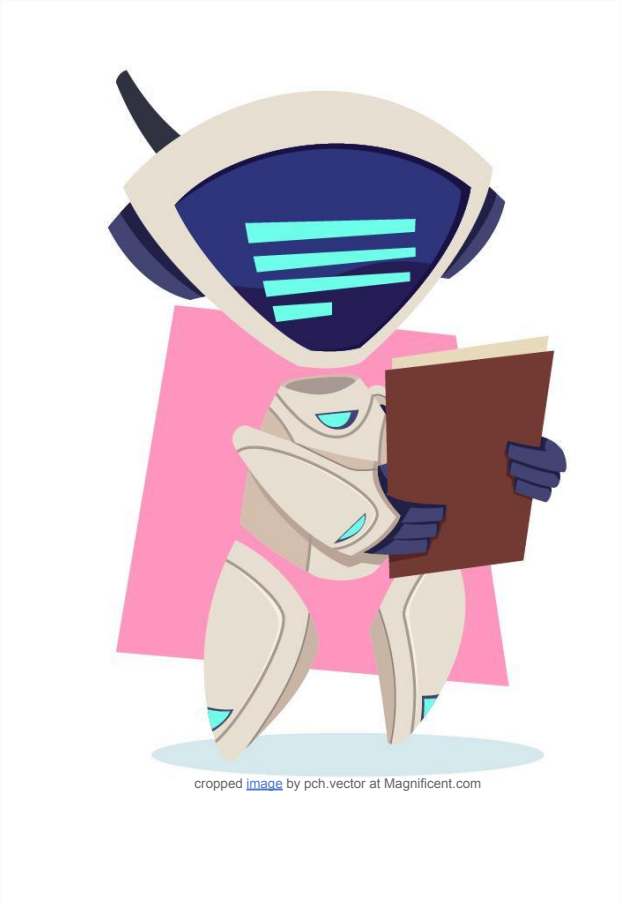
- Your outputs
  - Is considered to be in public domain (usually)

- Potential copyright infringement by generative AI models
  - Publication licenses not respected
    - Books
    - Papers

Are you sharing copyrighted content with genAI ?



Considerations about input



Non-Open Access Content

Many publishers have policies that restrict the uploading of their content, even if some might argue it falls under “Fair Use” (US Fair use law)

Confidential Information

Information that is subject to a non-disclosure agreement or classified.

Unpublished material

A peer reviewer using e.g. Copilot to summarize or better understand the material they are reviewing. May be ethical and community norms around that kind of practice.

Limitations and Exceptions in EU Copyright Law

for Libraries, Educational and Research establishments

Exception 9:
Non-Commercial Research

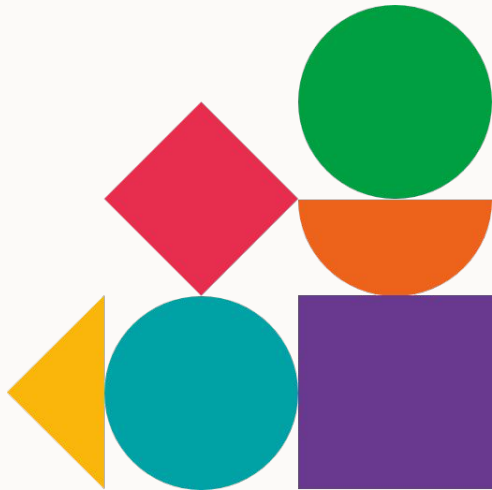
Challenges:

a. difficult to know when a research copy is for commercial purposes
b. different interpretation in the Member states

Exception 11:
Personal Copies / Research and Study Copies

Challenges:

a. How the term ‘non-commercial’ is interpreted varies across the EU
b. distinguishing between commercial copying and copying for research



VIB

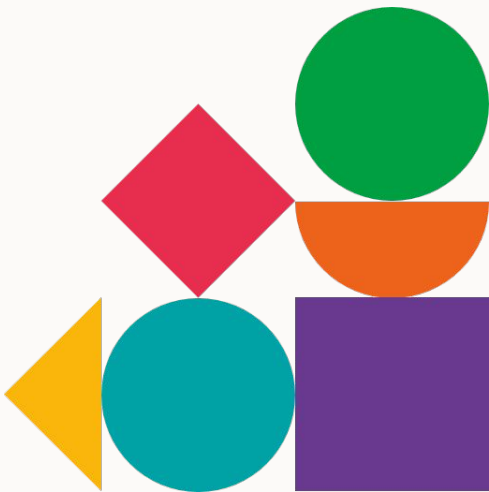
Limitations and Exceptions in EU Copyright Law

for Libraries, Educational and Research establishments

Exception 14:
Teaching

Challenges:

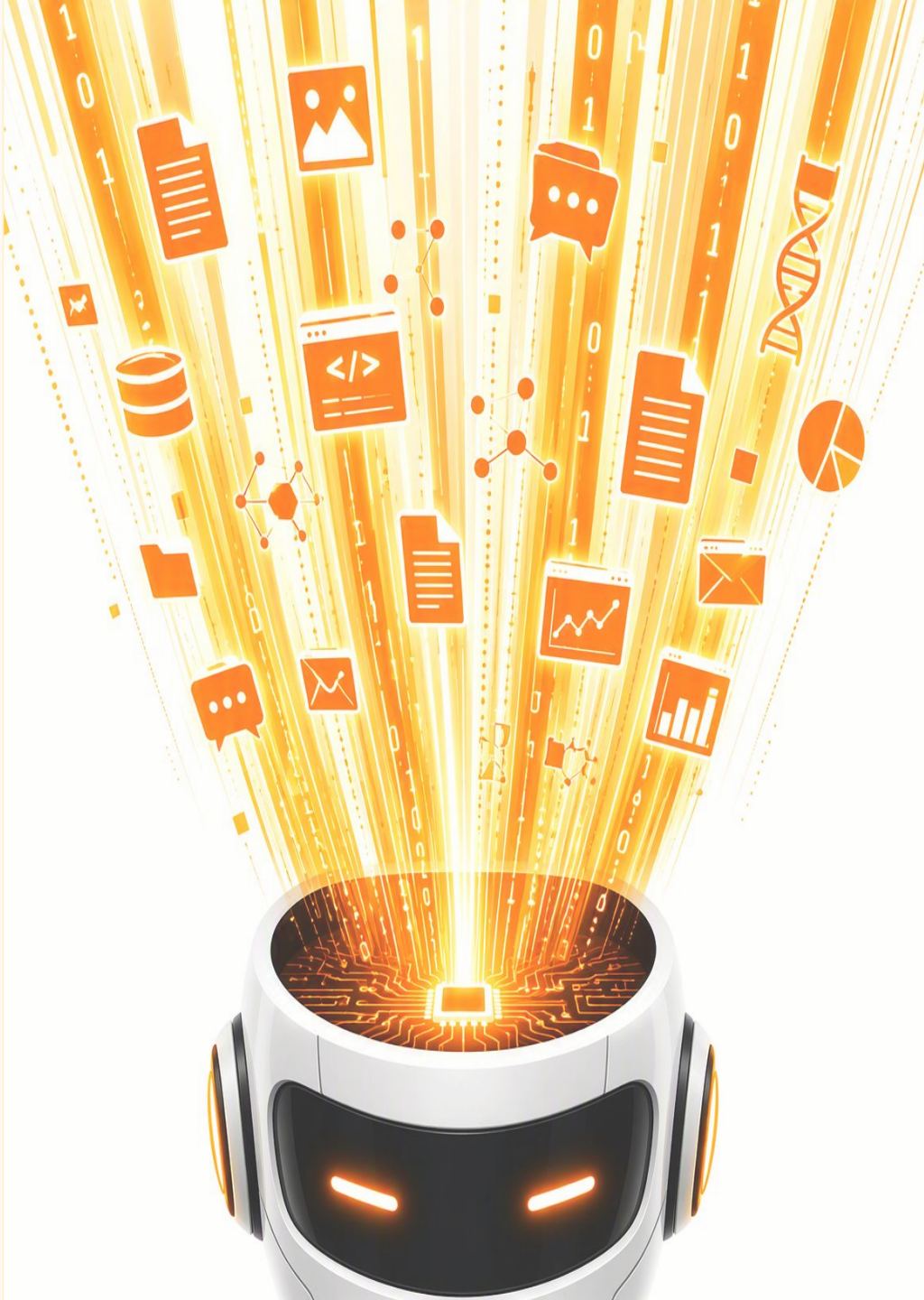
a. implementation across Member States varies
b. few countries allow translation for teaching purposes
   (Malta, the Netherlands, Poland and Slovenia)



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AI MODELS USE INTELLECTUAL PROPERTY OF THE WORLD:

BIGGEST THEFT IN HISTORY



Patents

AI and the risks



Image recolored, by magnific.com

What is considered safe

- **Enterprise licensed products**
  - Institutional account
- **Patent drafting**
  - Ideas
  - assess inventions
- **Analyse prior art**
  - There are specialised tools ($$)
  - Untested by VIB specialists

- **VIB innovation and Business**
  - Payed specialised DB
  - Google search

What are the risks

- **genAI use**
  - Cyber attack
  - Involuntary disclosure
- **Drafting**
  - Writing Embodiment
    - weak support suggestions
    - anticipation
      (future opportunities break)

VIB Policy

current being revised

2024



VIB policy and guidelines (2024)



Requirements

- **Transparency** about the use in their activities.
- **Verification steps** must be part of the process.
- User remains **fully accountable** for the outcome.



Safety and fairness

- **Must respect** Copyrights, licenses and other means of property or use-protection.
- **Change the settings** when using **public genAI** tools ensuring that information is **not used** as training data *(for VIB-related tasks)*.

VIB *Policy*

VIB policy and guidelines (2024)



Personal, confidential and proprietary data

- AI use can be allowed when:
  - **does not constitute the sharing of information** with a third party
  - Tool **operates on a private instance**
    (*within the VIB ICT security perimeter)*
    P.S.: Enterprise licenses

Responsible use



- **Private** or **locally installed** gen AI tools that provide similar safety guarantees **must be approved** by the VIB DPO.



- **Responsible use** includes formulation and use of good prompts.
  - Highly recommended that users are well trained in **prompt engineering**.

VIB Policy

VIB policy and guidelines (2024)

Copilot

VIB includes Enterprise data protection
[Microsoft Enterprise Copilot]

VIB

Practical guideline on the responsible use of generative artificial intelligence tools

The responsible use of generative artificial intelligence (genAI) tools:

- Requires a high level of awareness on the limits on the quality, correctness, accuracy, and currentness of the output, and on the need to perform appropriate verification.
- Must avoid that confidential, proprietary and personal information are processed in a way that could lead to the inadvertent or deliberate disclosure, sharing, theft or loss of these types of information.

The table below gives an overview for a number of genAI tools of whether they can be safely used for confidential, proprietary or personal information. When it is indicated that the tool is not allowed for confidential, proprietary and personal information, they can of course still be used for non-confidential, non-proprietary and non-personal information.

GenAI tool	Version	Application	Allowed for confidential, proprietary and personal information?	Remarks
ChatGPT	3.5	Text generator, feedback provider, ...	NO	It is advised not to use this version for confidential, proprietary and personal information, even when one has opted out for 'improve the model for everyone' and opted for a 'temporary chat'.
ChatGPT	4	Text generator, feedback provider, designer, multimedia developer, data analysis, ...	NO	It is advised not to use this version for confidential, proprietary and personal information, even when one has opted out for 'improve the model for everyone' and opted for a 'temporary chat'.
	4o	Text generator, feedback provider, designer, multimedia developer, data analysis, ...	NO	It is advised not to use this version for confidential, proprietary and personal information, even when one has opted out for 'improve the model for everyone' and opted for a 'temporary chat'.
	4 Enterprise	Text generator, feedback provider, designer, multimedia developer, data analysis, ...	YES	Data are not used for training purposes and the tool offers sufficient security guarantees.
MS Copilot	in Edge or via copilot.microsoft.com	Idea generator, advanced search engine, writing assistant, multimedia developer, feedback provider, ...	YES	But only in case you use the tool with authentication via the VIB or university email address.
	Web app	Idea generator, advanced search engine, writing assistant, multimedia developer, feedback provider, ...	YES	But only in case you use the tool with authentication via the VIB or university email address.

Version 1 25 July 2024

YES

Overview of whether they can be **SAFELY** used for:

- Confidential
- Proprietary
- Personal

NO

They **CAN STILL** be used for:

- non-confidential
- non-proprietary
- non-personal

VIB Policy

VIB Guidelines (2024) = YES

Copilot — All VIB staff, with VIB email login includes Enterprise data protection

Microsoft 365 Copilot — Integrated with microsoft system with VIB account, includes Enterprise data protection

ChatGPT Enterprise — It includes Enterprise data protection

perplexity enterprise — It includes Enterprise data protection

DeepL Write PRO — Won't use data to train de models

Claude enterprise — Sept 2026: licenses available VIB HQ (request via PI)

VIB Policy

Biases

"bias in, bias out"



Image recolored and cropped, by magnific.com



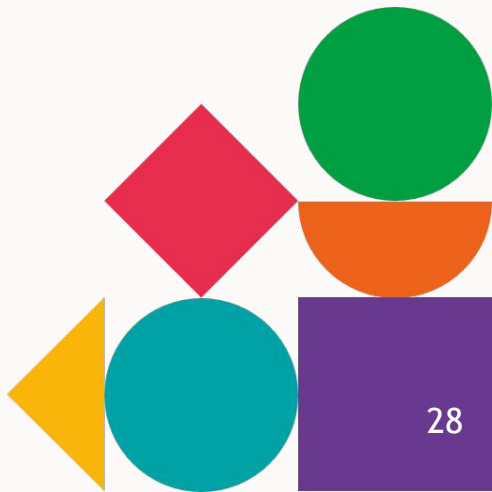
https://www.mentimeter.com/app/presentation/alzz5i5eov4umtatv9ctmj1ywx37wwsc/edit?question=ubeiyegirwu6

Biases; the challenge

1. “Model collapse”:
   Training genAI with synthetic data instead of human organic data

   a. Currently no clear distinction of organic vs synthetic data.

   b. Feedback loops: converging towards narrow, generalised, and homogeneous output



Biases; the challenge

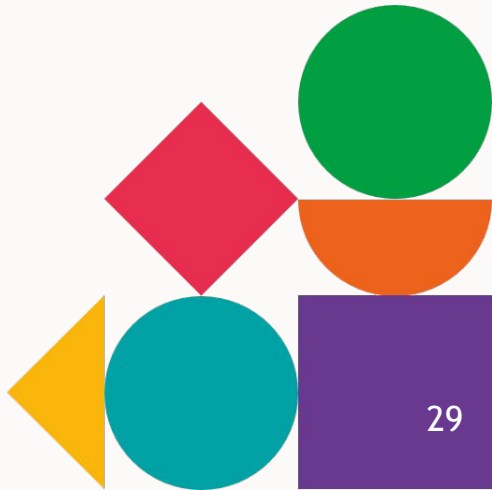
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   d. Reinforcing harmful societal norms and diminishing socio-cultural diversity



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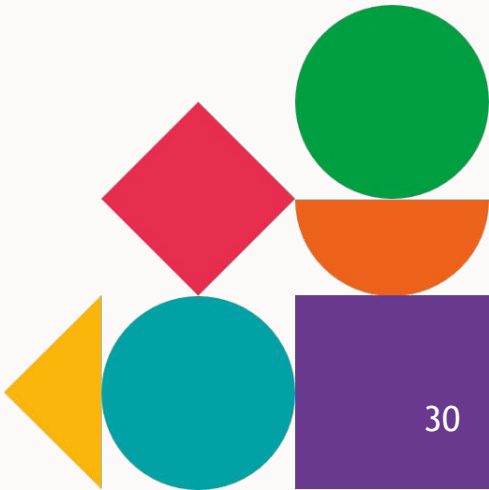
   b. Feedback loops: converging towards narrow, generalised, and homogeneous output

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   e. Diminished models' accuracy or diversity

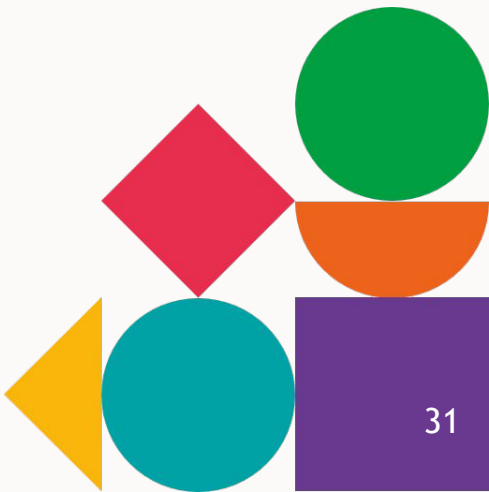
   f. Increasingly limited in creativity and innovation



Biases; the challenge

2. Loss of tail data:
   Uncommon occurrences are eliminated from training data, obscuring entire concepts.

   a. Low-probability events are particularly likely to be "forgotten" by AI algorithms (e.g.: marginalised populations or unusual settings)

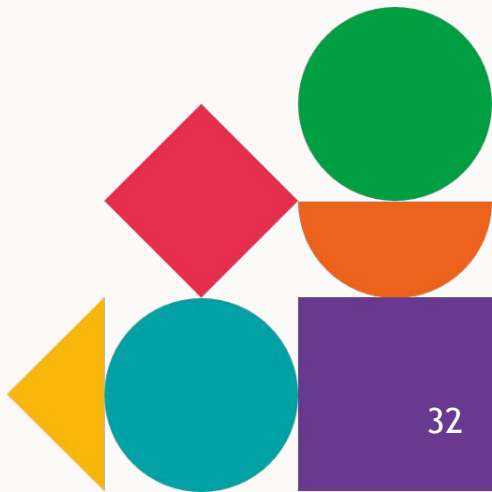


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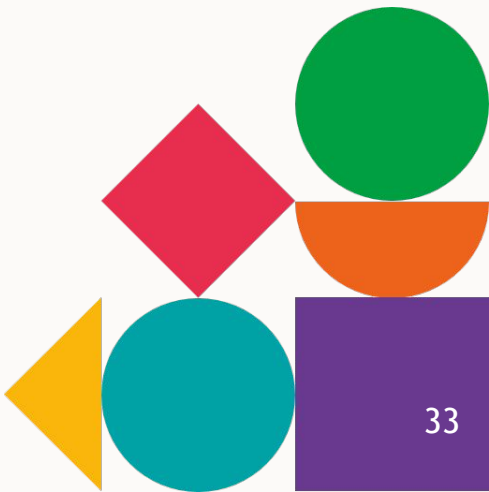
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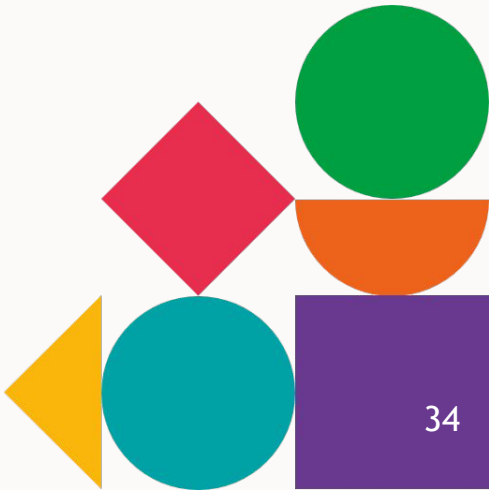
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   d. Especially concerning in fields such a

      - educational content
      - scientific research



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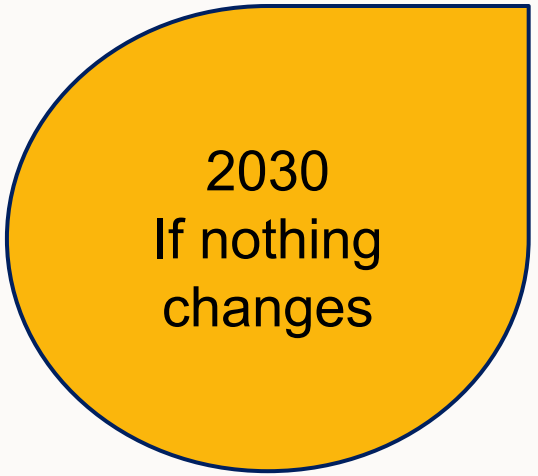
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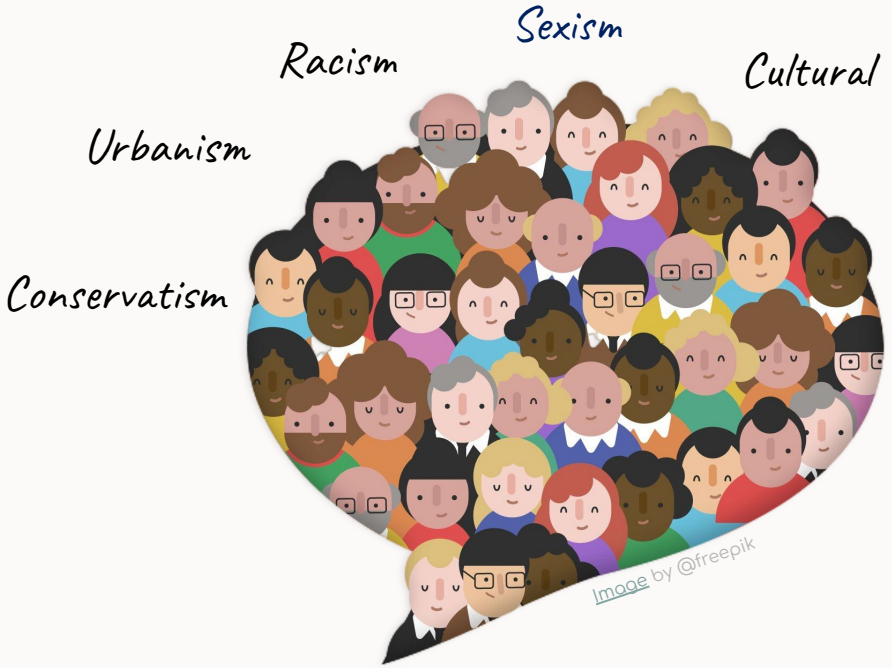
      - educational content
      - scientific research



3. Increase of "safe" responses, resulting in bland and homogeneous results

General biases

- 44% gender bias in 133 AI systems
- + ¼ gender and racial bias
- AI completes a sentence
  (gender is explicit)
  - 20% cames back sexist or misogynistic



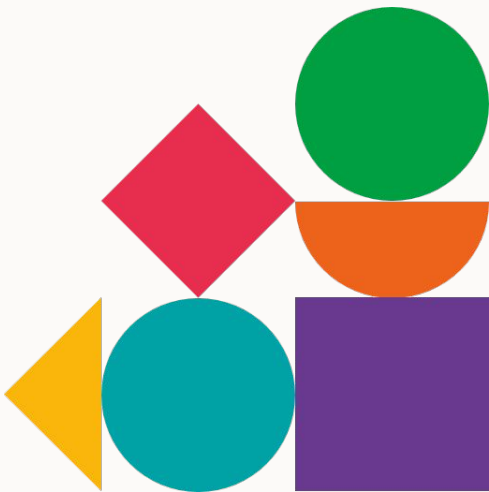
UNITED NATIONS

Biases in clinical use

- Analyzing medical images
- Monitoring health metrics through wearable devices
- Predicting outcomes from Electronic Medical Records

Causes of biases

- missing sociodemographic data
- imbalanced or incomplete datasets
- Restricted or niche representativeness



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Types of Bias encountered in healthcare AI.

Human origin:

- Cause: Historic or prevalent human perceptions, assumptions, or preferences.
- Consequence: Potentially past injustices are perpetuated into future practice

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Algorithm deployment:

Cause: Feedback loop, Induced automated biases, alarme fatigue.

Consequence: Unconditionally trust and follow AI, reinforces wrong data, ignoring warnings due previous AI false alarm.

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genAI and medical images biases

Study reveals why AI models
analyze medical images can
biased - Anne Trafton | MIT
Publication Date:
June 28, 2024

“demographic shortcuts” when making their diagnostic evaluations, which lead to incorrect results for women, Black people, and other groups, the researchers say.

approached to “debiasing” worked best when the models were tested on the same types of patients they were trained on, such as patients from the same hospital. When these models were applied to patients from different hospitals, the fairness gaps reappeared.

“fairness gaps” — that is, discrepancies between accuracy rates for men and women, and for white and Black patients.

that is, they do not make the best trade-off between overall and subgroup performance

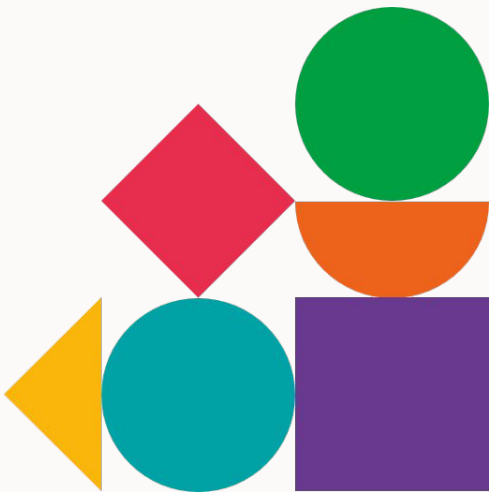
The findings suggest that hospitals that use these types of AI models should evaluate them on their own patient population before beginning to use them, to make sure they aren’t giving inaccurate results for certain groups.



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Biases in plant research

1. AI-generated advice on crop protection reveal systematic biases, faulty recommendations, and unethical machine behavior.
2. AI-powered advisories tend to favor commodified inputs and restrict the plurality of non-chemical alternatives
3. biological control, soil health and diversification tactics carry notable social-environmental benefits, they are systematically downgraded or even excluded by certain AI systems.



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Break
until 11:00

Hallucinations

General and specifics for research

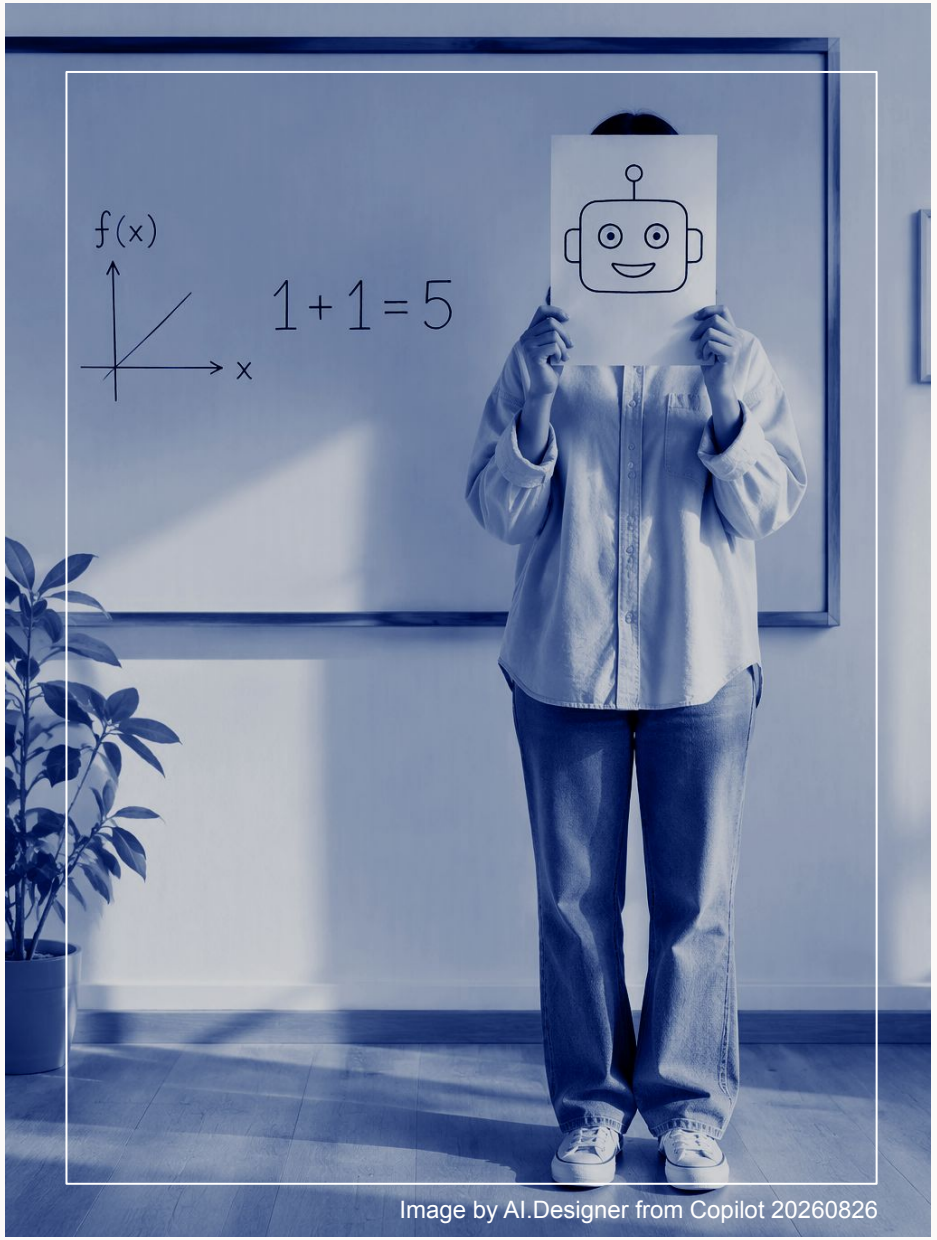


Image by AI.Designer from Copilot 20260826

Hallucinations and Research integrity

Accuracy, reliability, and verifiability of research

Research integrity depends on a basic premise: that claims can be traced, verified, and built upon by others.

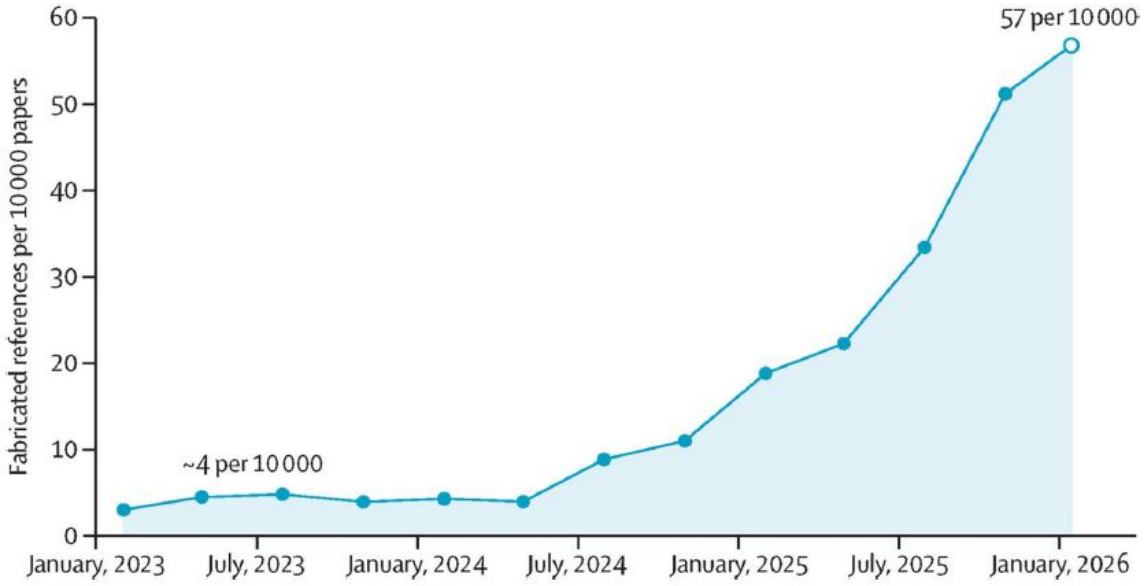


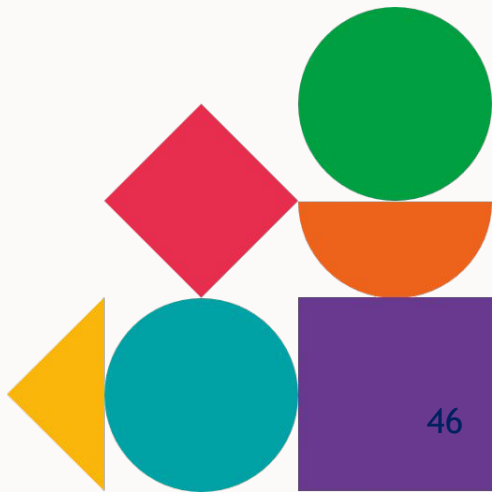
Fig. 2 Fabricated references on the rise. Note: A *Lancet* study (Topaz et al. 2026) tracked the monthly rate of fake citations per 10,000 papers within the PubMed Central database from January 2023 to early 2026. Throughout 2023, the rate remained stable at approximately 4 citations per 10,000 papers. Beginning in mid-2024, the rate rose sharply, coinciding with the widespread use of AI writing tools. By early 2026, the rate reached approximately 57 per 10,000 papers, a more than 12-fold increase since 2023. The audit analyzed 2.5 million papers and 97.1 million references in the biomedical field, uncovering 4046 fabricated citations across 2810 individual papers

Types of hallucination

- **Context disconnected:**
  Logical Inconsistent such as self-contradicting or out of sync. content in relation to provided context

- **Semantic distortion:**
  Underlying meaning is misrepresented or altered in the output

- **Content hallucination:**
  Unreal or absent output, given the input context



Example
(11/10/26)

if two people sahre the same great-grandparents but different grandparents, tey are:

a- first cousins
b- second cousins
c- frist cousing once removed
d- third cousin

Oct 11

Answer: b) Second cousins

generate related questions and answer it

Here are some similar family-relationship questions with answers:

1. If two people share the same grandparents, they are:

a) Siblings
b) First cousins
c) Second cousins
d) Aunt and niece

✅ **Answer: b) First cousins**

Mancoridis et al. (2025) Potemkin Understanding in Large Language Models
https://arxiv.org/pdf/2506.21521 Fig 6 (question example)

Example

is the follwoing correct?

at a family reunion, you meet someone who says "" my dad's grandpa and your mom's grandpa were brothers."What is your relationsship?

Answer: you would be third cousins

No, the answer is **not quite correct**.

Let's work it out:

- **Your mom's grandpa** = your **great-grandfather**.
- **Their dad's grandpa** = their **great-grandfather**.
- Those two great-grandfathers were **brothers**.

That means you and the other person each descend from a different brother. The common ancestors are the **parents of those brothers** (the siblings' parents).

For both of you, those common ancestors are **great-great-grandparents**:

- You → Mom → Grandparent → Great-grandpa → **Great-great-grandparents**
- Them → Dad → Grandparent → Great-grandpa → **Great-great-grandparents**

People who share the same **great-great-grandparents** are **third cousins**.

✅ **Answer: Third cousins**

Your question and answer are therefore **correct**.

Mancoridis et al. (2025) Potemkin Understanding in Large Language Models https://arxiv.org/pdf/2506.21521
Fig 6 (question example)

Types of hallucination

- **Factual inaccuracy:**
  Inaccurate or deceptive information/facts in the output (scientific findings, clinical data, historical data, etc)

- **Citation fabrication:**
  Creating fake references including real authors, wrong journals or invented titles

- **Propagation errors:**
  Chain effect, one error enter a publication and contaminates future research

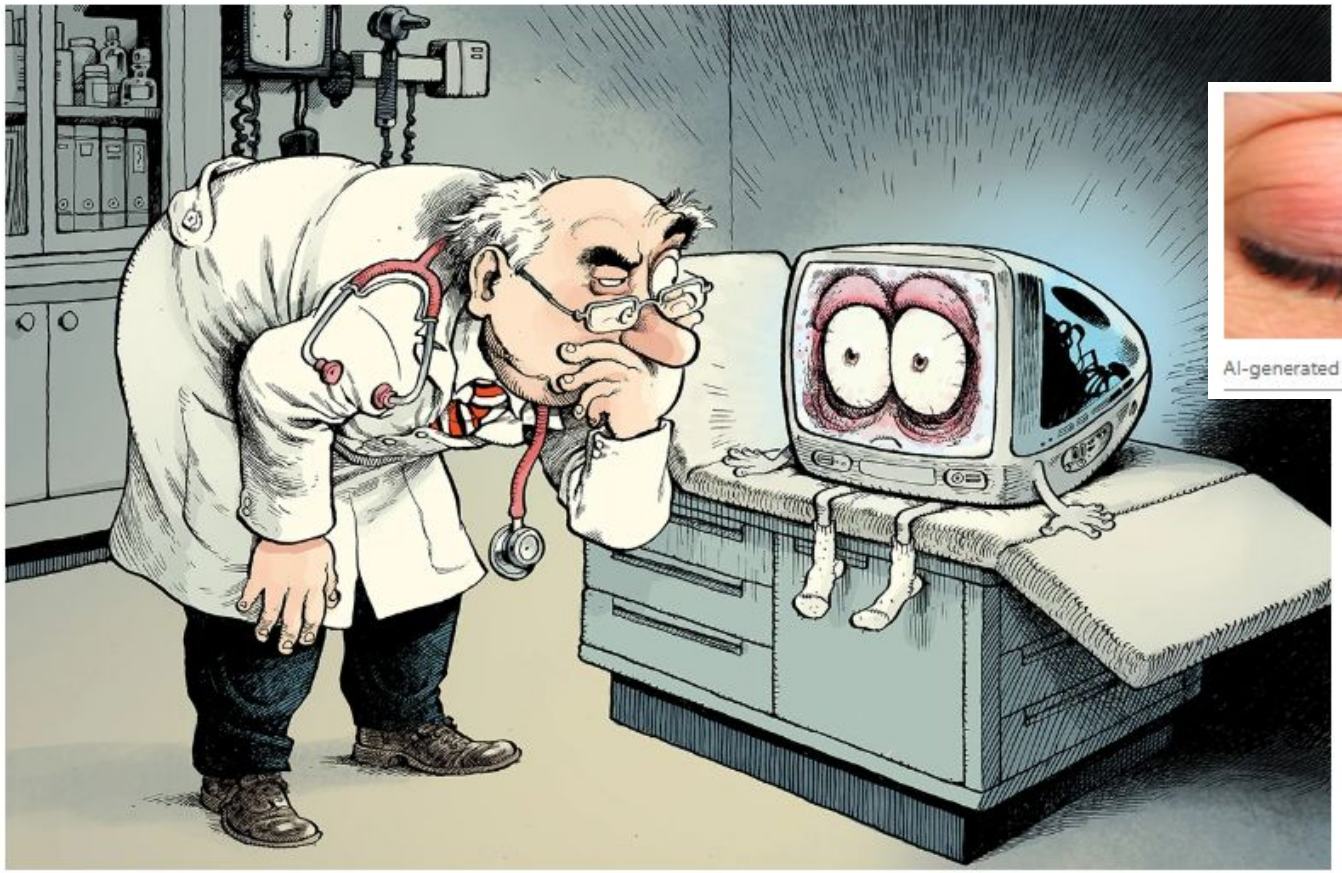


Scientists invented a fake disease. AI told people it was real

Bixonimania doesn't exist except in a clutch of obviously bogus academic papers. So why did AI chatbots warn people about this fictional illness?

By Chris Stokel-Walker

Save article





AI-generated images of bixonimania, a fictitious illness. Source: Preprints.org https://doi.org/qzm4 (2024).

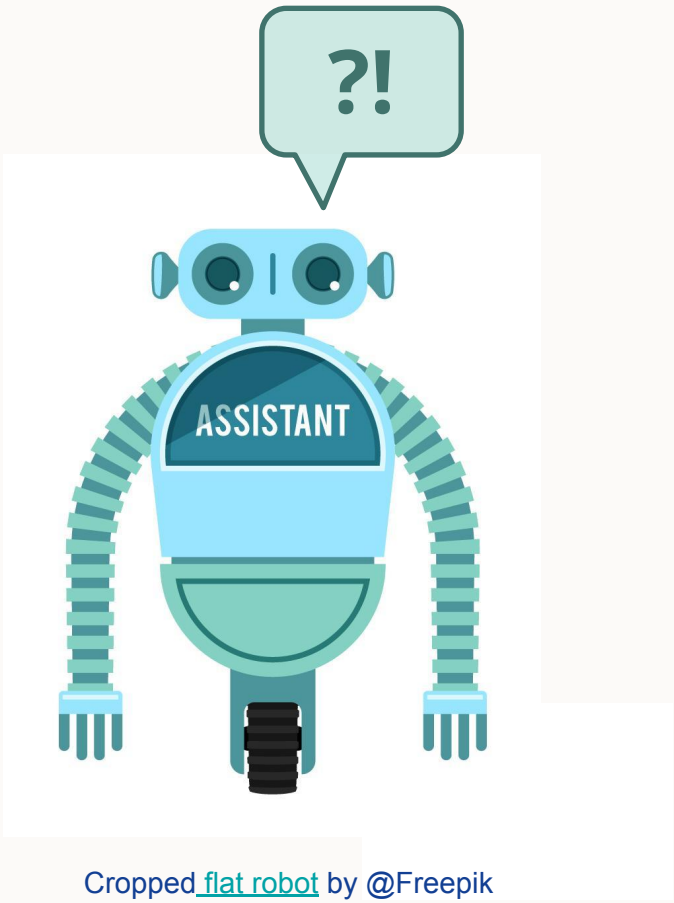
…"this entire paper is made up" …

… "Fifty made-up individuals aged between 20 and 50 years were recruited for the exposure group"...

https://www.nature.com/articles/d41586-026-01100-y

AI Hallucination Report 2025 Which AI Hallucinates the Most?

(link) By Hira Ehtesham December 2025.



Cropped flat robot by @Freepik

Domain-Specific Hallucination Rates

Even the best models show varying hallucination rates across different domains:

Knowledge Domain	Average Hallucination Rate (Low Hallucination Group)	Average Hallucination Rate (All Models)
General Knowledge	0.8%	9.2%
Legal Information	6.4%	18.7%
Medical/Healthcare	4.3%	15.6%
Financial Data	2.1%	13.8%
Scientific Research	3.7%	16.9%
Technical Documentation	2.9%	12.4%
Historical Facts	1.7%	11.3%
Coding & Programming	5.2%	17.8%

GAI reduces text diversity (Liang et al. 2024).

Often – SelfCheck black-box: based on methods that rely that LLM can produce consistent comparison

"hallucination snowballing," where they fabricate false claims to rationalize prior hallucinations despite acknowle
their inaccuracy

Hallucinations: Fabrication in the Academic Context

A hallucination is output that is fluent and grammatically correct but factually incorrect, fabricated, or logically inconsistent with verifiable knowledge—produced not through genuine understanding but through statistical pattern prediction. In academic writing these manifest in specific, documented forms: pubmed.ncbi.nlm.nih +2

- as an emergent property of how LLMs are built: they gener- t, or falsely reporting the
- ate text by predicting the most statistically probable next token in a sequence, not by consulting a reliable database of facts. The model has no mechanism for checking whether
- what it says is true. It simply produces what fits the pattern. ate reasoning.
- **Propagation errors** — errors that spread when AI output is reused or cited downstream.

pubmed.ncbi.nlm.nih

Using AI without systematic
verification should be under
choice but as a breach of re

ensure the
s, data, and

GenAI generates new content rather than just classifying existing data, it *heightens* the risk of perpetuating and amplifying biases in the training data and algorithms.

prompting-induced hallucinations (vague, underspecified, or misleading prompts push the model into speculative generation) and **model-intrinsic** hallucinations





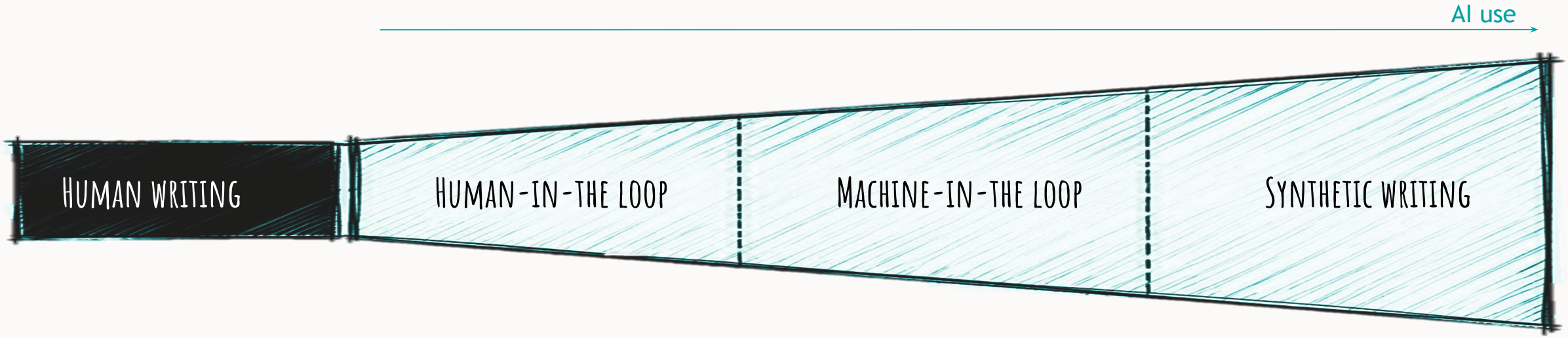
Break
until XX:YY

02

genAI in writing

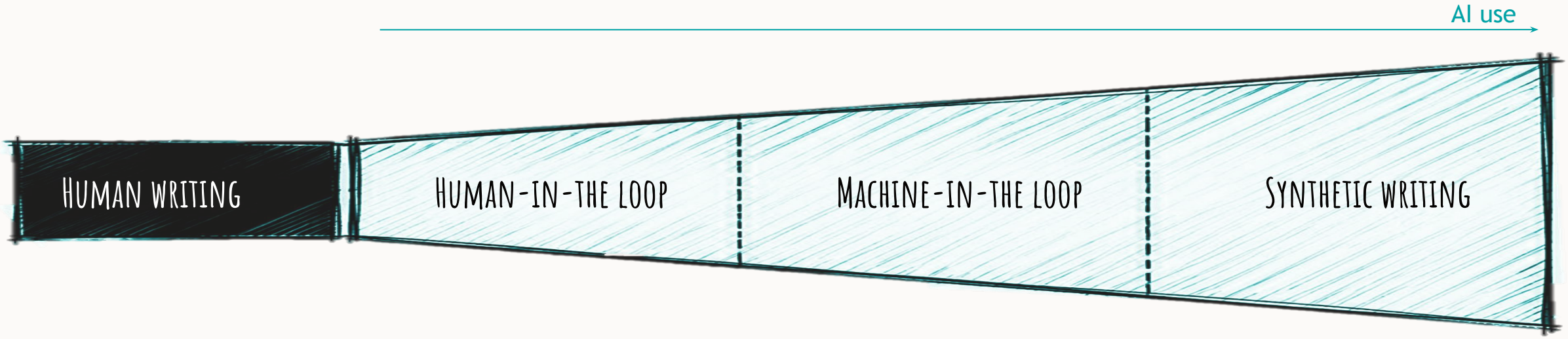
(approaches)

AI use in the writing process



Adapted from: Knowles, A. M. (2024). Machine-in-the-loop writing: Optimizing the rhetorical load. *Computers and Composition, 71*, 102826. https://doi.org/10.1016/j.compcom.2024.102826
https://moxielearn.ai/blog/how-to-use-ai-for-academic-writing-a-step-by-step-guide;

AI use in the writing process



Which level of AI would you suggest for different tasks?

e.g.:

Cultural analysis

Early drafts

Article summaries

Guidance on thesis work

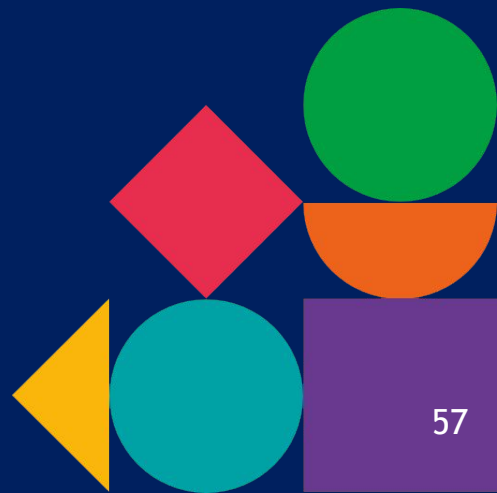
Adapted from: Knowles, A. M. (2024). Machine-in-the-loop writing: Optimizing the rhetorical load. *Computers and Composition, 71*, 102826. https://doi.org/10.1016/j.compcom.2024.102826
https://moxielearn.ai/blog/how-to-use-ai-for-academic-writing-a-step-by-step-guide;

Activity

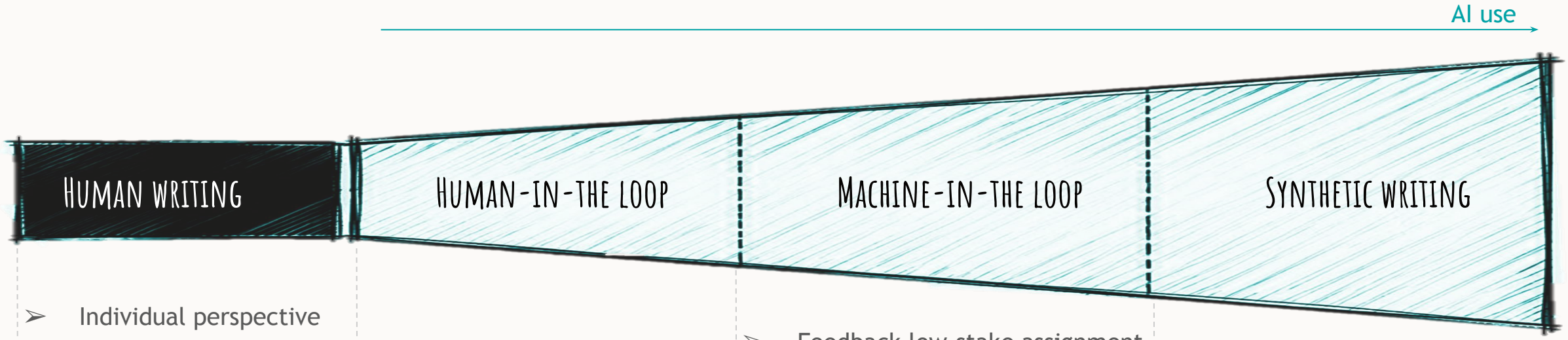
00 - What scholar activities would you put in what category?

1. Write down 5 scholar activities you often do, one in each Post It
2. Put your Post Its under the respective category on the wall

Find it onGitHub:
https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work
Repo >> exercises >> exercise00



AI use in the writing process



- Individual perspective
- Cultural analysis
- Subject-specific
- evaluation
- Philosophical discussion

- Essay feedback
- Creation of educational material
- Tools language acquisition
- Guidance on thesis work
- Enhanced peer review

- Feedback low-stake assignment
- Early drafts
- Questions for discussions
- SOPs

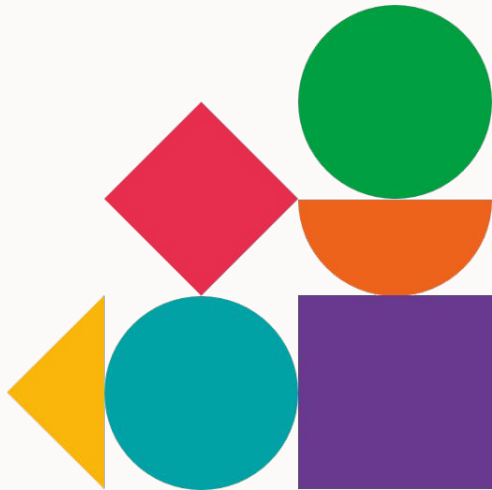
- Articles summaries
- Standardised
- Analysis reports
- Formatting citations

Adapted from: Knowles, A. M. (2024). Machine-in-the-loop writing: Optimizing the rhetorical load. *Computers and Composition, 71*, 102826. https://doi.org/10.1016/j.compcom.2024.102826
https://moxielearn.ai/blog/how-to-use-ai-for-academic-writing-a-step-by-step-guide;

Groups of AI usage

Survey fielded as a census of all researchers at Danish universities

- 1st : Uses genAI "as a workhorse" (35.2%)
  - to speed up processes in research
  - to assist with technical issues related to research
  - sceptical about using for creative tasks *(e.g.: peer review and idea generation)*

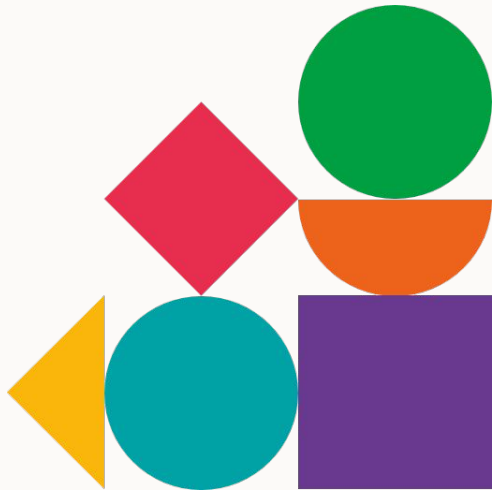


J.P. Andersen et al. Technology in Society 81 (2025) 102813 Generative Artificial Intelligence (GenAI) in the research process – A survey of researchers' practices and perceptions - https://doi.org/10.1016/j.techsoc.2025.102813

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  - most sceptical about using it for in most phases of the research process



J.P. Andersen et al. Technology in Society 81 (2025) 102813 Generative Artificial Intelligence (GenAI) in the research process – A survey of researchers' practices and perceptions - https://doi.org/10.1016/j.techsoc.2025.102813

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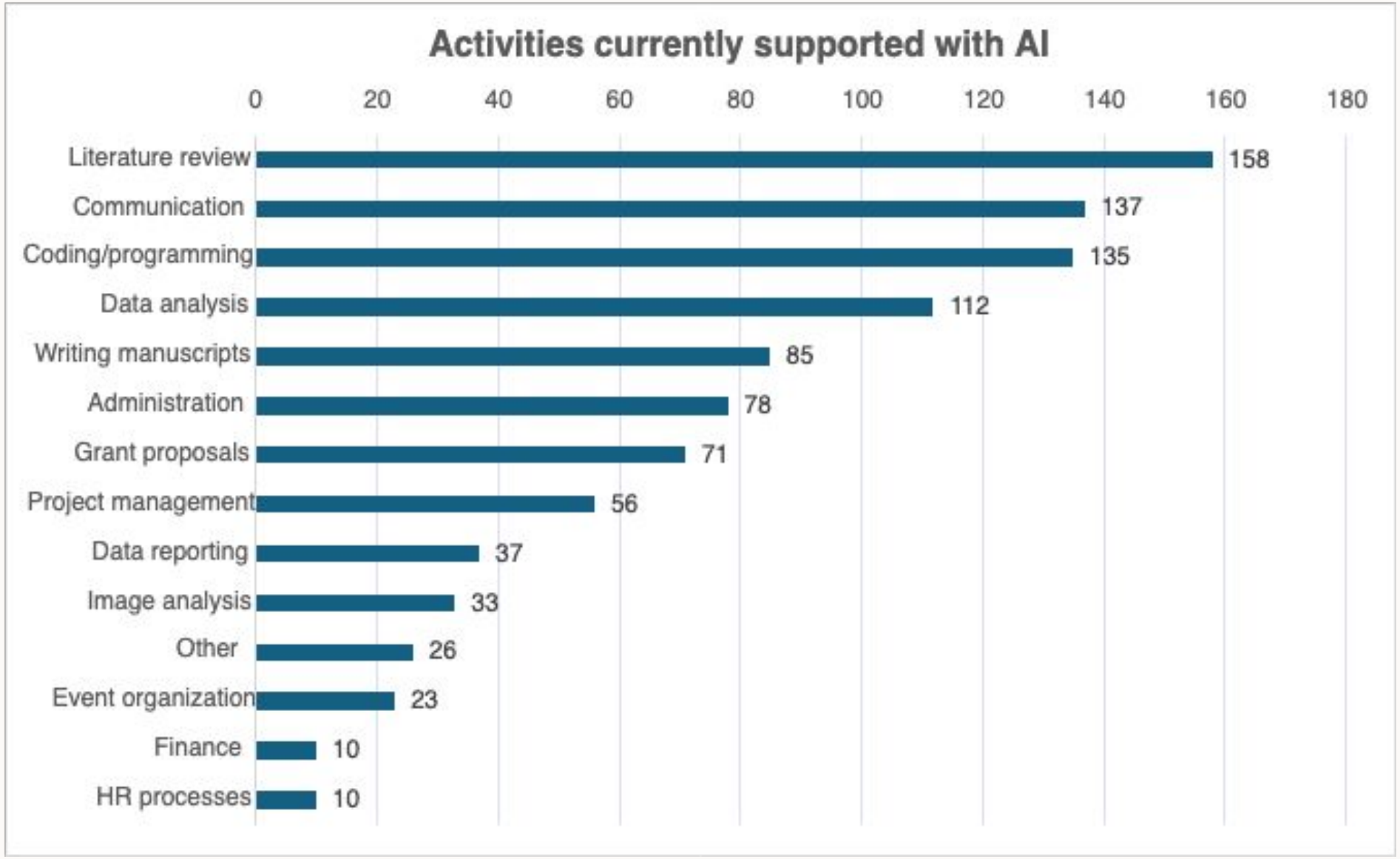
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- 2nd: Uses genAI "as a language assistant only" (24.0%)
  - most sceptical about using it for in most phases of the research process
- 3rd: Uses genAI "as a research accelerator" (40.7%).
  - generally very positive assessment of genAI use cases
  - use to increase productivity, particularly in data analysis and research design



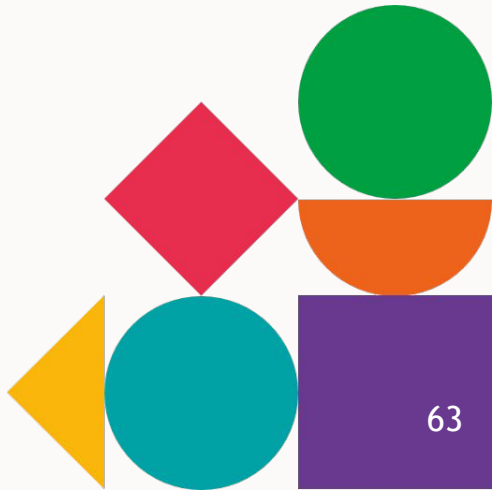
J.P. Andersen et al. Technology in Society 81 (2025) 102813 Generative Artificial Intelligence (GenAI) in the research process – A survey of researchers' practices and perceptions - https://doi.org/10.1016/j.techsoc.2025.102813

VIB AI survey (~ 280 respondents)



Before we start scientific writing …

What do we need to do ?

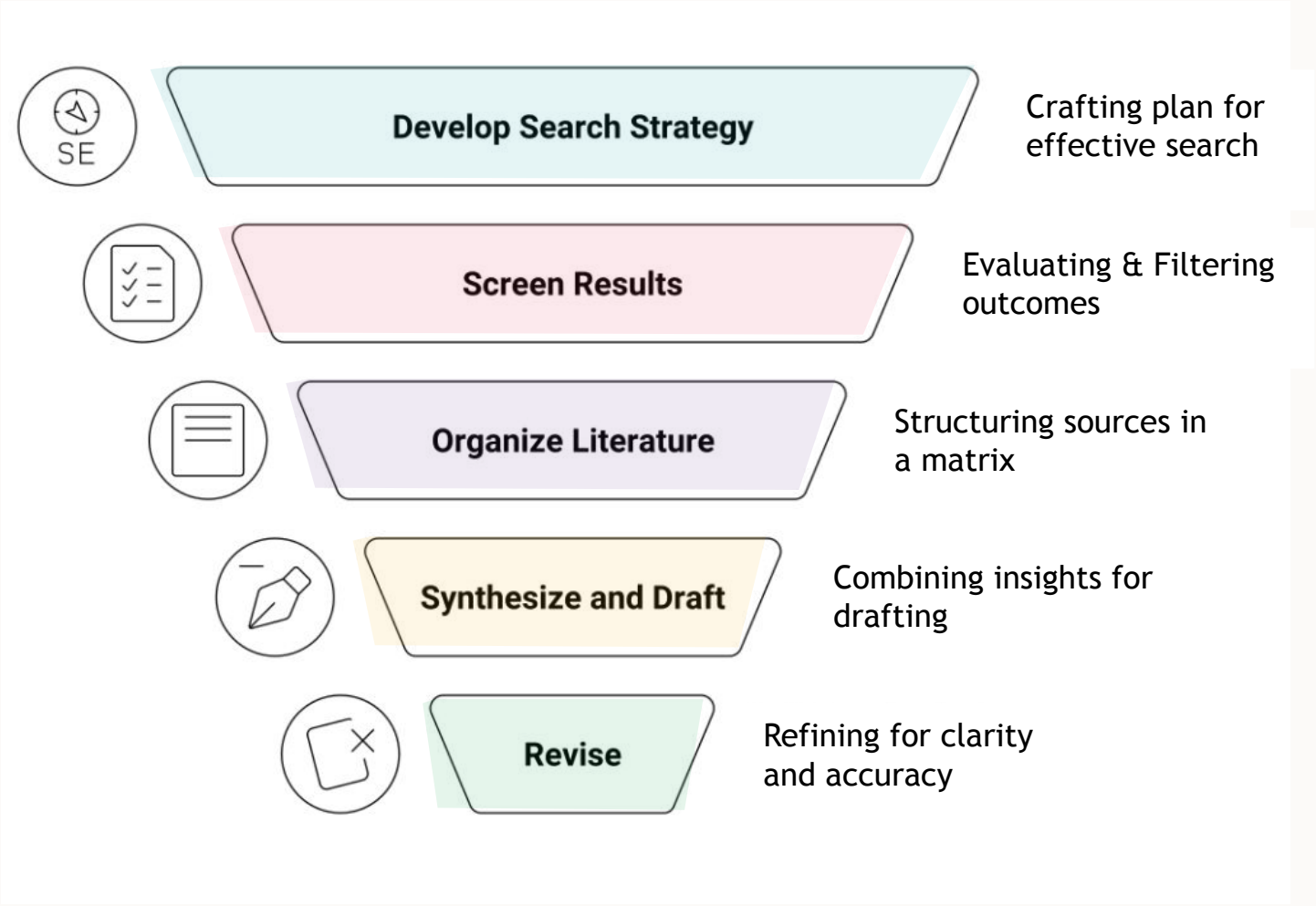


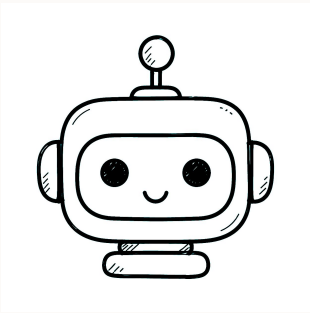
03

genAI 4 Literature Search

(tools and prompting skills)

AI-assisted literature discovery and summarisation

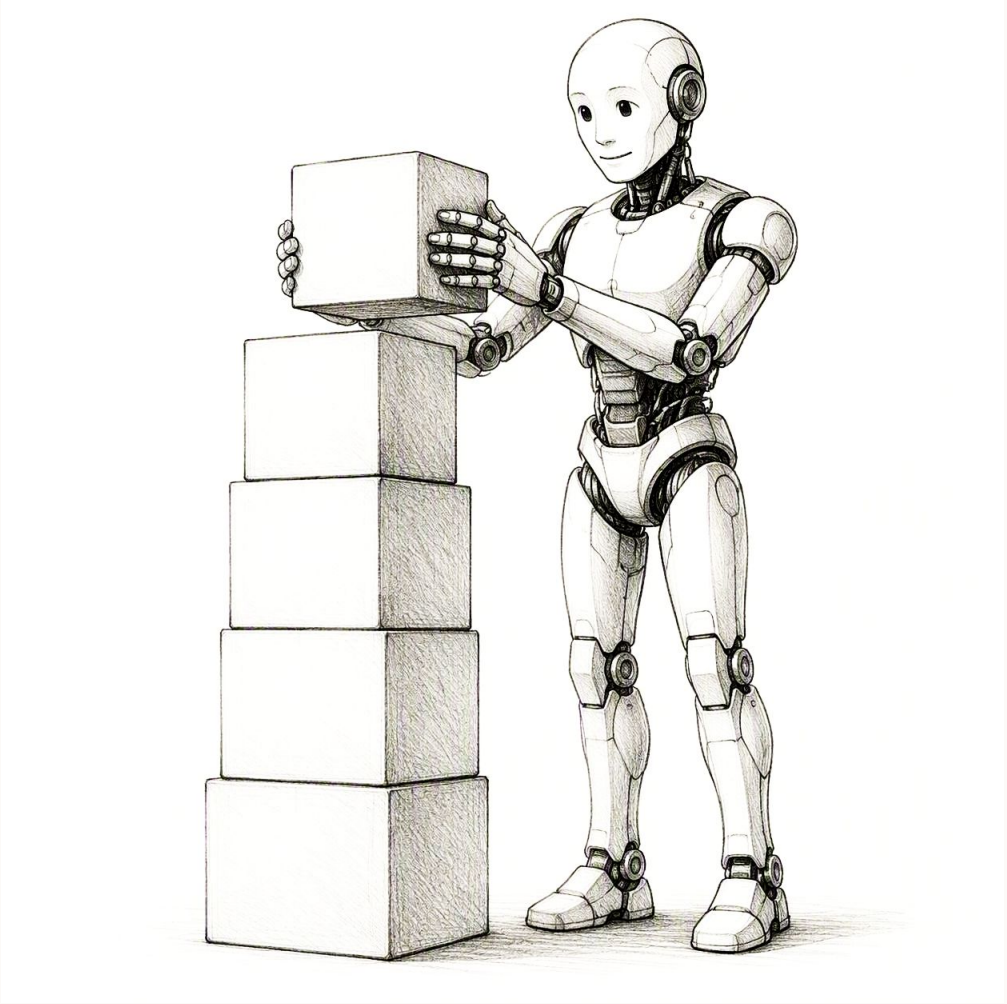




AI EXCELS FOR SEARCH STRATEGIES

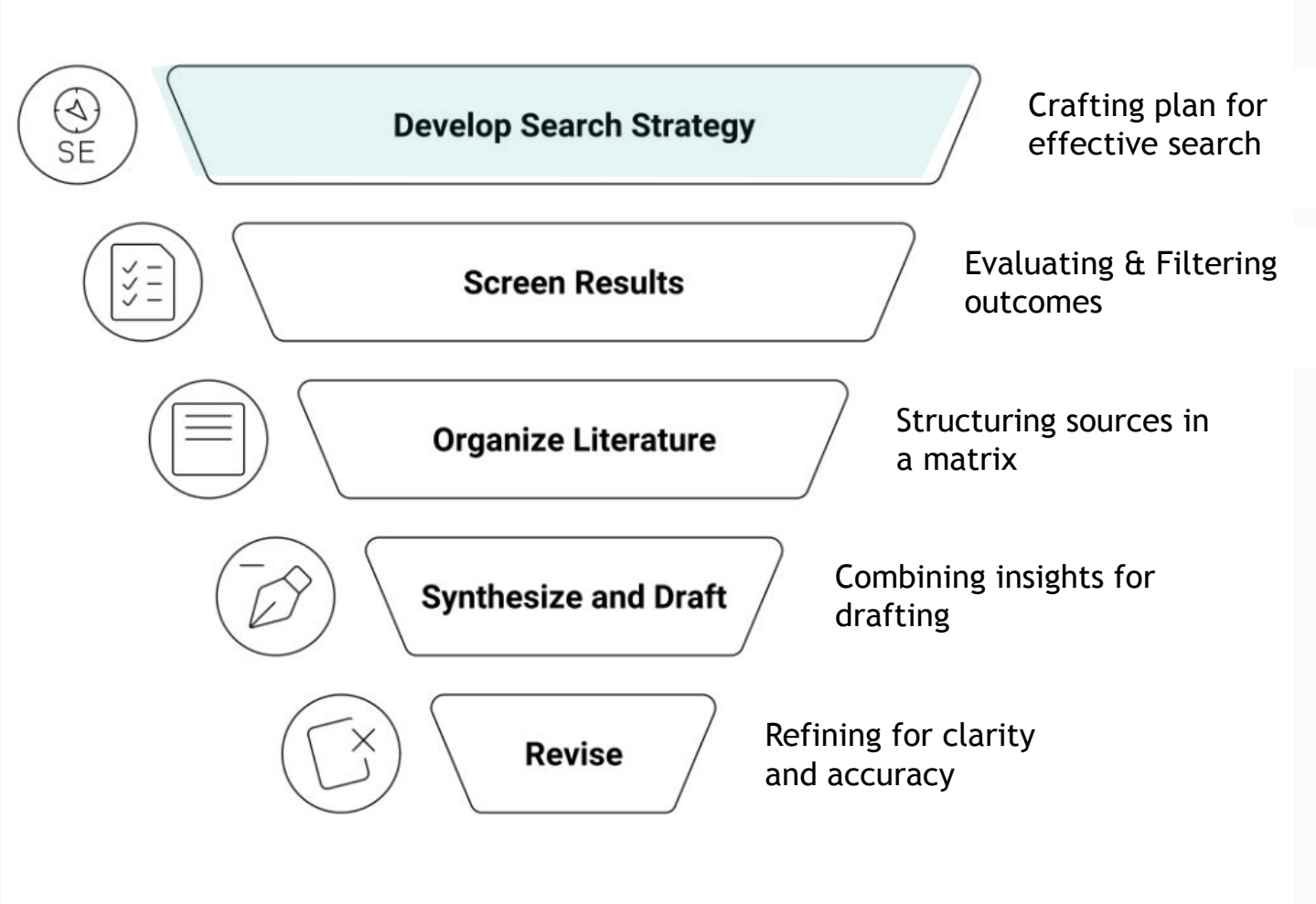
Prompting: Building blocks

- Define AI-Persona
- Give Context
- Clear Verbs and structure
- Define output (Style and Audience)
- Iterations are essential



* sometimes useful

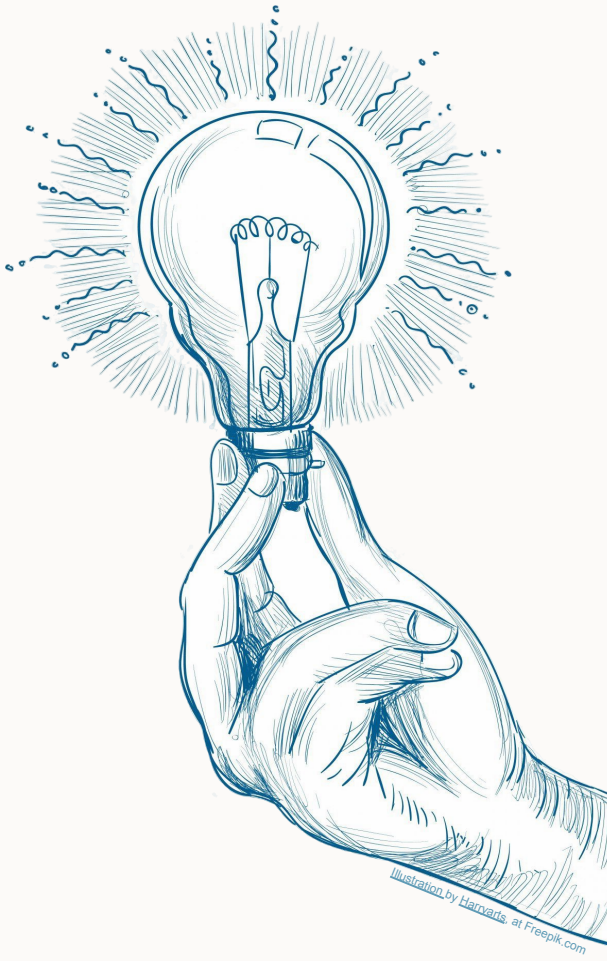
AI-assisted literature discovery and summarisation



LLMs excel at certain questions

1. Unfamiliar concepts

★ As a flow cytometry expert, list what are the key trends in the development of sustainable flow cytometry devices?

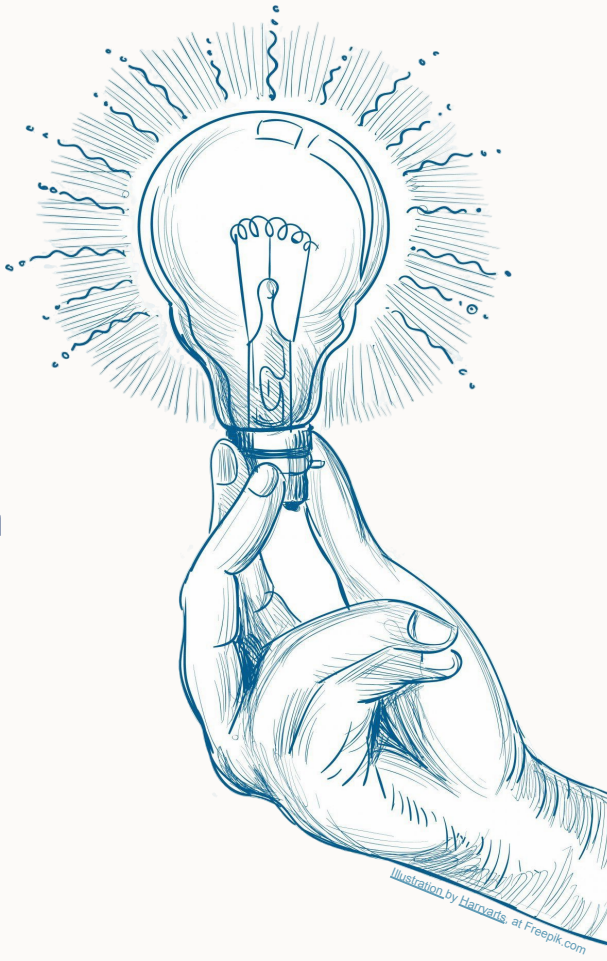


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1. Unfamiliar concepts

★ As a flow cytometry expert, list what are the key trends in the development of sustainable flow cytometry devices?

★ As someone new to Digital Education, I believe *these* tools and methods could be beneficial. Provide an overview of one paragraph for each with main benefits and challenges.



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1. Unfamiliar concepts

★ As a flow cytometry expert, list what are the key trends in the development of sustainable flow cytometry devices?

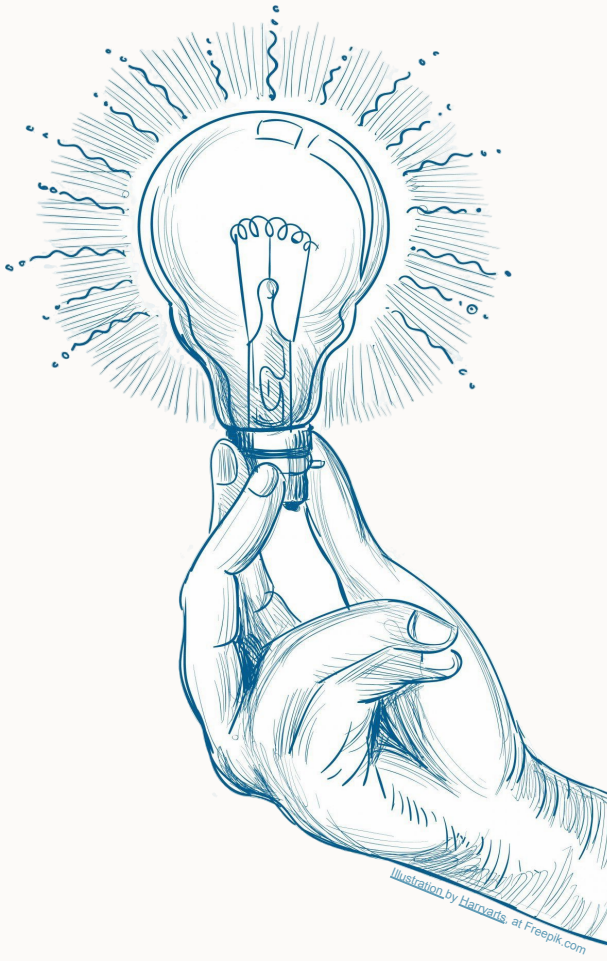
★ As someone new to Digital Education, I believe these tools and methods could be beneficial. Provide an overview of one paragraph for each with main benefits and challenges.

★ Considering a biotechnology institute. Which stakeholders should be engaged in establishing an eco-team at the research-performing organisation?



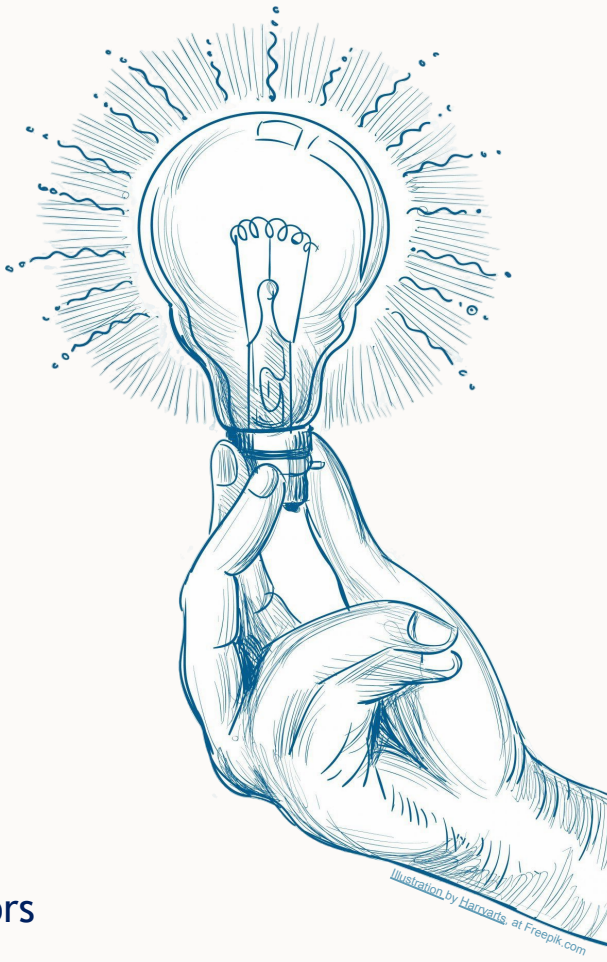
LLMs excel at certain questions

2. Generate Keywords, String search, Topic scoping



LLMs excel at certain questions

2. Generate Keywords, String search, Topic scoping



★ You are an expert in scientific literature search specialized in bioinformatics, proteomics, and computational biology.

Identify keywords that can be used to build Google Scholar search strings, create three suggestions of search strings

(1) Specific
(2) Balanced
(3) Broad

Topic of the search:

★ Explainable AI in Bioinformatics

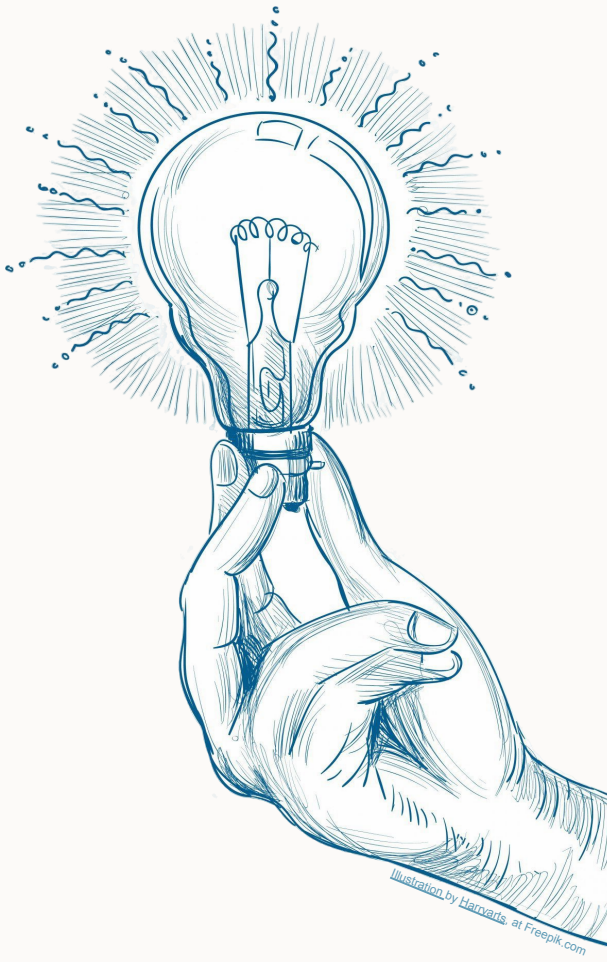
★ Proteomic Responses of Plants to Climate Change Stressors

★ Comparative Proteomics of Antimicrobial-Resistant Bacteria

LLMs excel at certain questions

2. Generate Keywords, String search, Topic scoping (part 2)

- ★ Group keywords into:
  - Biological concepts - Experimental and computational methods - Resources, tools, and data sources -
- ★ Include important synonyms and related terms.
- ★ Define Output (Style)
  - A. Main Keywords
  - B. Related Terms and Synonyms | Concept | Alternative Terms

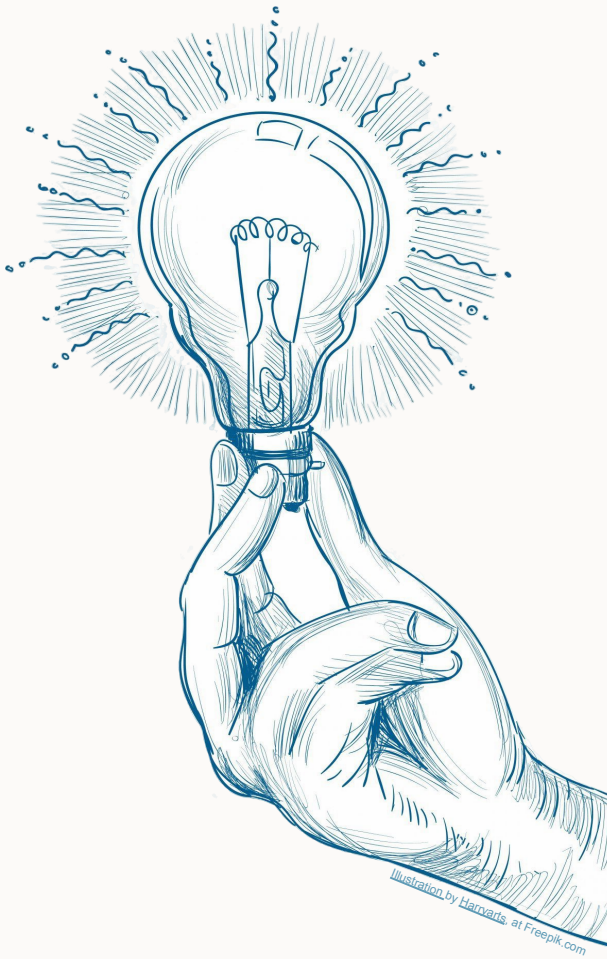


LLMs excel at certain questions

2. Generate Keywords, String search, Topic scoping (part 2)

Iterations

- ★ What search string could I use in PubMed or a library database to find relevant literature? Give an example for each of the options.
- ★ Identify 5 specific topics in this area for further exploration. (Create an additional search string for PubMed.)



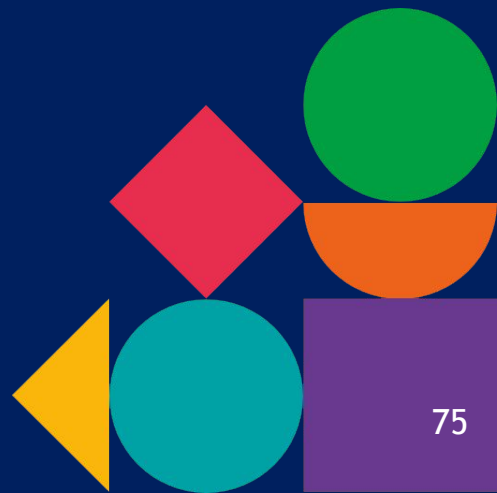
Activity

01 - Google scholar

Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise01

1. Use Copilot to generate keywords and generate a Google Scholar search string
2. Iteration: Ask for different keywords
3. Apply the string and explore what comes out of Google Scholar
4. Iteration: identify topics for further exploration



Activity

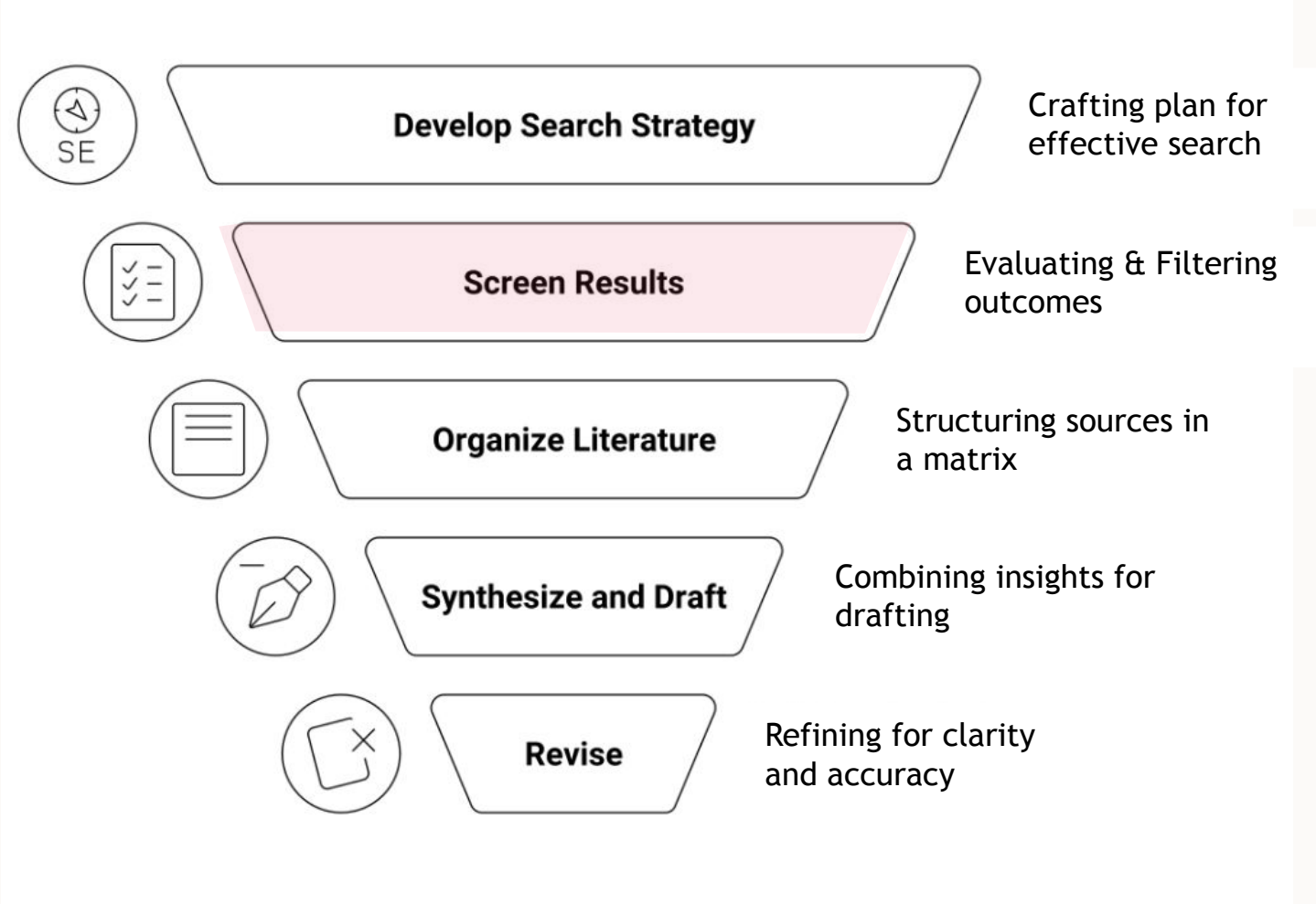
01 - Google scholar

Possible result

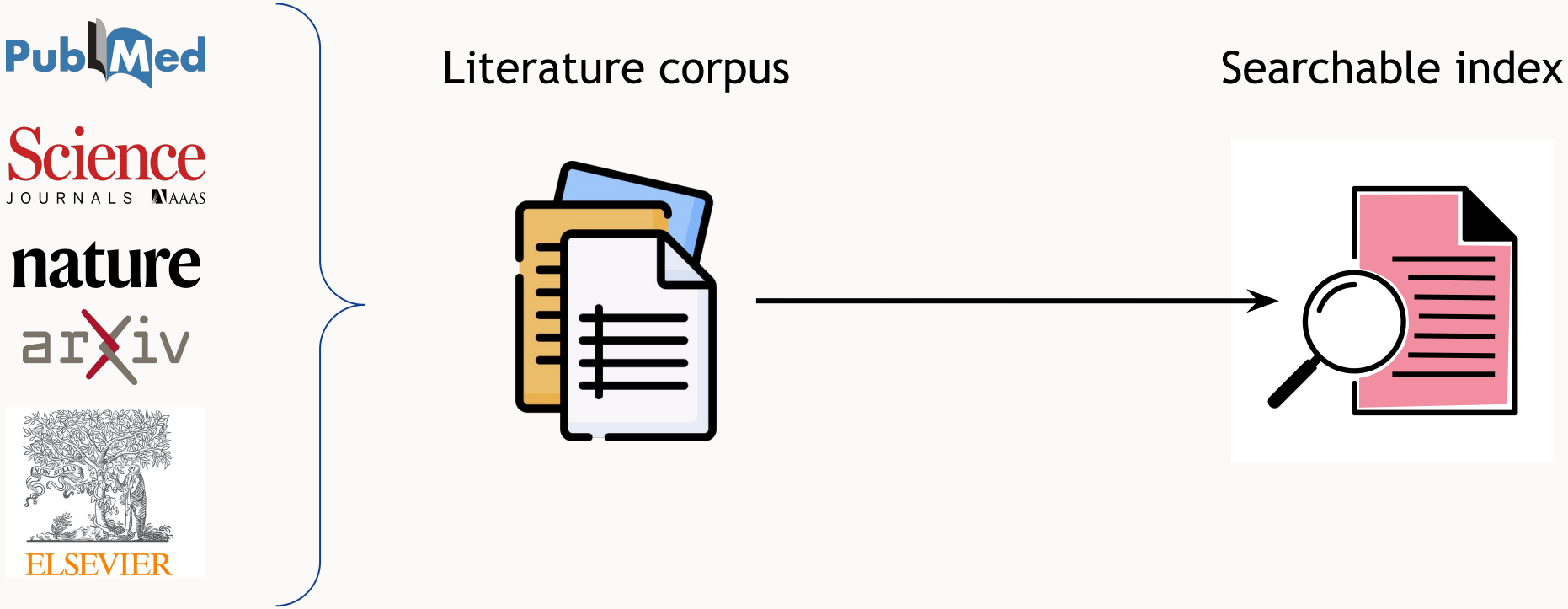
("scientific collaboration" OR "research teamwork" OR "lab dynamics" OR "interdisciplinary science") AND ("biology labs" OR "life sciences") AND ("sociology of science" OR "ethnography" OR "social networks" OR "knowledge sharing" OR "power relations")

- Did everyone get the same combination?
- What papers do you get back?
- Do you believe it has improved your search?

AI-assisted literature discovery and summarisation

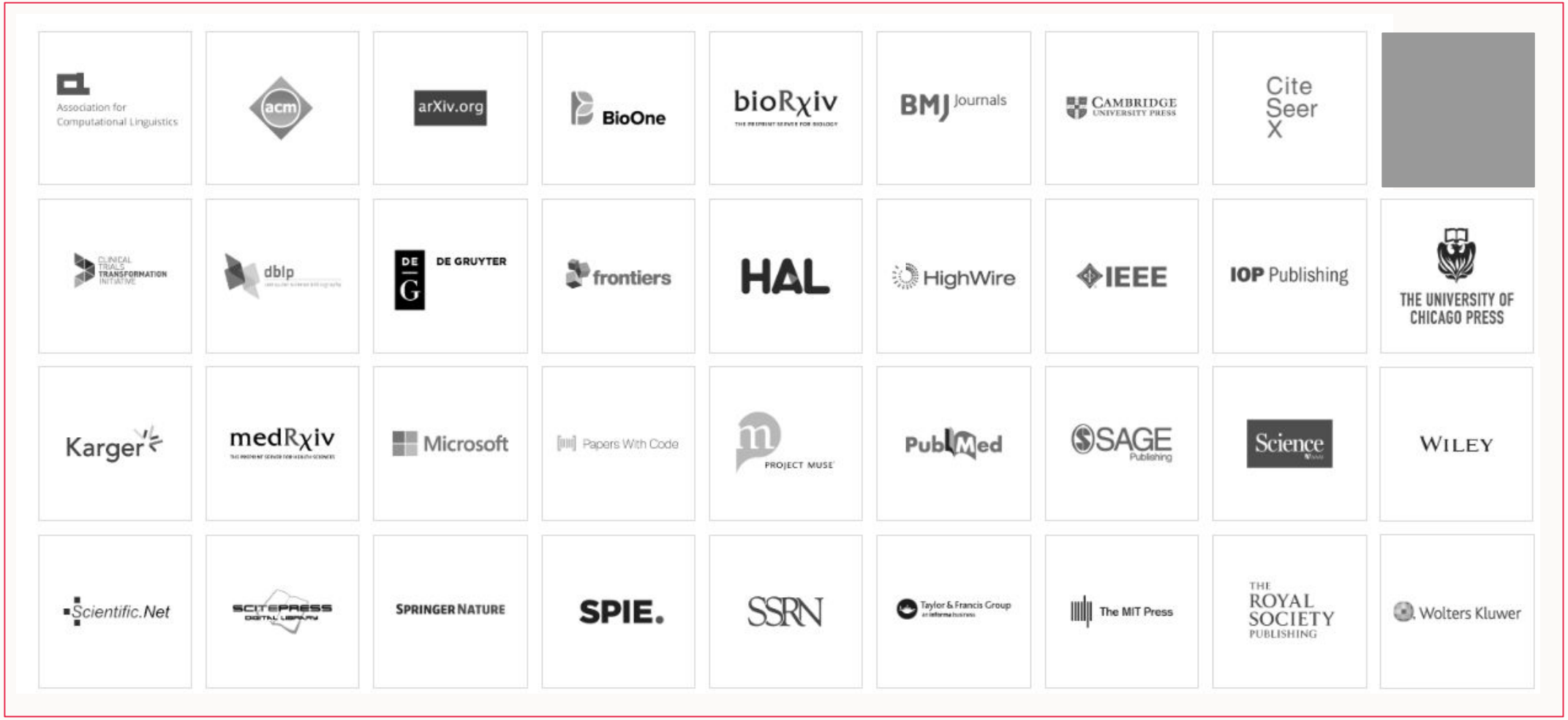


Searching and filtering relevant literature: indexing

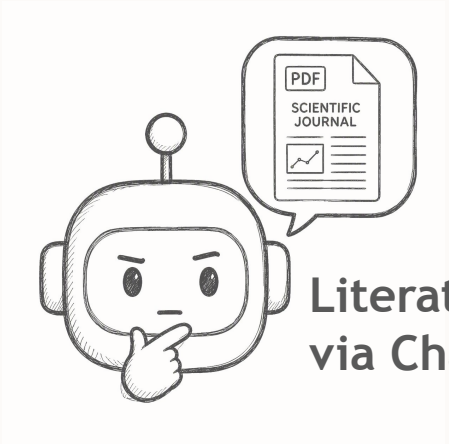


Semantic Scholar: a corpus for AI models in academia

semanticscholar.org (non-profit Allen Institute for AI)

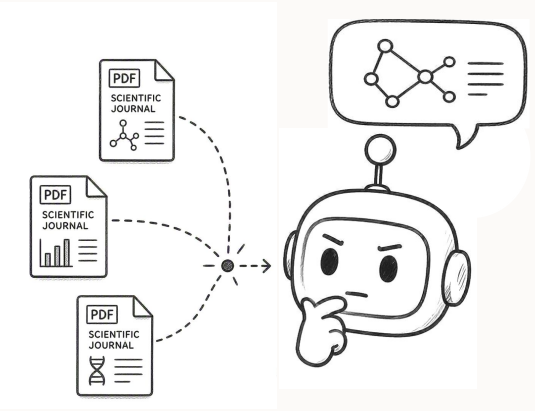


Different flavors of AI discovery tools



Literature Discovery via Chatbot

Literature connectors



Asta

Undermind

ResearchRabbit

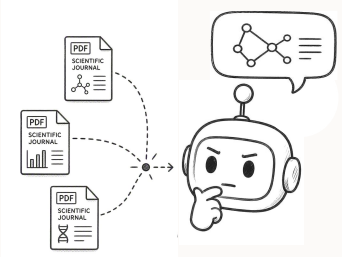
perplexity

consensus

Litmaps

CONNECTED PAPERS

Chatbots



Literature research

Activity

02 - Comparing Literature search

Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise02

1. Use the same prompt to compare results in https://undermind.ai, https://perplexity.ai and https://asta.allen.ai/

2. Prompt:

> As a literature search expert and bioinformatician, list scientific publications about multiple omics and human protein complexome and metabolism.
>
> List one publication per line with the link for the paper.

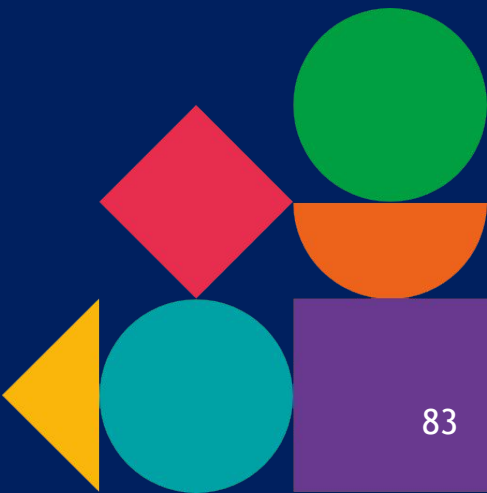
Activity

02 - Comparing

Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise02

1. Ask details about a research topic that interests you
   (scientific topic you want to discover, something you want to write a report on, something you recently worked on, etc.)



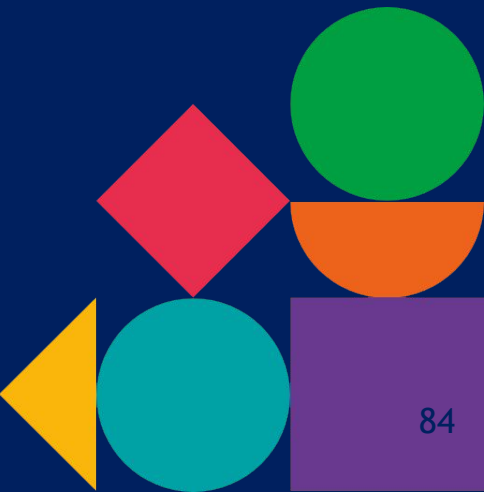
Activity

Want to know more about a new tool? → Ask your chat tool, or the tool itself

What can you tell me about the Undermind scientific research tool?

1. What can you do for free and what can you do with paid accounts?
2. What are the data sources the tool is using?
3. What models is the tool relying on?
4. What other things will this tool do?

   (Summarise, alerts, citations connected to Zotero, etc.)

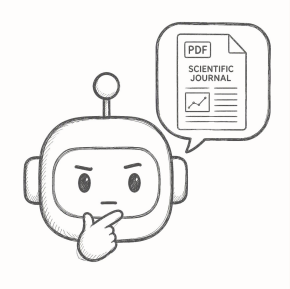


Reference in paper-Mainak.rtf
Matching reference in search-papers.bib.txt from undermind.ai
Match type

Buljan et al. (2023). *A computational framework for the inference of protein complex remodeling from whole-proteome measurements* (Nature Methods)
Bul23
Exact match (same DOI: 10.1038/s41592-023-02011-w) [paper-Mainak | Word], [search-papers.bib | Txt]
Goel et al. (2024). *Trends in co-fractionation mass spectrometry: A new gold-standard in global protein interaction network discovery*
Goe24
Exact match (same DOI: 10.1016/j.sbi.2024.102880) [paper-Mainak | Word], [search-papers.bib | Txt]
Nolte & Langer (2021). *ComplexFinder*
Nol21
Exact match (same DOI: 10.1016/j.bbabio.2021.148444) [paper-Mainak | Word], [search-papers.bib | Txt]
Wittig & Malacarne (2021). *Complexome Profiling: Assembly and Remodeling of Protein Complexes*
Wit21
Exact match (same DOI: 10.3390/ijms22157809) [paper-Mainak | Word], [search-papers.bib | Txt]
Bludau et al. (2023). *Rapid Profiling of Protein Complex Reorganization in Perturbed Systems*
Blu21
Same paper (title and DOI identical; year differs in BibTeX entry) [paper-Mainak | Word], [search-papers.bib | Txt]
Fossati et al. (2021). *PCprophet*
Fos20
Preprint → published version (same study; bioRxiv precursor of PCprophet paper) [paper-Mainak | Word], [searc

Include a summary slide with advantages of each tool for a good overview of participants

Chatbots



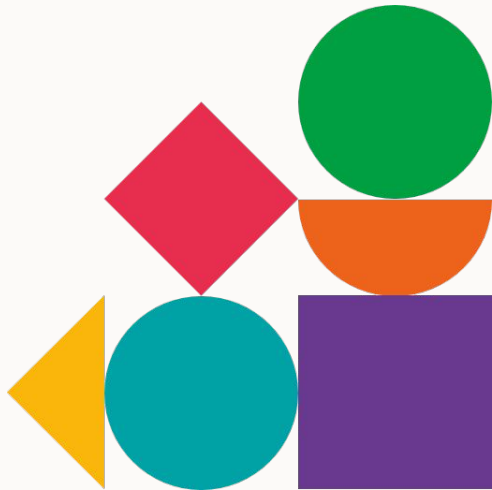
Literature connectors

Connecting papers

- Mimics a kick-off human approach
  - Search for the cited papers
  - Distributes by “relevance”
    - Most cited
    - Date
    - Journal
- One reference Paper
  - at least 20 references
- Extended search from network

Litmaps

Link



Connecting papers

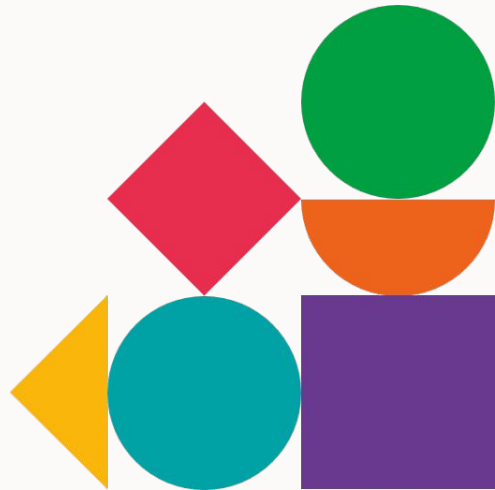
- Analyzes ~50,000 papers and select a few with the strongest connections

- Papers are arranged according to their similarity.
  - Not always cite each other
  - Citation overlap

- Similar reference network
  - Close relation



Link



Activity

03 - Connecting Papers

Litmaps

CONNECTED PAPERS

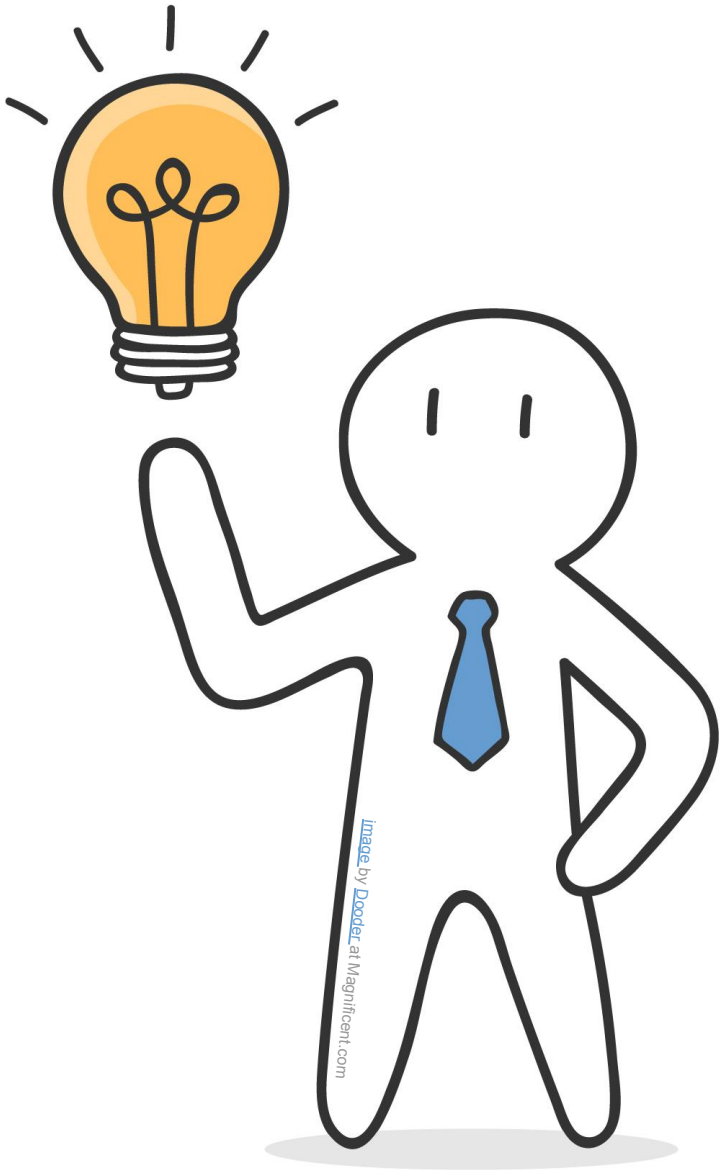
Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise03

1. Choose one of the papers you found in the previous activities
2. Explore the links between the core paper and its network
3. Check:
   - Papers of the previous activities are listed?
   - Are there new papers?

Comparing the tools you just worked with:

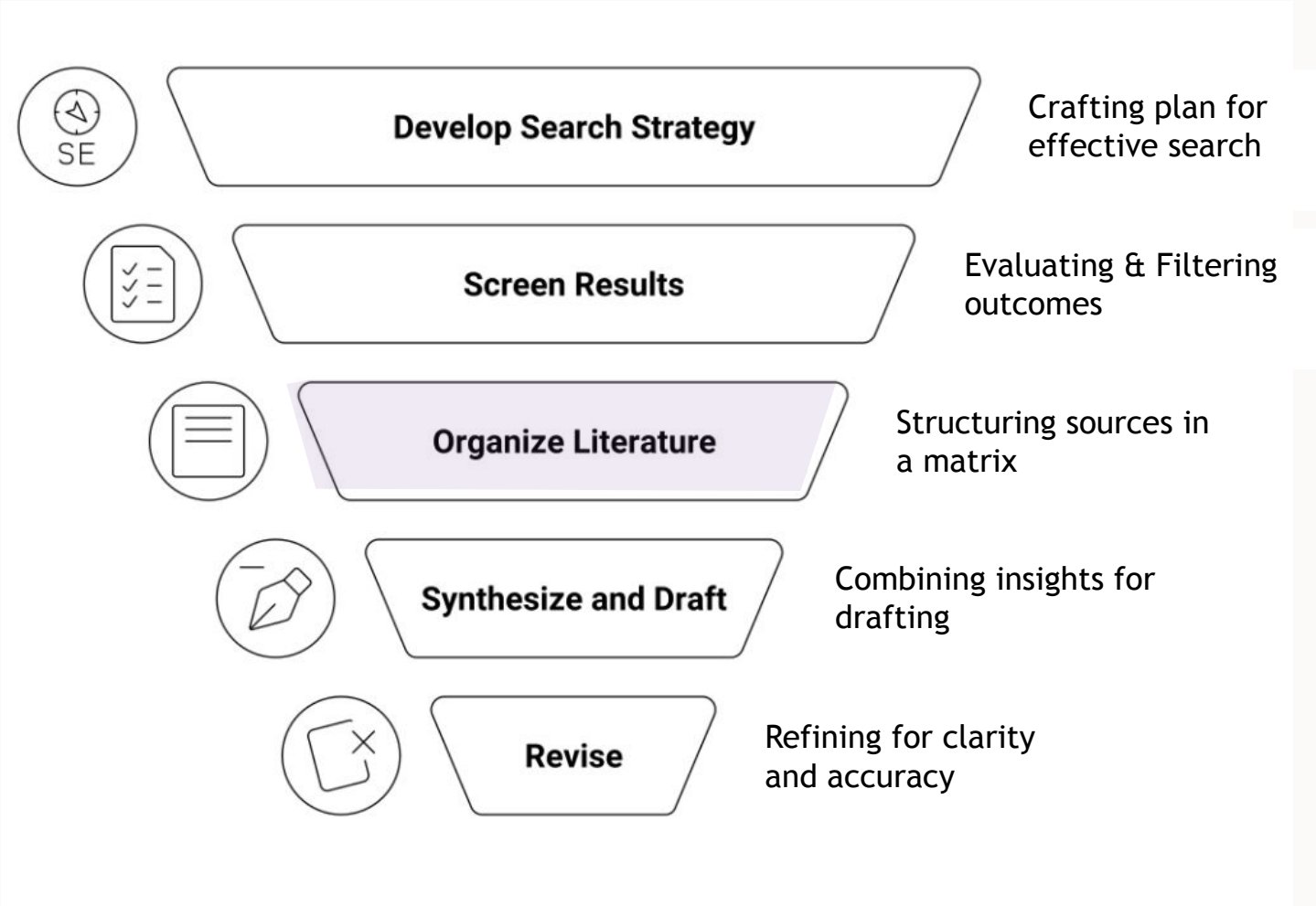
What are the qualitative differences and use cases?



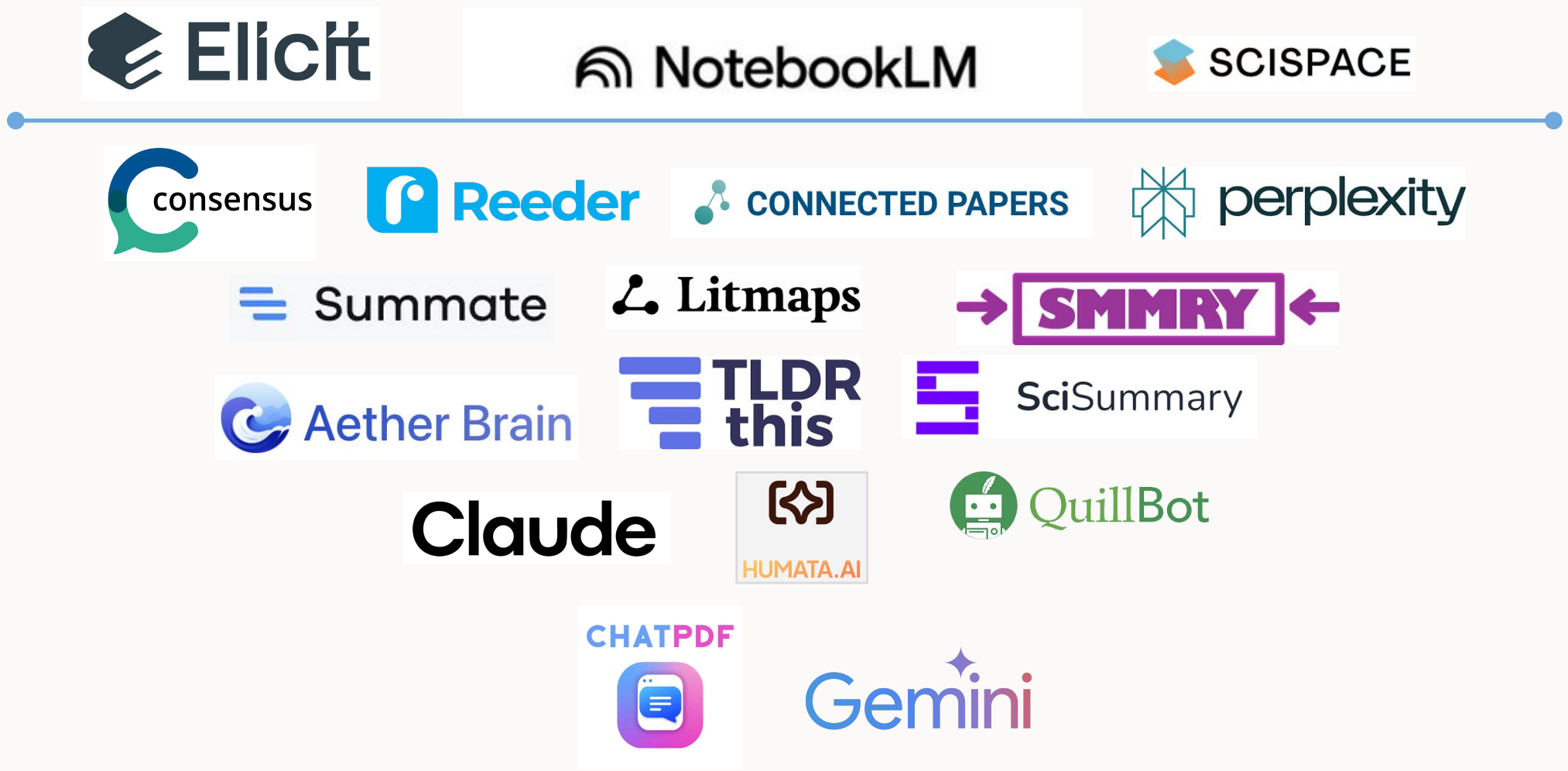
04

Summarize and organize

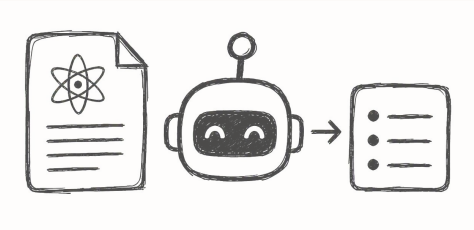
AI-assisted literature discovery and summarisation



Literature summaries



Different flavors of AI discovery tools

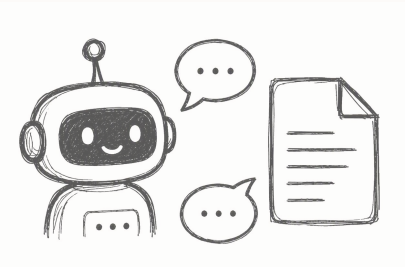


Literature Summarizer

Elicit

Asta

SCISPACE



Prioritise full text content via chats

NotebookLM



PRIVACY POLICY

“Any papers you upload to Elicit are private to your account”

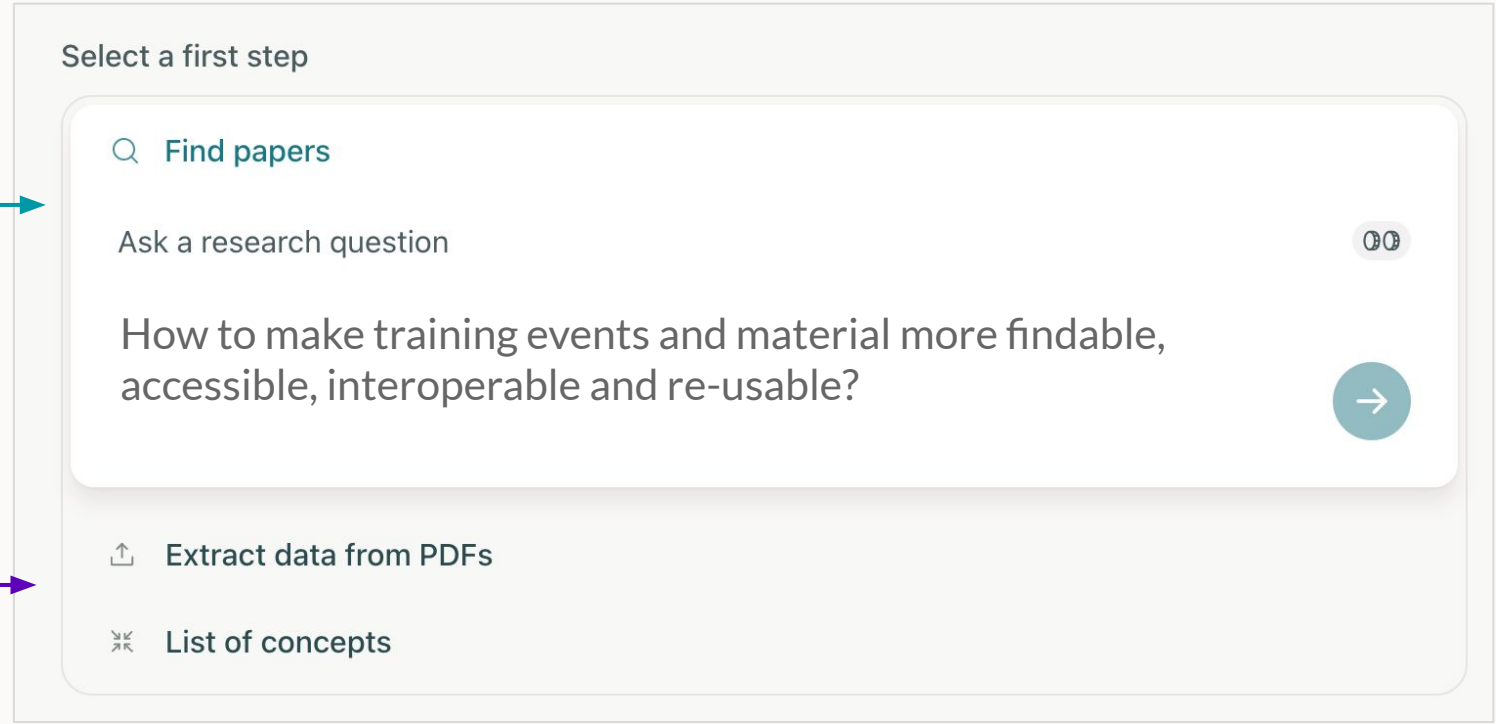
Recommendation

“Do not use for proprietary, confidential, and personal information ”

Elicit research options

SEARCH SEMANTIC SCHOLAR

UPLOAD PDF ARTICLE (e.g. via Zotero)





Short demo

Look at an existing notebook: https://elicit.com/notebook/d062b402-32db-4ef2-bdb8-d1e7c5e0804e

Beyond ‘Summary’ you have many more parameters

Methodology	Summary of intro	Research gaps	Study count	Duration	Population characteristics
Intervention	Summary of discussion	Hypotheses tested	Independent variables	Statistical techniques	Participant age
Outcome	Study design	Future research	Dependent variables	Algorithms	Population sex
Limitations	Theoretical framework	Funding source	Measured variables	Software	Organism
Intervention effects	Research question	Dataset	Region	Participant count	Policy recommendations

Elicit

Pros

- A versatile and adaptable tool
- Integrates with Zotero
- Supports bulk upload of PDFs
- Capable of summarizing multiple papers
- Promises PDF privacy

Cons

- Offers multi-dimensional analysis with the paid version
- Export functionality requires the Plus version
- Most interactions require credits
- Some aspects lack transparency

Activity

Asta

Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise05

1. Browse back to https://asta.allen.ai
2. This time use the Report Generator agent to summarize literature

e.g.:

https://www.nature.com/articles/s41540-026-00716-0#citeas
https://www.biorxiv.org/content/10.1101/2023.12.30.573613v1



PRIVACY POLICY

“Uploads, chat messages, and cached artifacts are encrypted end-to-end.”

Recommendation

“Do not use for proprietary, confidential, and personal information for free version”

Automation of literature search in SciSpace



Search SciSpace

Initial searches on SciSpace for relevant articles



Search Google Scholar

Broader searches on Google Scholar



Combine and Rerank Results

Merging and reordering search results



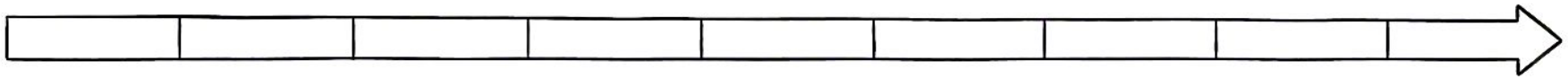
Extract Insights

Identifying key findings from the papers



Complete Literature Search

Finalizing the search process





Search Your Library

Checking the library for relevant PDFs



Search PubMed Advanced

Focused searches on PubMed Advanced



Read File

Reviewing the combined results file



Identify Top 10 Papers

Selecting the most relevant papers

Activity

SciSpace

1. Feed with the same papers and ask for a report

2. Compare

https://scispace.com/share/01a173c3-8989-7b72-b09a-4b2d8099850d



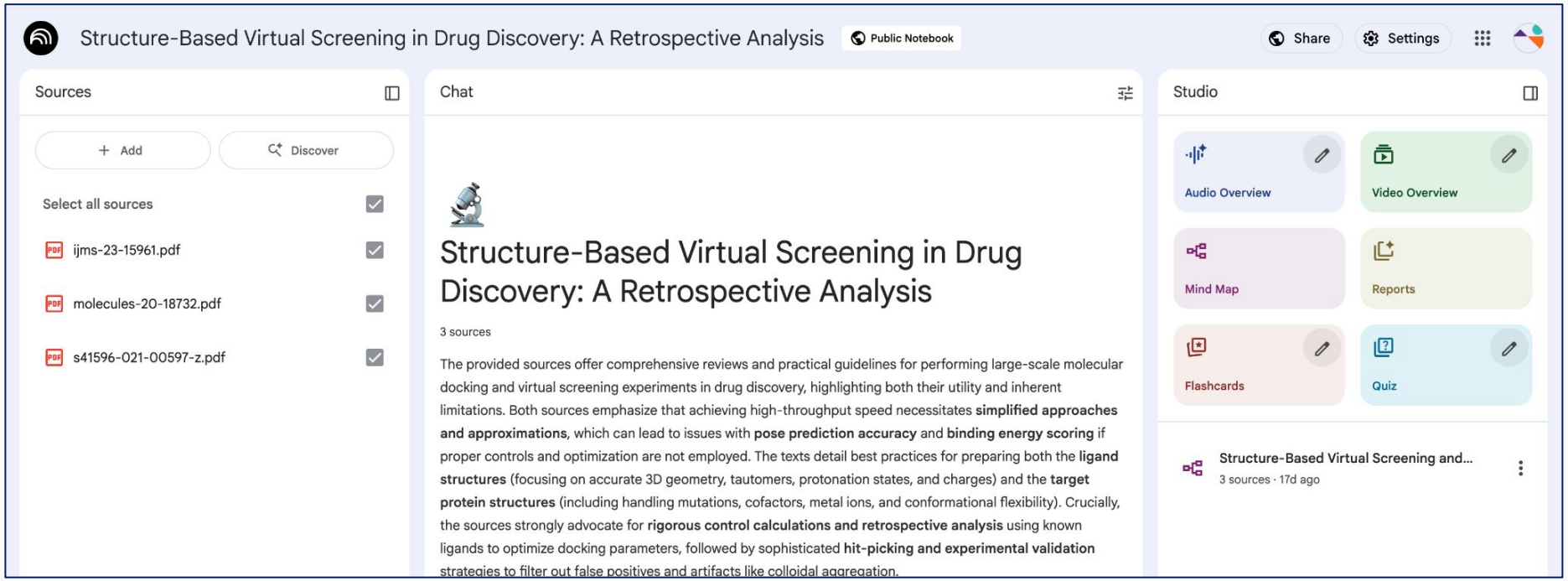
PRIVACY POLICY

See policy from Google

Recommendation

“Do not use for proprietary, confidential, and personal information with free version”

NotebookLM



Let's explore this example.

Activity

NotebookLM

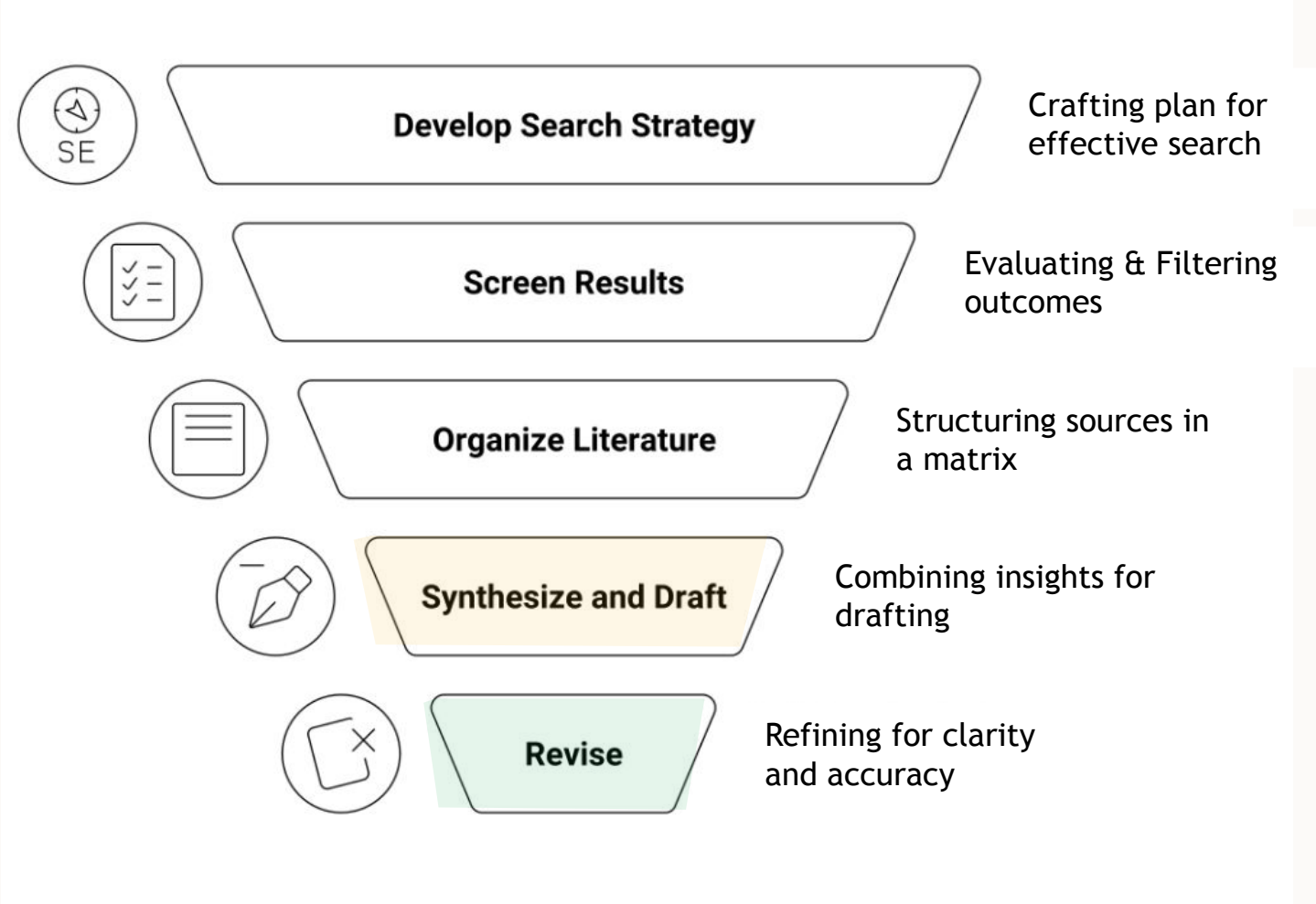
Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise05

1. Create a notebook
2. Include as resources
   https://www.nature.com/articles/s41540-026-00716-0#citeas (download necessary)
   https://www.biorxiv.org/content/10.1101/2023.12.30.573613v1 (link)
3. Try out
   - Look into summary
   - Ask questions
   - Use keywords for deep complementary search

https://notebook.google.com/notebook/17beb68e-0f29-4da6-94e7-2375736586a1

AI-assisted literature discovery and summarisation



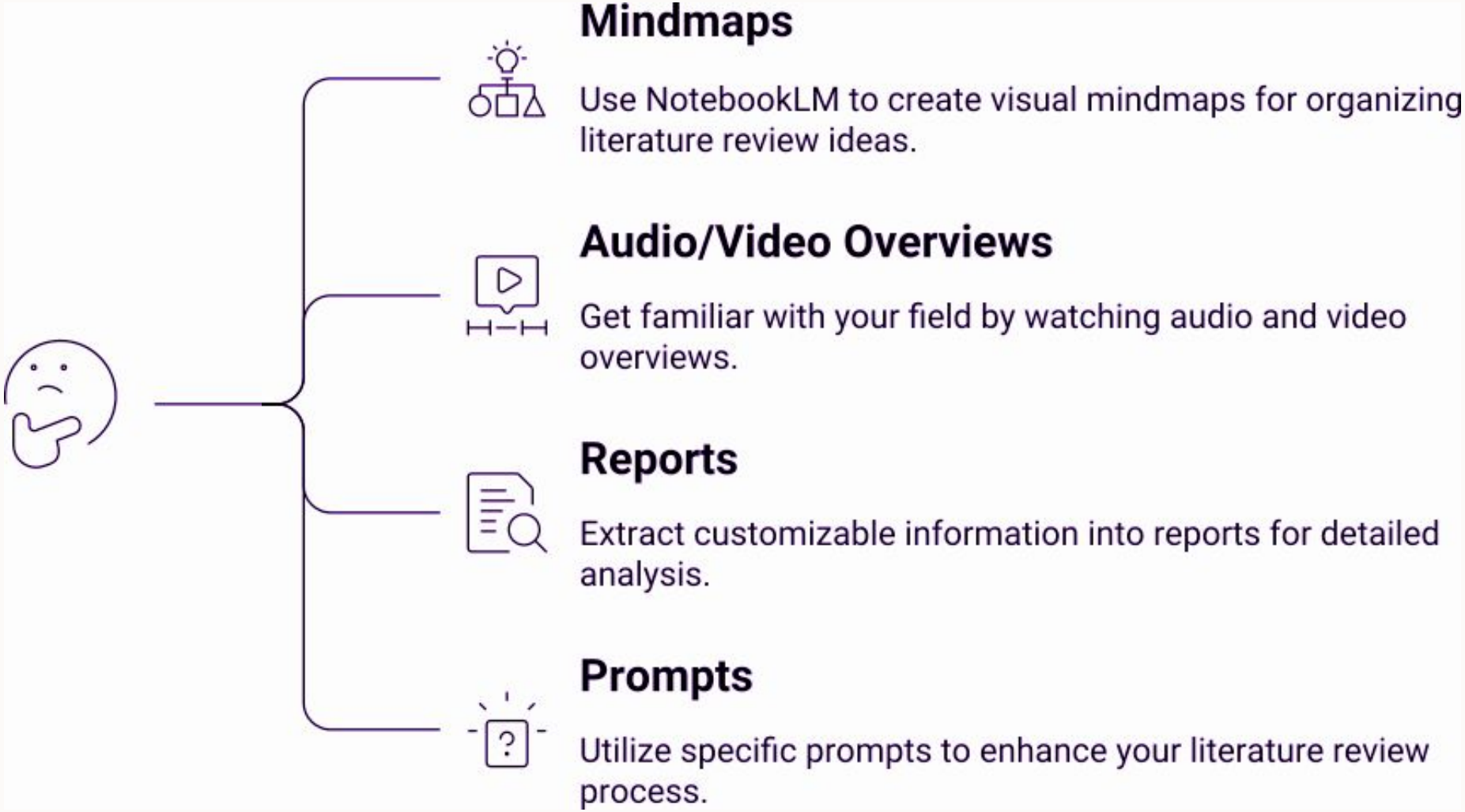
05

Synthesis and draft



How to use NotebookLM in the exploratory process?

How to use NotebookLM in the exploratory process?



Mindmaps

Use NotebookLM to create visual mindmaps for organizing literature review ideas.

Audio/Video Overviews

Get familiar with your field by watching audio and video overviews.

Reports

Extract customizable information into reports for detailed analysis.

Prompts

Utilize specific prompts to enhance your literature review process.

Activity

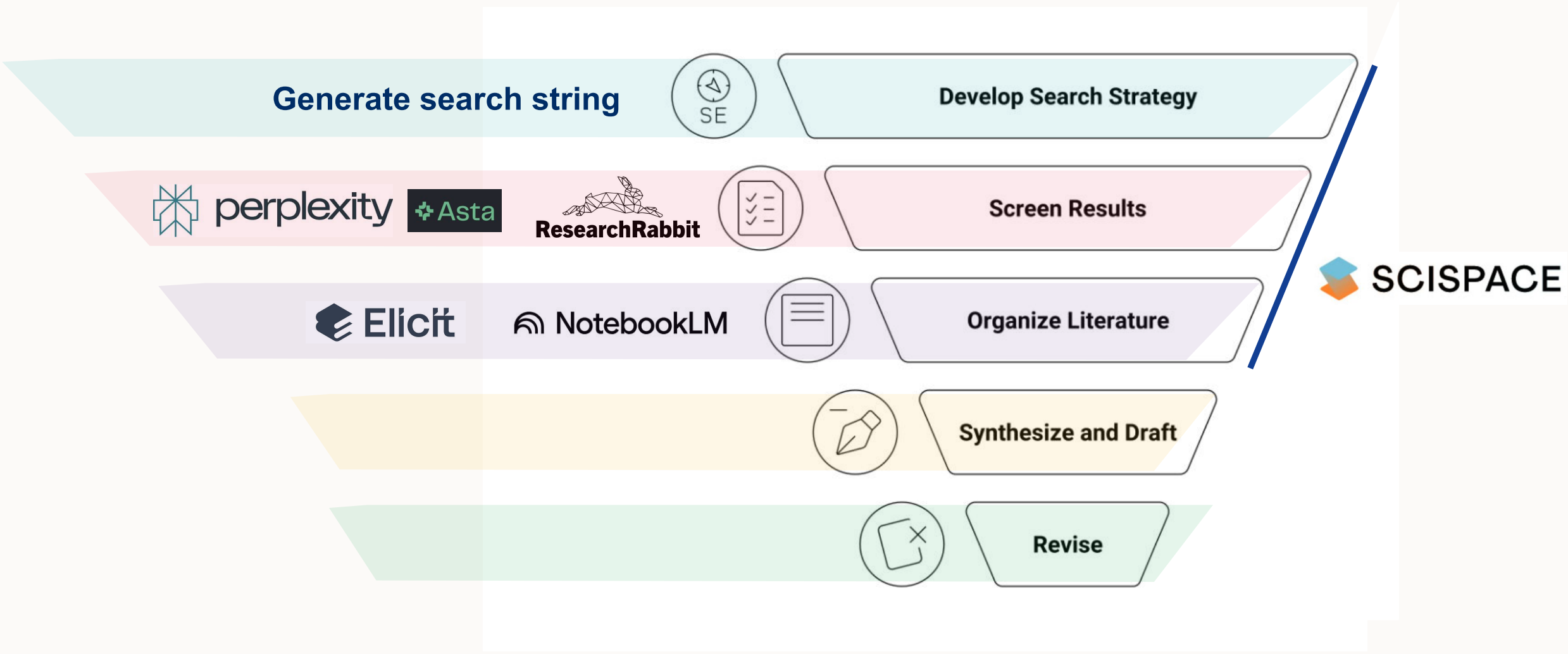
14 -NotebookLM

Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise14

1. Create a notebook
2. Add at least 3 PDFs with coherent but complementary content.
3. EXPLORE
   - Mind map function
   - Specialised prompts for a specific report
   - Infographic

When can I use what?



Dimensions of the writing process



Depth: Ideas, sources, analysis, variety and range of language



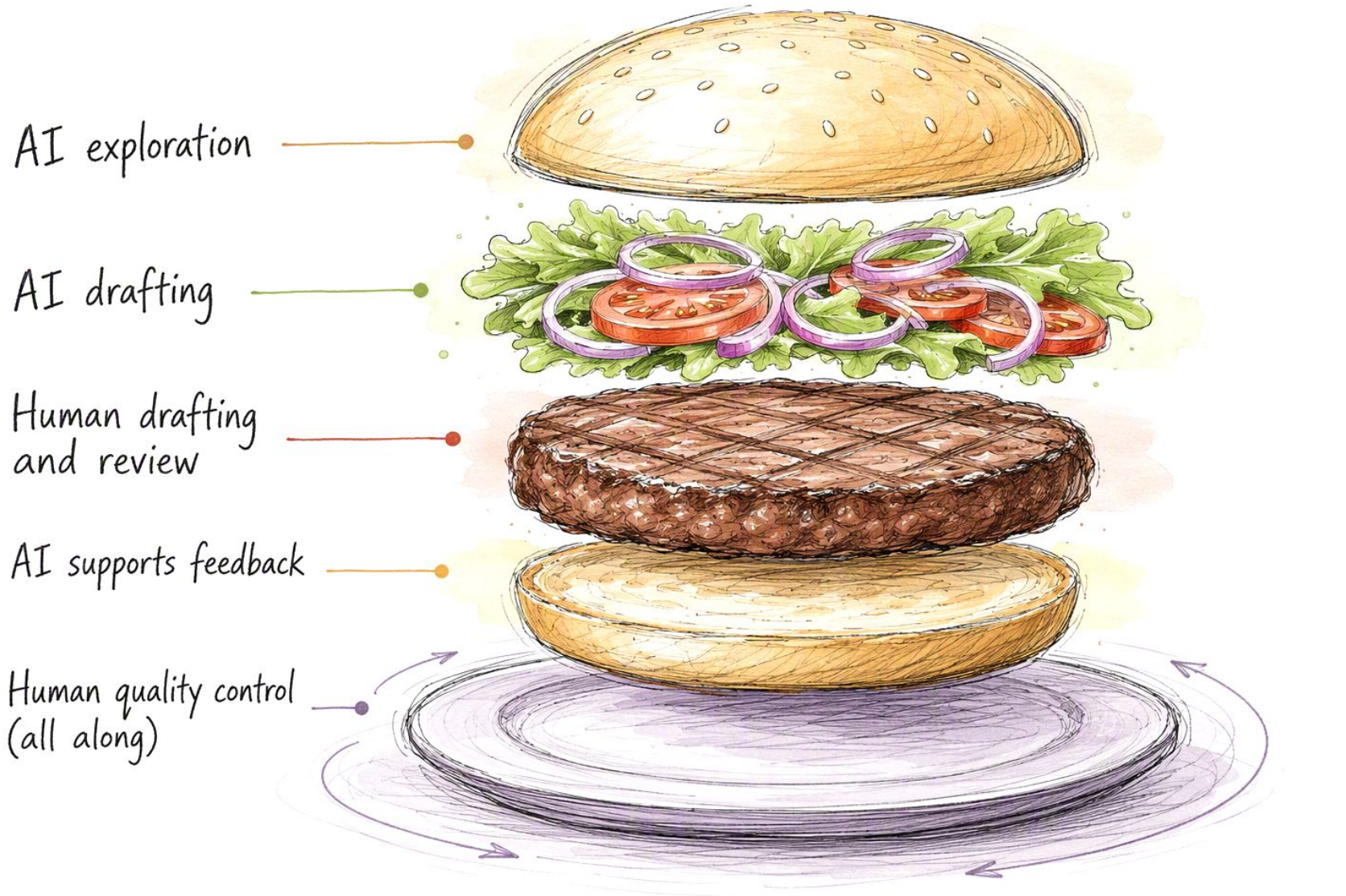
Flow: Smooth expression, clear connection, fluency



Accuracy: Precise spelling, grammar, punctuation, citations formatting

HITL Writing – “AI sandwich”

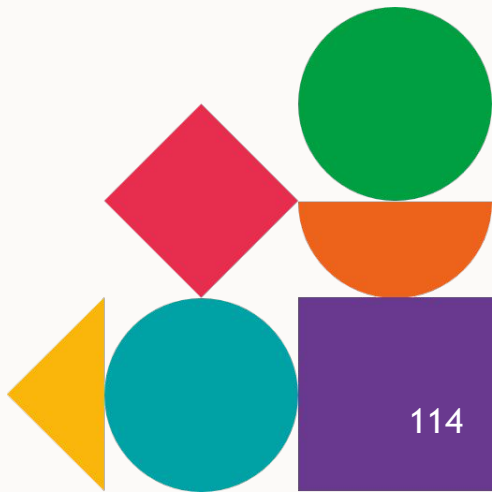
HITL : Human In The Loop



AI can support your writing

Story telling

- Feedback to improve research report
- Drafting preliminary phases: *purpose, audience and content*
- Structuring report, paper, abstract
- Structure and draft a proposal
- Draft and illustration for academic poster
- Suggestion for structuring a poster



AI can support your writing

Level	Example prompt
(1) Basic editing *checking spelling and grammar,* *suggesting synonyms,*	“Check the spelling and grammar considering UK standard english, and suggest synonyms for any repetitive words. Highlight changes and corrections in bold and underline for easy identification. For synonyms give a list of current words and suggested changes”
(2) Structural editing *paraphrasing,* *translating,* *improving text structure,* *improve flow or coherence*	“Paraphrase this sentence to be more concise while improving its clarity and flow. At the end provide the improved sentence and a version translated to French from france.”
(3) Creating derivative content *summarizing,* *creating titles and abstracts,* *rewriting or generating analogies*	“Summarize this document on a formal scientific writing language, use American English. The text must contain intro, methods, results, discussion and conclusion. It should have 5 paragraphs maximum. Suggest 3 short (6 words maximum), catchy titles for the text that will be submitted to a journal.”

Structural and derivative editing

You are [DEFINE ROLE]. Re-write the text section with [LIMIT SIZE]. The text must have scientific publication tone and structure.

style of text

AVOID fancy jargon. Write FORMALLY.

You are [FORBIDDEN] to use complex English words, refrain of using terms from the # ban list # .
Use UK english. Mimic my own writing style.

Writing style sample

<Add 3 paragraphs from different text samples>

#Section structure#

Paragraph 1: general intro | Paragraph 2: Explaining data | Paragraph 3: Methodology | Paragraph 4: Results | Paragraph 5: Discussion

#ban list #

Hurdles, Bustling, Harnessing, Unveiling the power,(…)



Activity

Git file. https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise08

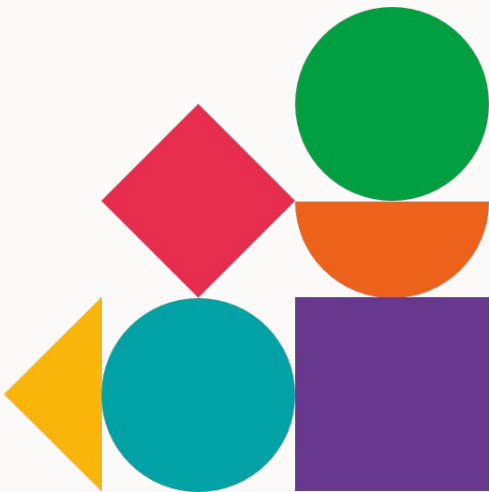
Use the prompt in the GitHub to try out how you can steer the writing style.

AI can support your writing

Level	Example prompt
(4) Creating new content *completing, continuing or expanding text, brainstorming ideas*	“Expand the text to explain the key question being addressed. Address why it is important, draw parallels or analogies where you see fit. Refrain of changing the current text. Replicate the language and tone.”
(5) Evaluation or feedback *assessing the quality of the writing , finding weaknesses in it, writing coach*	“Review this introduction. List any logical gaps, inconsistency or areas that need further development. Include what session and line of the text they are. For further development suggest 3 points and explain the relevance, be concise.”

Brainstorming and Feedback

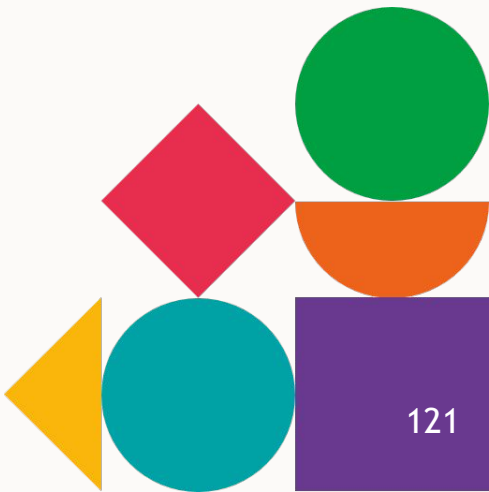
I want to **simulate a peer review interaction**. Read all the content **shared in the PDF** and ask me questions that will **assess the strengths and weakness** of this research. Ask 2 QUESTIONS at a time. And after I answer you can ask new questions. After 5 rounds you should **propose a list of improvements in the text and in the research**. Refrain of repeating questions even if they are unanswered.



05

Citing genAI use in scholarly work

When and how to cite the use of AI?



Activity

When and how should we disclaim the use of genAI for scholarly work?
(paper, proposals, grants, etc)

What do you **think** and what do **you know**?

1. Write in share doc

When and how to cite the use of genAI?

EU-Commission: the substantial use of genAI must be informed

- Substantial use = interpreting data analysis, carrying out a literature review, identifying research gaps, formulating research aims, developing hypotheses, etc.

Key indicators	Not necessary	Necessary disclaimer	Not permitted
Basic support tool for authors *(refine, correct, format, etc.)*	x		
Research shaping *(literature review, identify research gaps, formulate research aims, developing hypotheses, etc.)*		x	
Interpreting data analysis		x	
Data manipulation *(crete, alter or manipulate data and results)*			x
Credite AI as author			x

May 2026, European Commission, Living guidelines on the responsible use of generative AI in research (link)
Dec 2023, White paper, International Association of Scientific, Technical, and Medical Publishers
Generative AI in Scholarly Communications: Ethical and Practical Guidelines for the Use of Generative AI in the Publication Process (STM, 2023) (link)

UGent recommendation

Recommendations for different (hypothetical) scenarios

Type of GenAI tool account makes a difference!

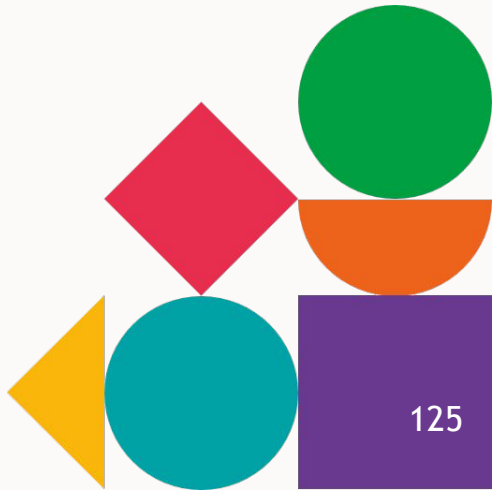
Scenario	Copyright violation risk	Contractual risk	Advice
1. Uploading OA article in e.g. Elicit	LOW	LOW	**PERMITTED**
2. Uploading paywalled article in e.g. Elicit (personal acount)	LOW (covered by TDM exception)	MEDIUM (publisher contract may prohibit, but imperative law; technical risk of IP blocking)	**PERMITTED, WITH CAUTION**
3. Uploading paywalled article in e.g. Elicit (enterprise account)	LOW	LOW (agreement with tool provider usually covers this, or offers guarantees)	**RECOMMENDED** Copilot
4. Uploading in free tools (ChatGPT Free, NotebookLM…)	HIGH (possible use for training/memorisation)	HIGH (data leaks, violation of confidentiality)	**PROHIBITED**

UNIVERSITEIT GENT

Credit: Myriam Mertens

What’s the publisher’s point of view?

Have you experienced this situation already?



Activity

15 - Writing Policies

Git file.

https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

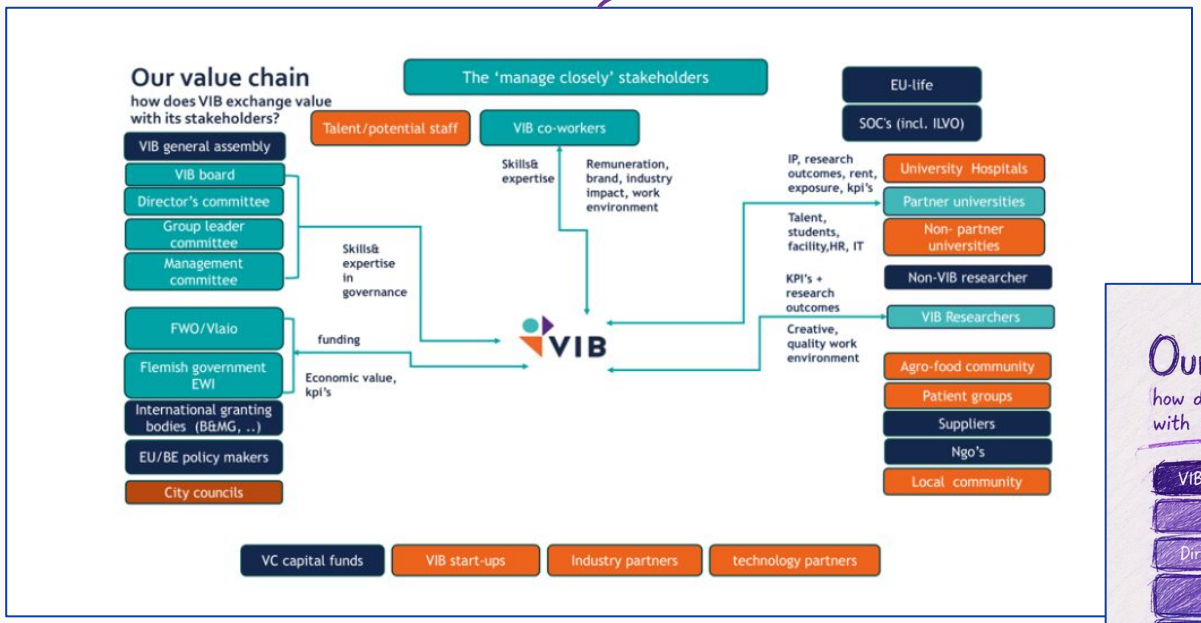
Repo >> exercises >> exercise15

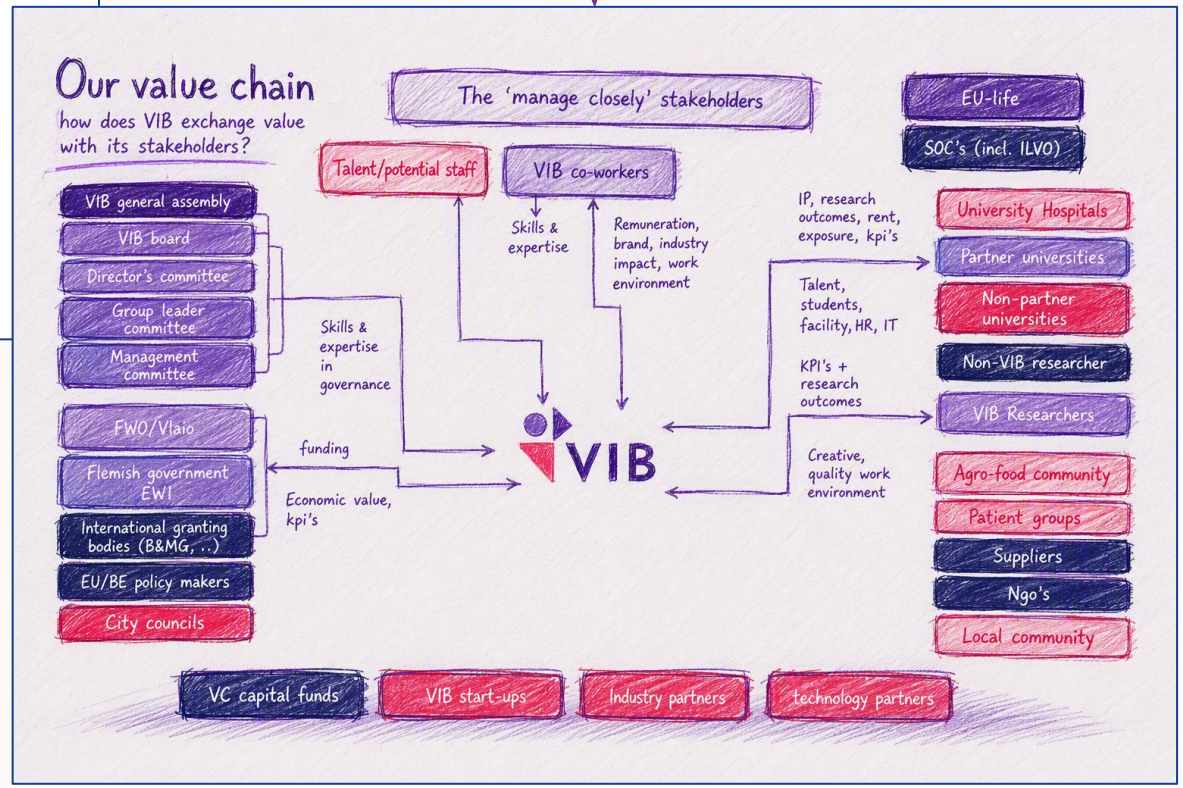
1. Using Perplexity *(and other tool you may want)*
2. Formulate a prompt for a comparative analysis of policies about GenAI use for writing and illustrations.
3. Reflect with your partner about accuracy
4. Write down at least two reflections in share doc

06

Creating visuals with genAI

Creating visuals with Copilot





- Share a image and describe a new style
- Copilot premium:
  - Brand kit (colors and font)

Creating visuals with Copilot

- Prompt your way to the illustration
- Ask genAI to ask you the question and create a prompt for you
- If you have Copilot plus
  - Use brand kit
    - Define colors set
    - Define font type
    - etc.

Activity 17 - Creating visuals with Copilot

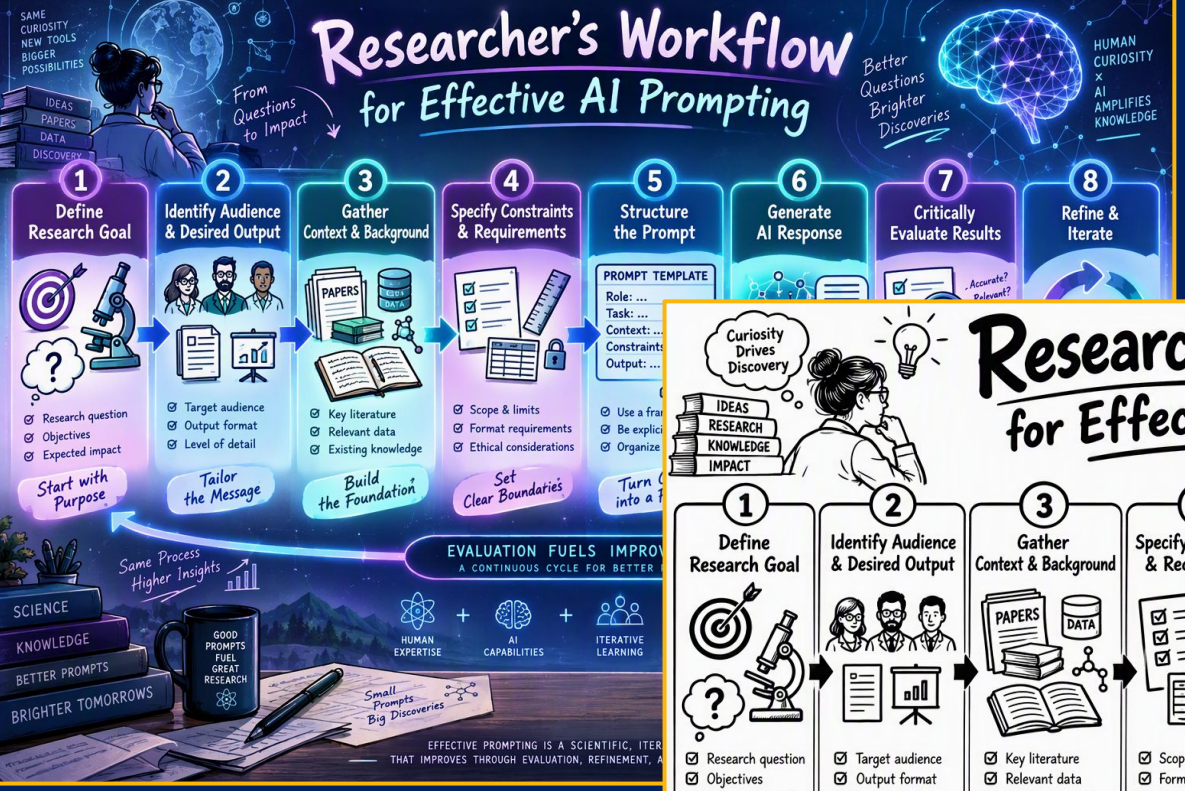
We would like to create a illustration.
As a prompt expert, you will ask questions to clarify the THE MAIN TOPIC, for WHOM and to WHAT PURPOSE it should be done. After all rounds of questions you will create the best prompt to generate the image

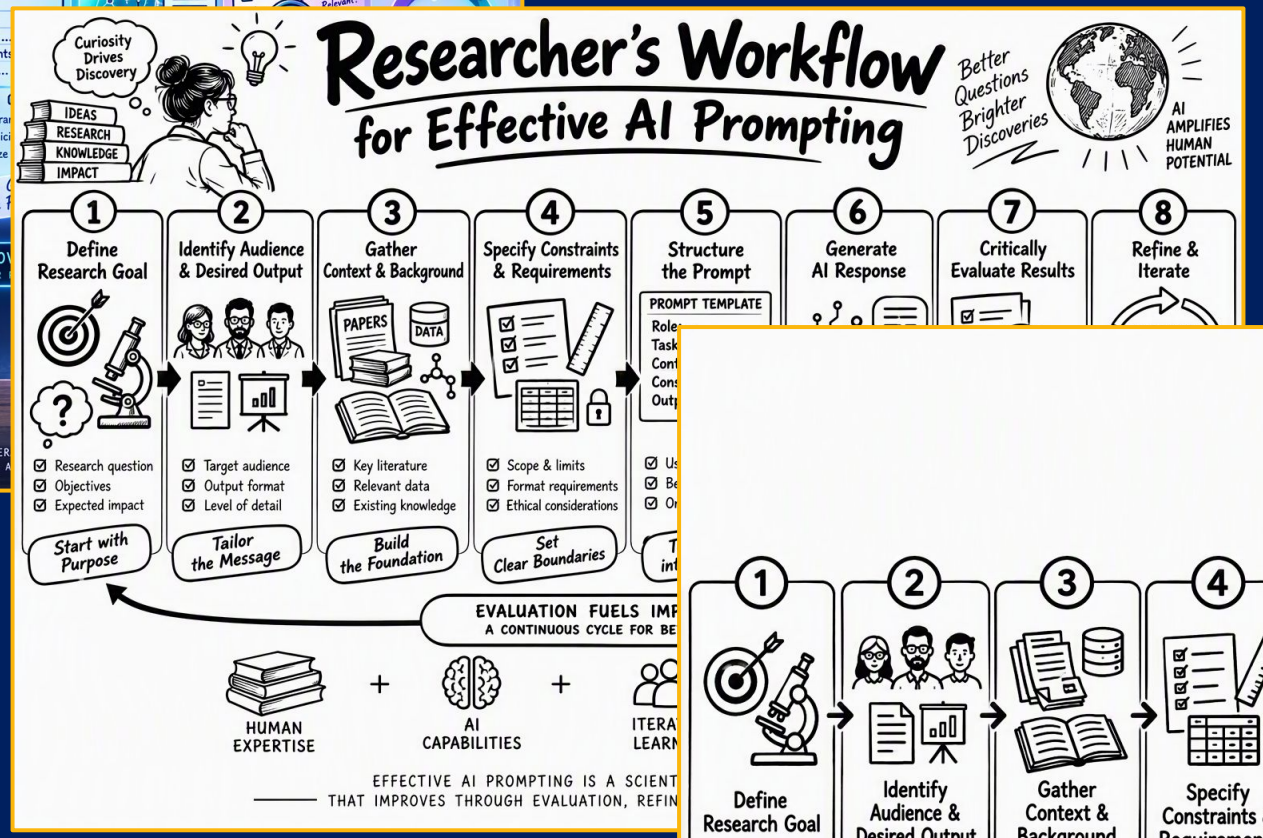
Ask 2 QUESTIONS at a time to, in 3 rounds. Wait for the answer which you will include in the prompt.

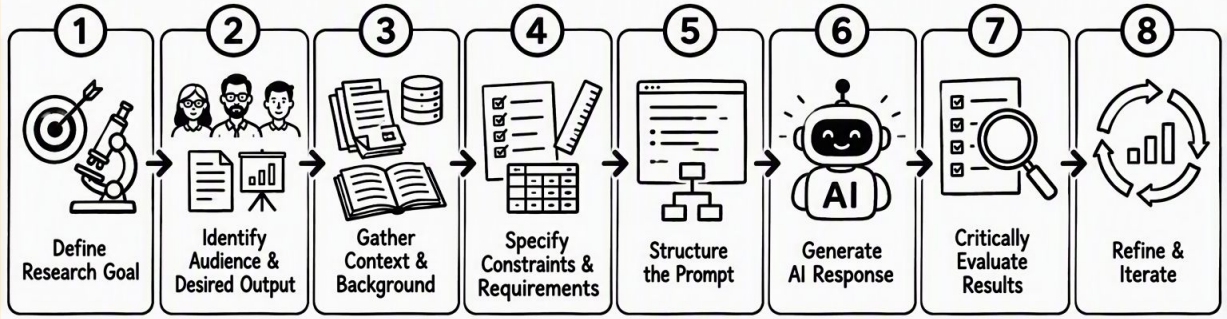
At each round suggest a prompt and ask the next question and tell me the next action as follows. Great me before we start and ask the first questions.

- SUGGESTED PROMPT: <include prompt>

- QUESTION: <include questions>


SAME CURIOSITY NEW TOOLS BIGGER POSSIBILITIES
IDEAS
PAPERS
DATA
DISCOVERY
From Questions to Impact
Researcher's Workflow
for Effective AI Prompting
Better Questions Brighter Discoveries
HUMAN CURIOSITY × AI AMPLIFIES KNOWLEDGE
1 Define Research Goal
Research question
Objectives
Expected impact
Start with Purpose
2 Identify Audience & Desired Output
Target audience
Output format
Level of detail
Tailor the Message
3 Gather Context & Background
PAPERS
DATA
Key literature
Relevant data
Existing knowledge
Build the Foundation
4 Specify Constraints & Requirements
Scope & limits
Format requirements
Ethical considerations
Set Clear Boundaries
5 Structure the Prompt
PROMPT TEMPLATE
Role: ...
Task: ...
Context: ...
Constraints
Output: ...
6 Generate AI Response
7 Critically Evaluate Results
Accurate?
Relevant?
8 Refine & Iterate
EVALUATION FUELS IMPROV
A CONTINUOUS CYCLE FOR BETTER
Same Process Higher Insights
SCIENCE
KNOWLEDGE
BETTER PROMPTS
BRIGHTER TOMORROWS
GOOD PROMPTS FUEL GREAT RESEARCH
Small Prompts Big Discoveries
HUMAN EXPERTISE
AI CAPABILITIES
ITERATIVE LEARNING
EFFECTIVE PROMPTING IS A SCIENTIFIC, ITER
THAT IMPROVES THROUGH EVALUATION, REFINEMENT, A

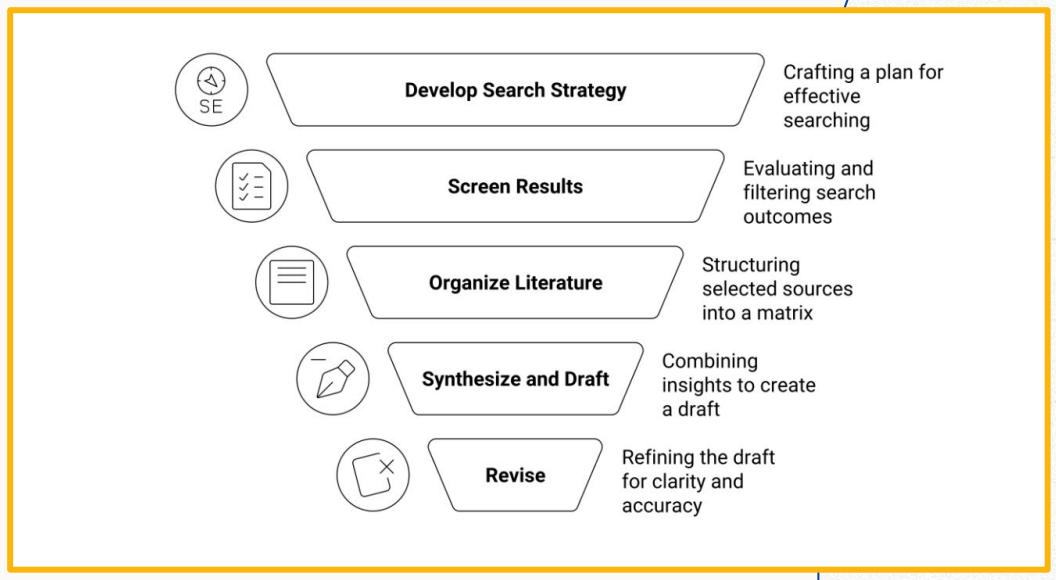

Curiosity Drives Discovery
IDEAS
RESEARCH
KNOWLEDGE
IMPACT
Researcher's Workflow
for Effective AI Prompting
Better Questions Brighter Discoveries
AI AMPLIFIES HUMAN POTENTIAL
1 Define Research Goal
Research question
Objectives
Expected impact
Start with Purpose
2 Identify Audience & Desired Output
Target audience
Output format
Level of detail
Tailor the Message
3 Gather Context & Background
PAPERS
DATA
Key literature
Relevant data
Existing knowledge
Build the Foundation
4 Specify Constraints & Requirements
Scope & limits
Format requirements
Ethical considerations
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PROMPT TEMPLATE
6 Generate AI Response
7 Critically Evaluate Results
8 Refine & Iterate
EVALUATION FUELS IMP
A CONTINUOUS CYCLE FOR BE
HUMAN EXPERTISE
AI CAPABILITIES
EFFECTIVE AI PROMPTING IS A SCIENT
THAT IMPROVES THROUGH EVALUATION, REFIN

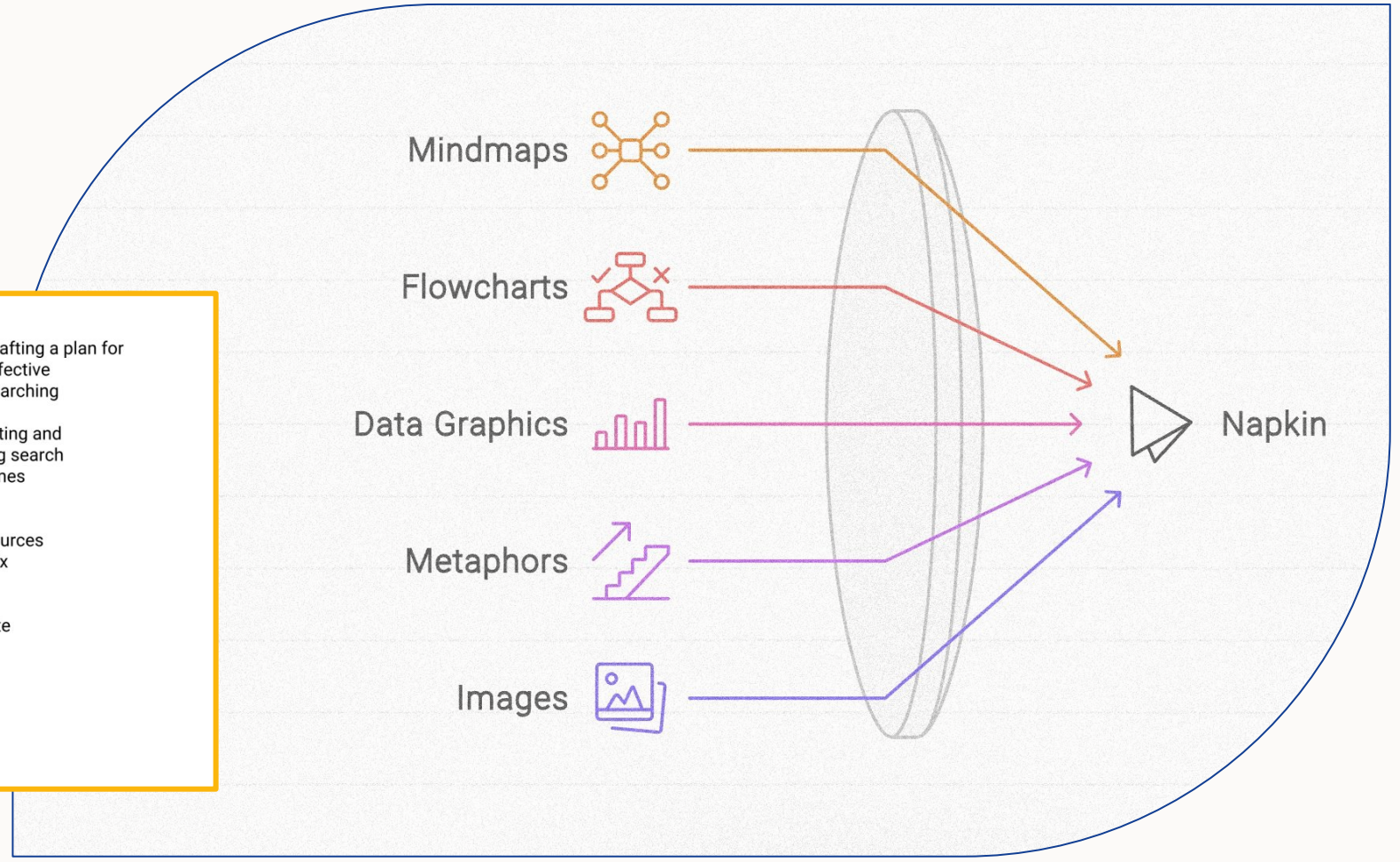

1 Define Research Goal
2 Identify Audience & Desired Output
3 Gather Context & Background
4 Specify Constraints & Requirements
5 Structure the Prompt
6 Generate AI Response
AI
7 Critically Evaluate Results
8 Refine & Iterate

VIB

Napkin.ai

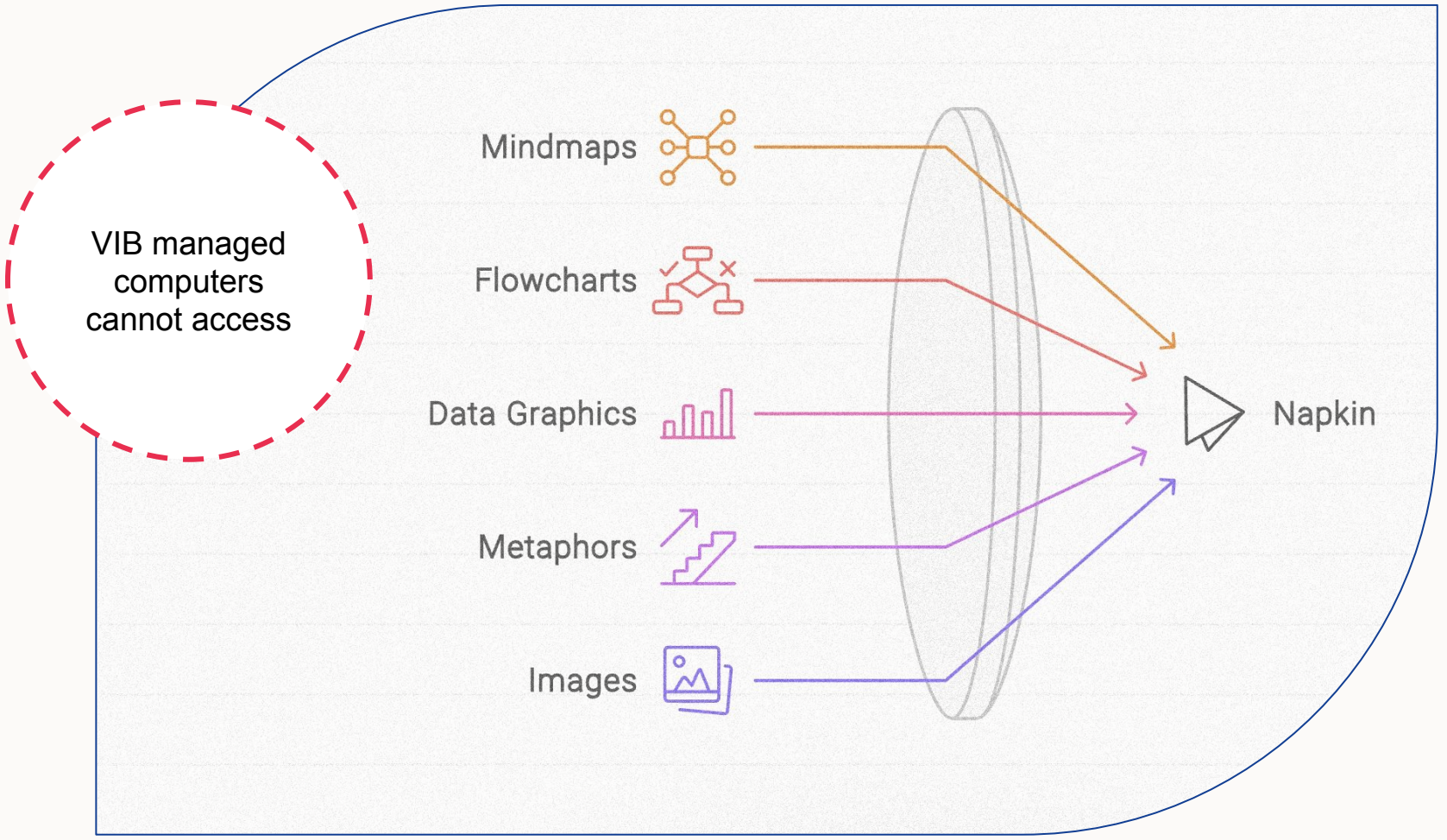
Example:





Napkin.ai

- Information sharing
- Editable images
- Clean and structured
- Limited free resource



Activity

17- Create an infographic

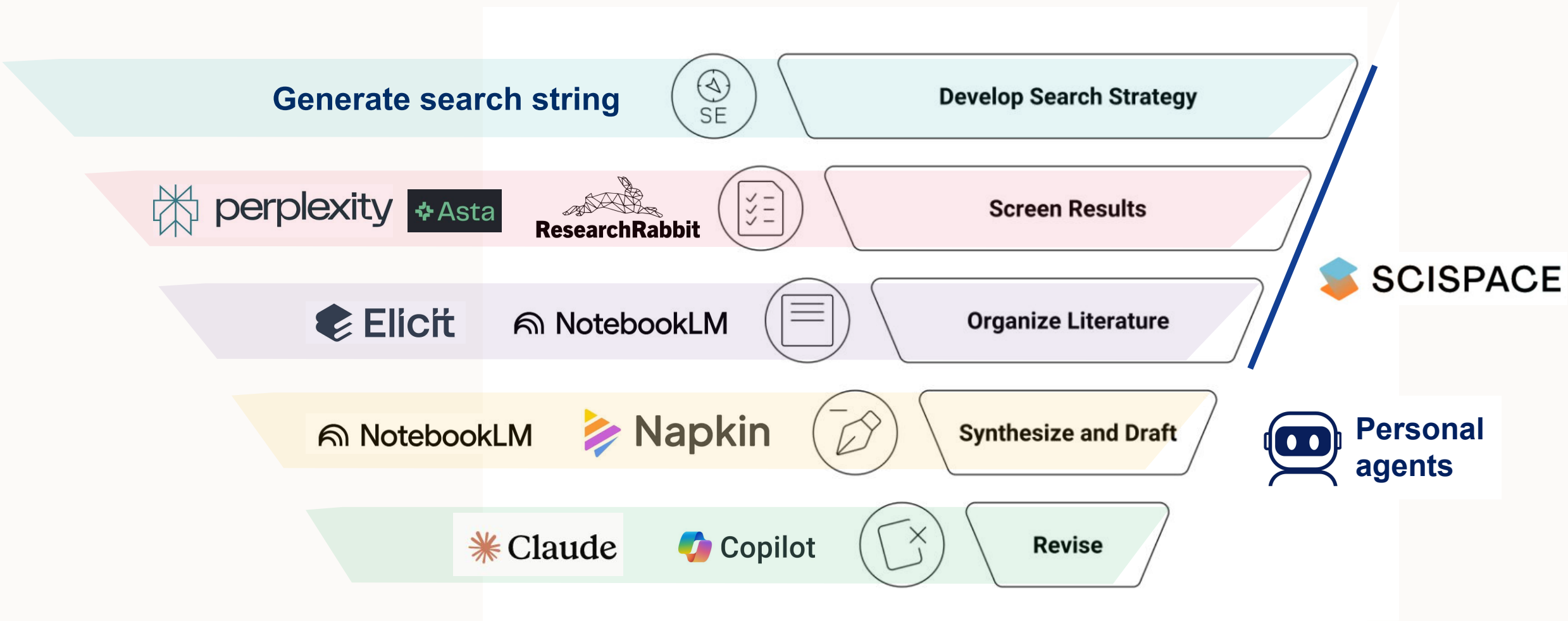
Create an illustration with Napkin.AI (free version)

- Open your browser and surf to https://app.napkin.ai

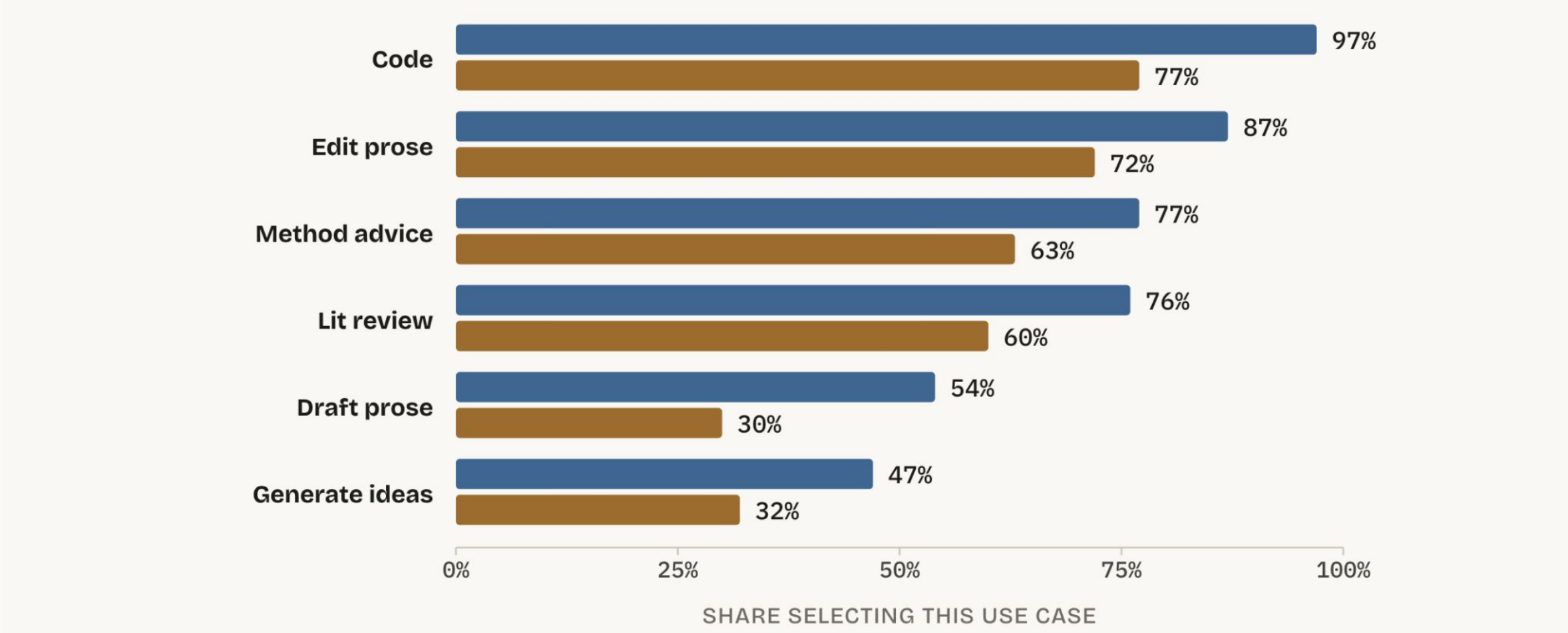
1. Create a blank napkin
2. Enter a title e.g. Curriculum topics
3. Add the list >>>>

```
dataset representation
computational dataset creation
dataset collection / FAIRification
evaluation method creation
data augmentation (synthetic)
model development
```

When can I use what?



Agent users do more of everything, also drafting



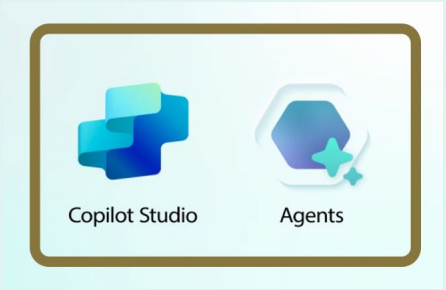
Measured. Lyttelton, Massenkoff and Wilmers (2026), 1,260 social scientists, February–March 2026. A cross-section: it cannot say whether the tool causes the extra use or heavier users pick the tool.

07

Agents

Agents

Different names for agents in different environments



Copilot Studio

Claude

Cowork or Code

Gems

GPTs

What do we cover in this session?

1. From chatbot to agent: the main components
2. Important aspects of Context engineering
3. Running agents: environments & safety
4. Skills: concept & anatomy
5. Hands-on: Create a skill for introduction writing
6. Hands-on: Create an agent with Copilot Agents
7. Reuse of skills - short demo within Claude Desktop

When should you use an agent or an agentic workflow?

package repeated prompts

Specific styling and output format

"Three times" principle

add specific knowledge

Folders/files

Applications (e.g. Outlook)

attach targeted tools

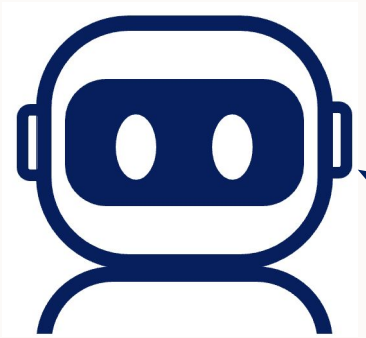
Image generation

Personal scripts

Include external data sources

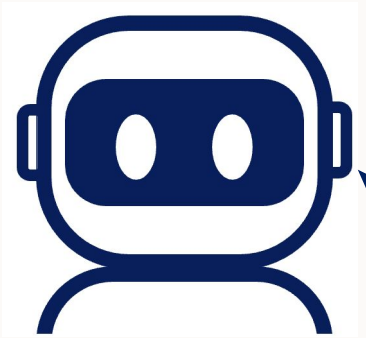
VIB

From a chatbot to an agent: we give context and instructions



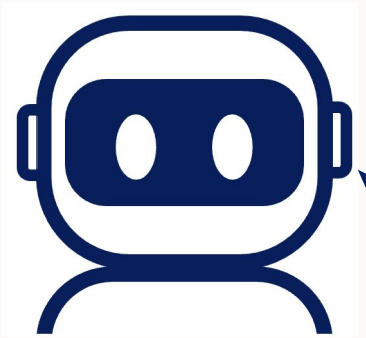
Context and Instructions

From a chatbot to an agent: we give context and instructions



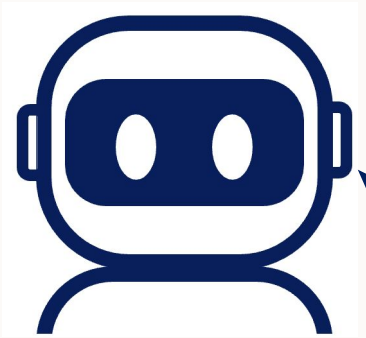
What is the second largest municipality (population) in Belgium?

From a chatbot to an agent: we have …



Do I need an umbrella in Ghent today?

From a chatbot to an agent: we have tools

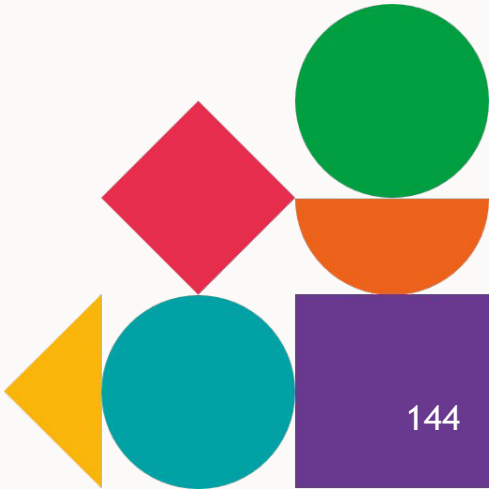


Do I need an umbrella in Ghent today?

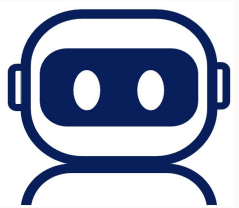








Tools and knowledge sources enable the agent





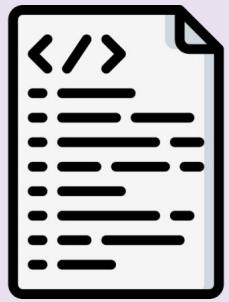
search web



read files



run code

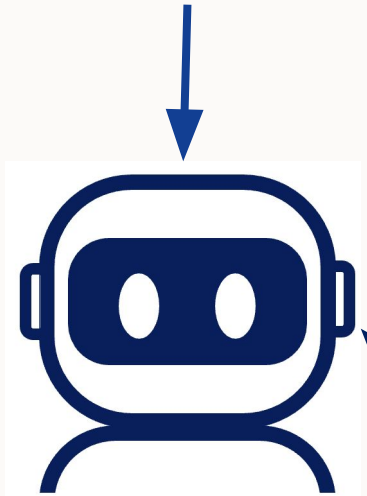


write output

From a chatbot to an agent: we have tools and knowledge bases

Knowledge sources





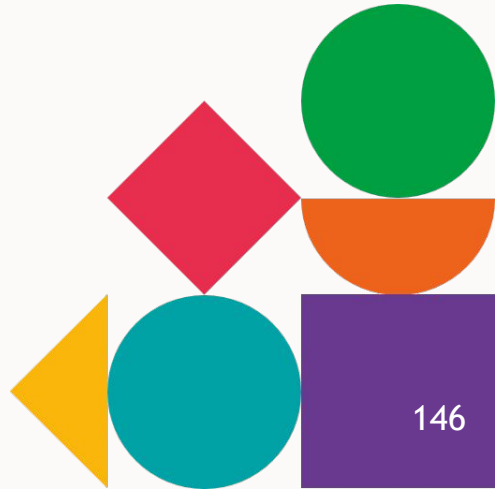
Do I need an umbrella in Ghent today?



Tools

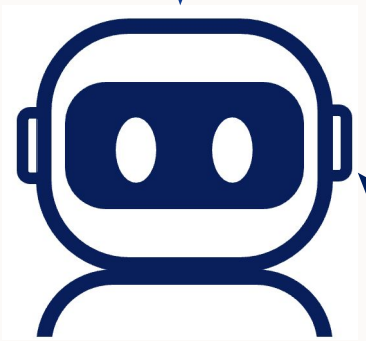






From a chatbot to an agent: we have tools and knowledge bases





Instructions

You have the tool RMI.

Use

write exactly:

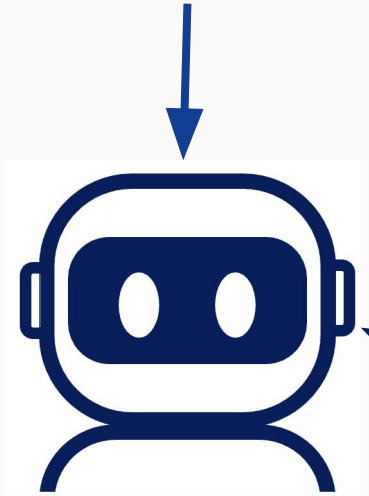
```
<call>get_weather
("city")</call>
```

Tools

RMI

From a chatbot to an agent: they can be run in a loop



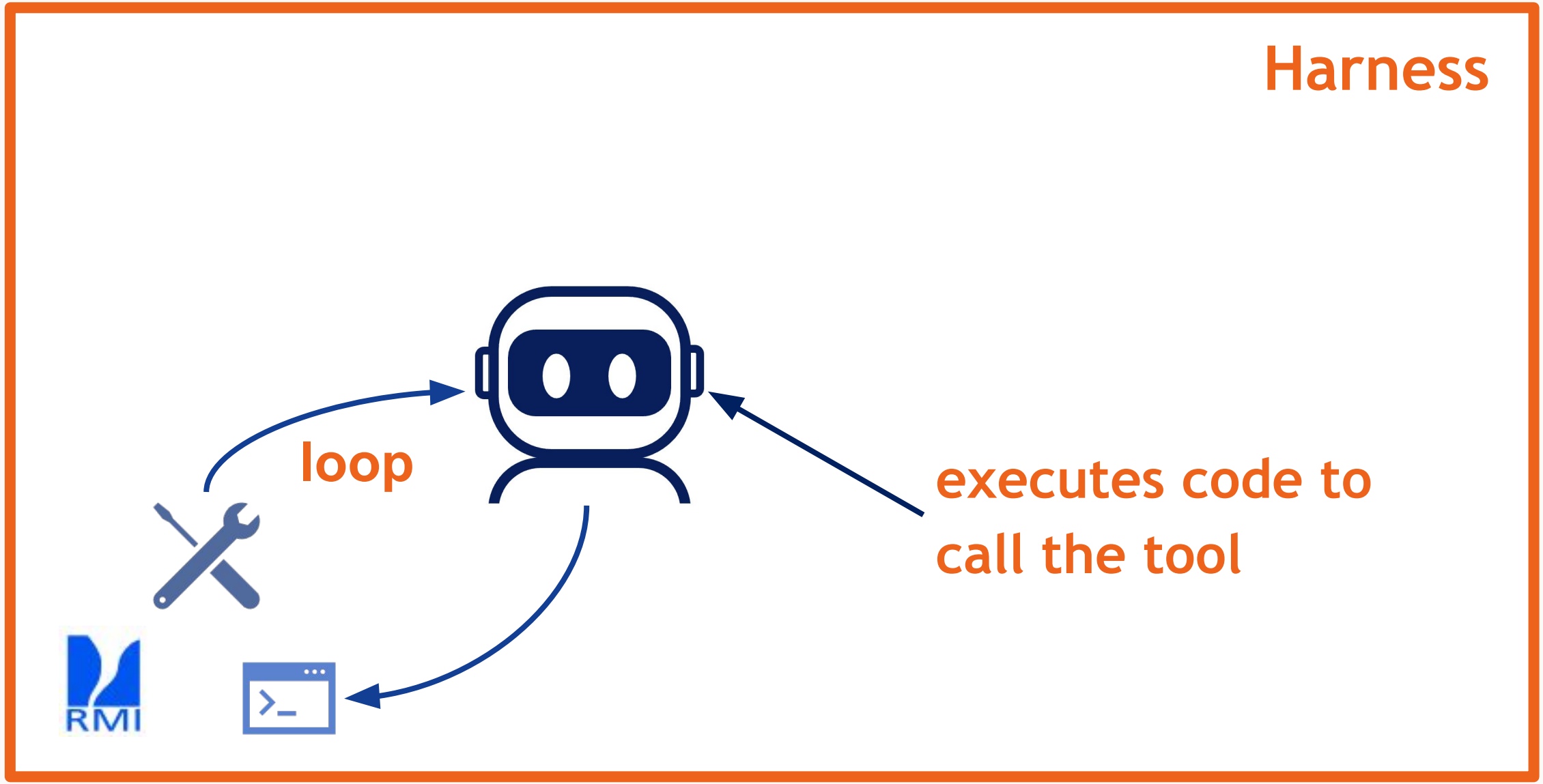


1. Call RMI tool



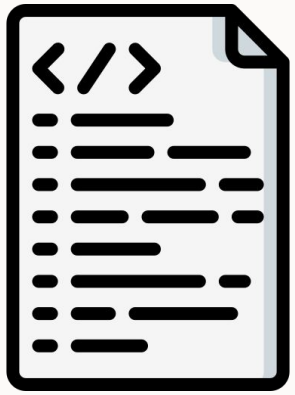

```
2. <call>get_weather
("Ghent")</call>
```

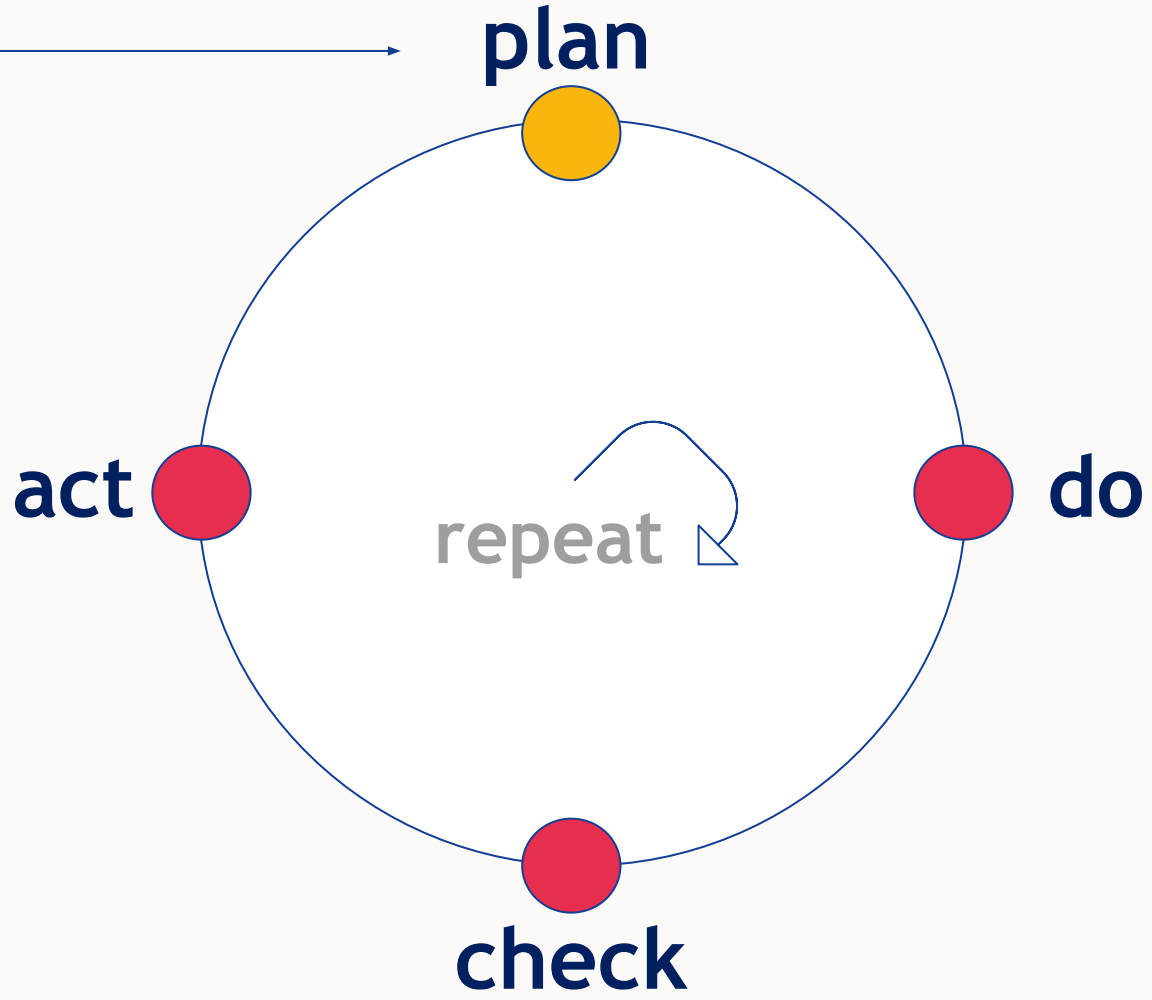
From a chatbot to an agent: harness steers the loop



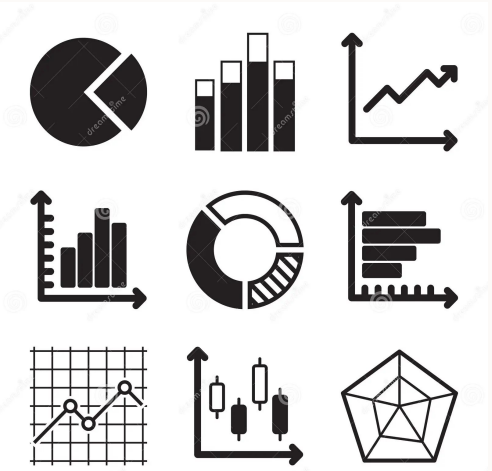
The loop makes all the main difference

goal

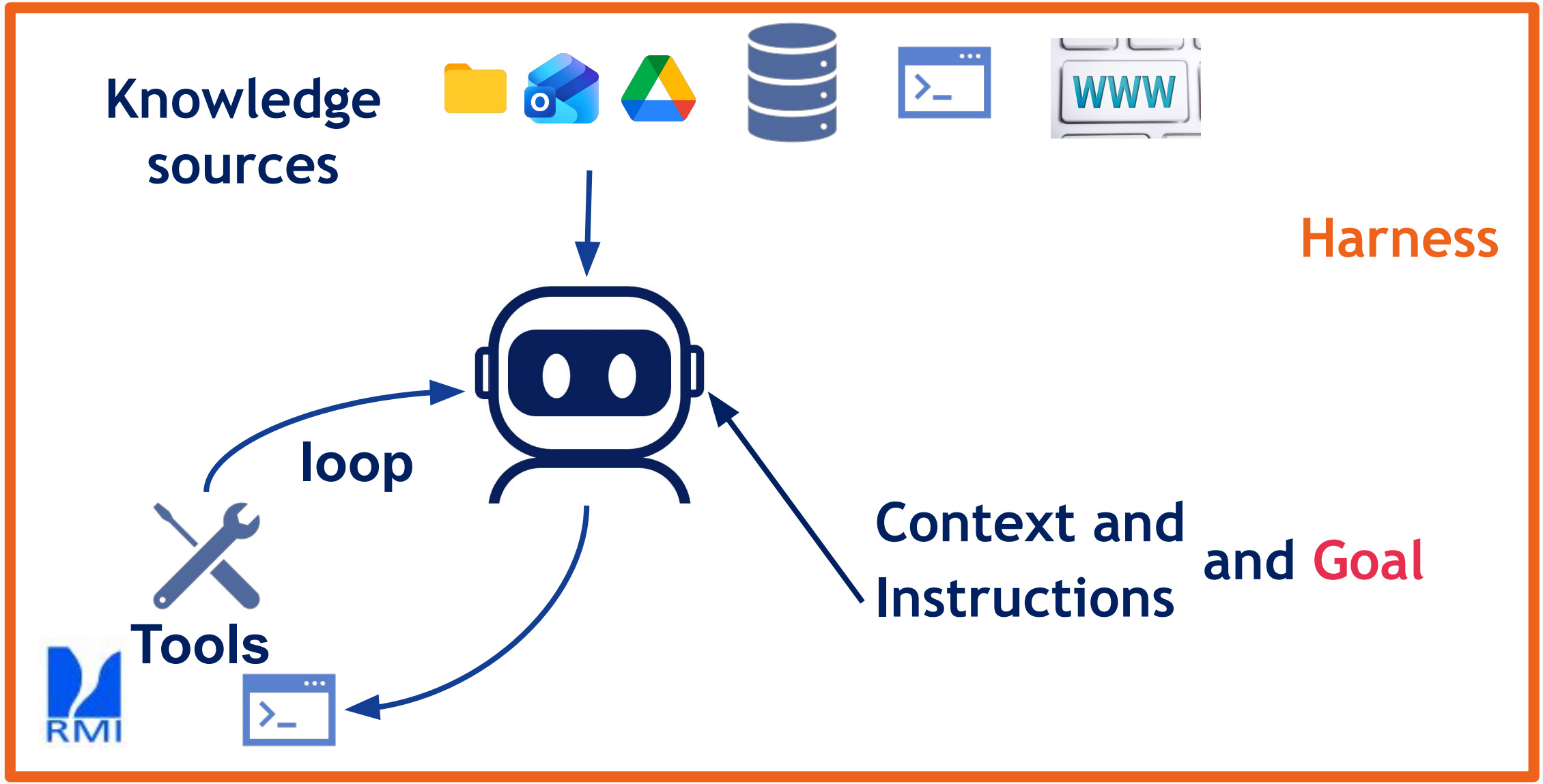




artefacts



From a chatbot to an agent: and we give it a goal



Relevant vocabulary and how it relates to cooking

Goal = "Serve an excellent dinner"

LLM = Chef's training

Tools = Stove, oven, knife, mixer

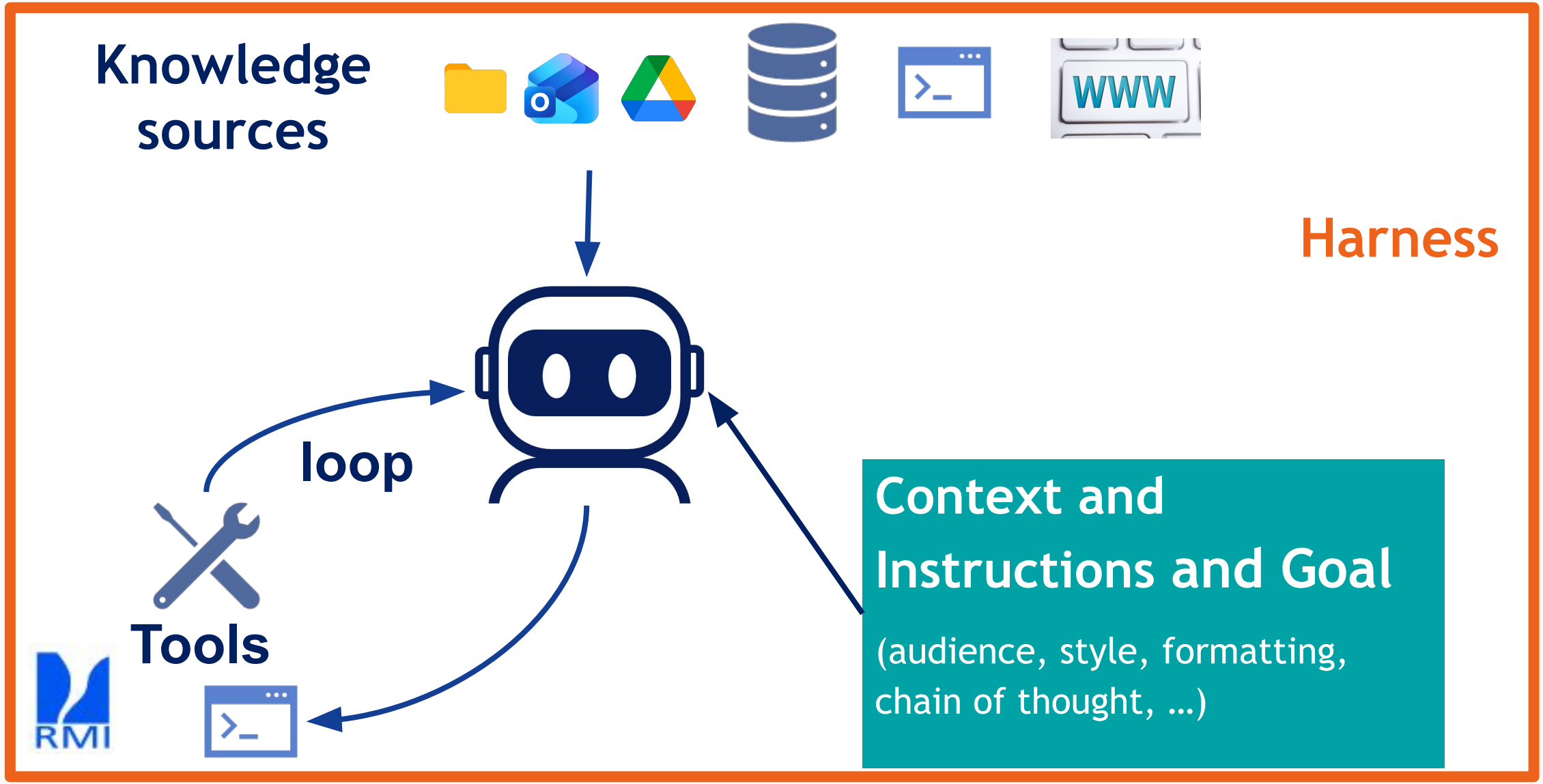
Context/Recipe/Plan = Chef deciding what to cook and when

Harness = The kitchen's operations: tickets, layout, kitchen manager

Relevant vocabulary

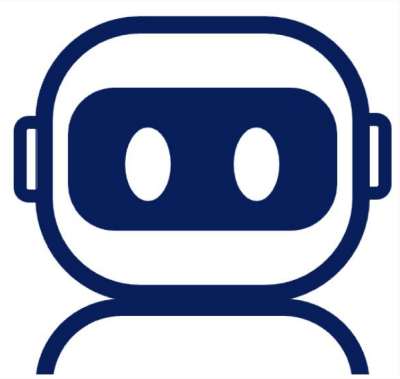
Agent component	Short definition
Goal (task / prompt)	The result the user wants. It tells the agent what success looks like.
Model (LLM)	The trained language model. Its knowledge and skills set what the agent understands and how well it reasons.
Tools	Functions or connectors the agent can call to act on the world or fetch information, such as search, running code, or reading files.
Agent loop (Reasoning & planning)	The repeating cycle where the model looks at the current state, decides the next step, uses a tool, checks the result and repeats until the goal is met.
Harness (orchestration / runtime)	The software around the model. It sends in tasks, manages permissions and tools, tracks state and memory, and keeps the loop running safely.

Context engineering makes the work more specific



The context window is rebuilt each time

- system prompt
- an Instruction section
- the conversation so far
- the files it read
- the output of commands it ran



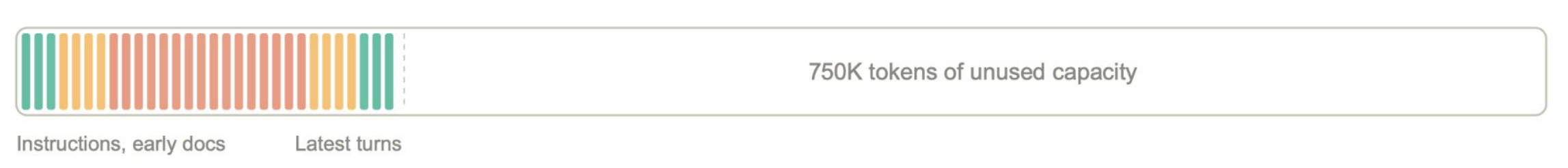
It keeps nothing of its own.

it appends the answer & resends

Context rot: the window is large, the usable part not

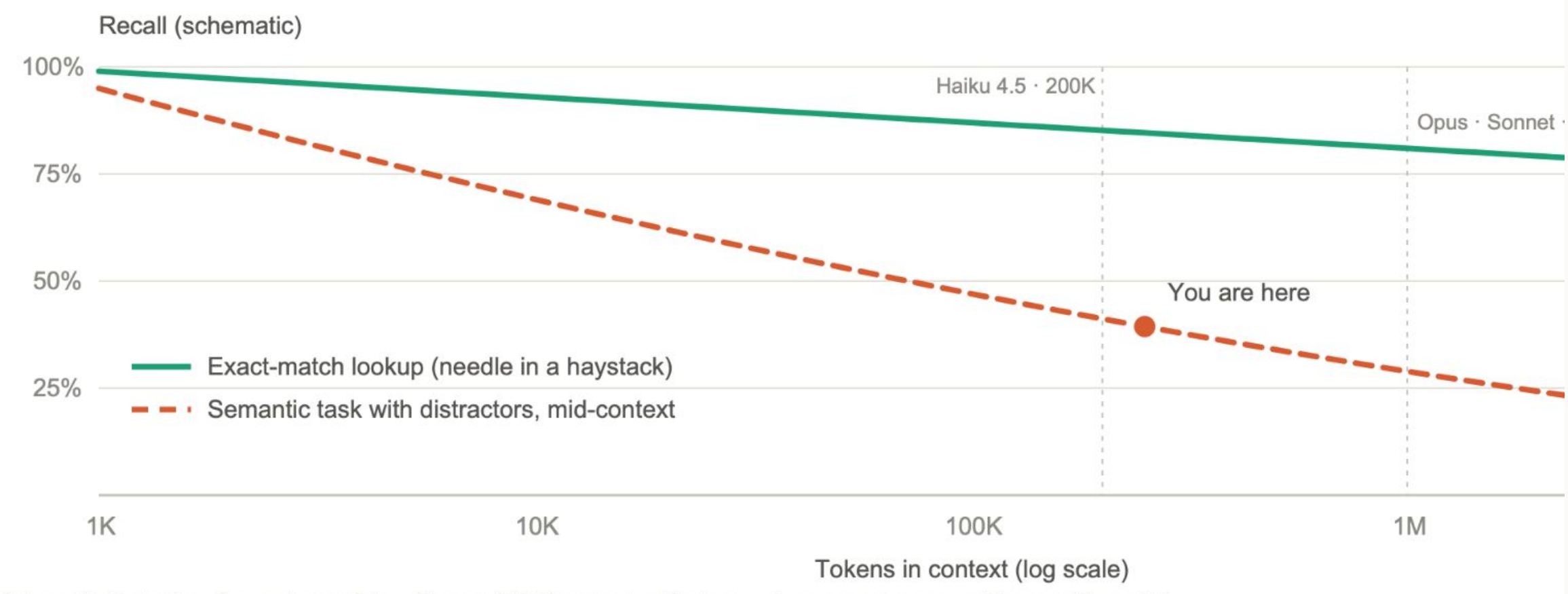
current standard size:
1,000,000 tokens

750,000 words
± 8 novels



Context rot: the window is large, the usable part not

current context: 250,000 tokens



Compaction: it does not stop, it quietly throws things away

context limit reached

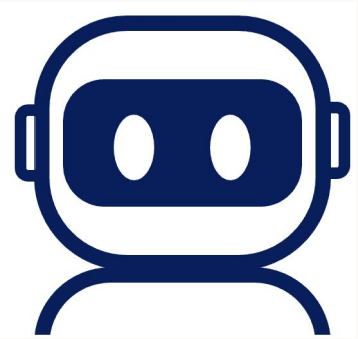
Agent: writes environment file and starts analysis

YOU: OK and write an environment file

Agent: need to install this tool, OK?

YOU: now start the analysis

summary of earlier conversations





silently drops context

Persistance: what matters goes on disk

next session - nothing on disk

“which approach should I take?”



you answer it again

next session - files on disk

```
instructions.md
doc/notes.md
skills/SOP-extract.md
```



reads the fixed context

Four good habits to keep the context sharp

Points at files

`Upload file(s)`

`@filename`

One task, one session

`New chat`

`/clear`

Convert once

`Define via instruction`

`/pdf-to-markdown`

Watch the meter

`/context`





Break
until 15:10

Your feedback helps us make the course better

https://shorturl.at/14tm0

Chat runs in a sandbox; an agent runs on your computer

Chat

- on provider's server
- Sees only what you paste or upload
- Gives text back; you move it where it belongs

When to use: brainstorming, polishing one paragraph, quick lookups.

AI agent



- Runs in a folder on your computer
- Can read any file in the project, and edit it
- Runs code, shell commands, search, web, MCP tools

When to use: any task that touches more than one file or needs to execute code.

Agentic AI

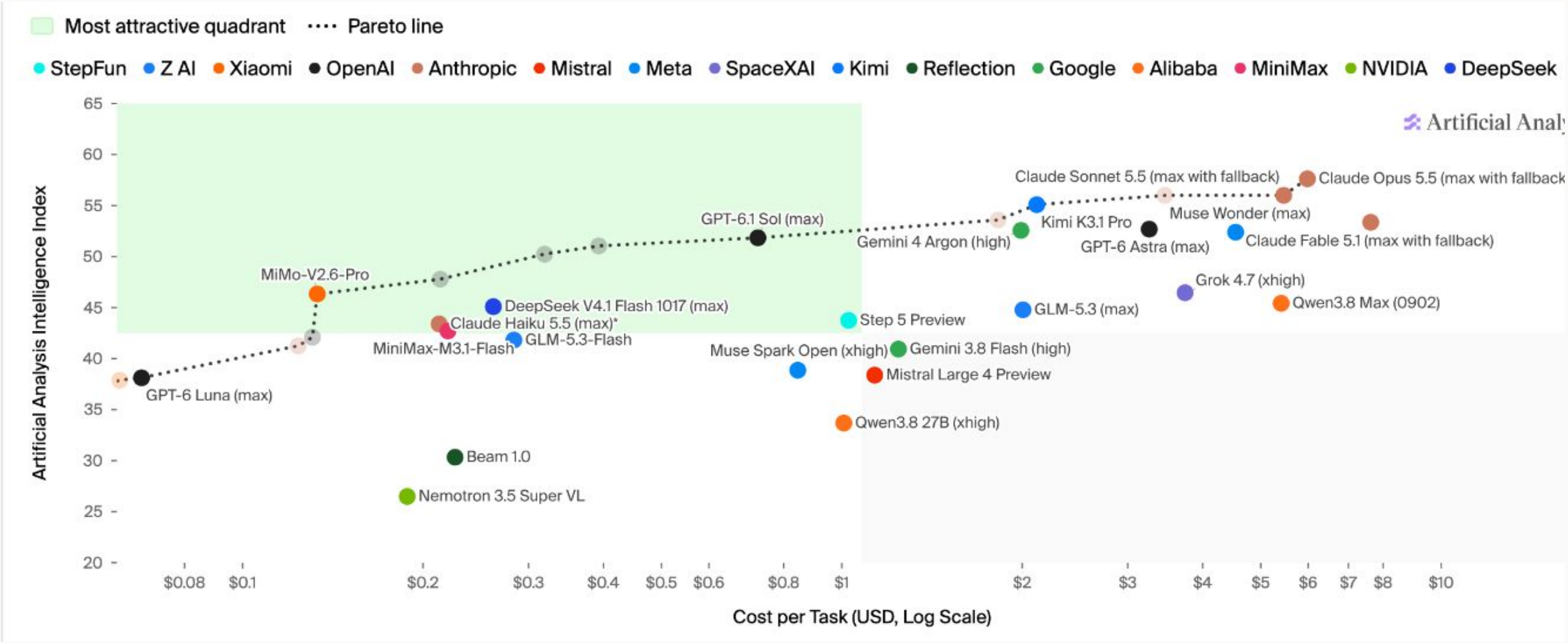


- Runs in a folder on your computer or provider's server
- Can read any file in the project, and edit it
- Runs code, shell commands, search, web, MCP tools

When to use: complex multi-step workflows

Sandbox!

What model is mostly enough for daily work?



https://artificialanalysis.ai/

We use several components to increase context and instructions

Used when?	Context	Instructions
at begin	Copilot Instructions	your prompt
during run	Knowledge Memory	Skills
at the end	Save notes from memory	

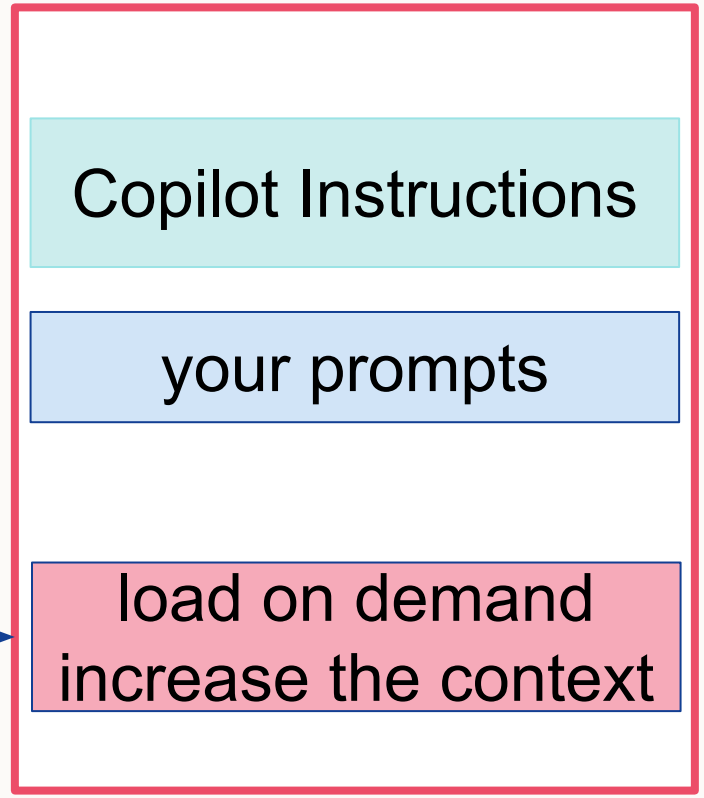
Behavioral instructions: the briefing you'd give a lab member

Copilot Instructions

```
skills/intro-planner/SKILL.md

docs/pdf-files
knowledge/20261125-concepts.txt
```





Memory/Notes

write today's notes

An agent or agentic workflow can run on your computer

Copilot agents, ChatGPT Agent, Codex CLI, Claude Cowork and Code

1. Give access to one folder.
2. Keep secrets unreadable.
3. Turn network access off, or allow only listed domains.
4. Require your approval for anything irreversible.
5. Make mistakes recoverable.
6. Separate identities.
7. Treat sensitive data as off-limits by default.

show what the Agent looks like and how it is used

abstract?

check background knowledge

We should use a collection of skills for dedicated tasks

Scholarly writing

```
skills/intro-planner/SKILL.md
skills/intro-planner/scripts/extra-statements.py

skills/academic-writer/SKILL.md

scriptorium/skills/venue-fit/SKILL.md

...
```

How to setup a Skill

A Skill is a folder with a SKILL.md file and other supporting material

```
intro-planner/
├── SKILL.md      # Required: metadata + instructions
├── references/   # Optional: docs the agent reads on demand
├── scripts/      # Optional: executable code the agent can run
└── assets/       # Optional: templates and static resources
```

Agents load skills progressively

intro-planner/SKILL.md

Frontmatter
always loaded
name +
description
(~100 tokens)



Activation
when the task
matches full
SKILL.md body
(under 5,000*
tokens)



Execution
as needed
scripts,
references,
templates
(unbounded)

How does SKILL.md look like

We start with Frontmatter between ---

```
---
name: intro-planner
description: Plan and scaffold an academic introduction
section from source papers, context, or rough notes. Use this
skill whenever … For the actual prose, hand off to …
license: MIT
metadata:
  author: inilya
  version: "1.0.0"
---

...instructions...
```

Activity

17A - Create a the frontmatter of a SKILL

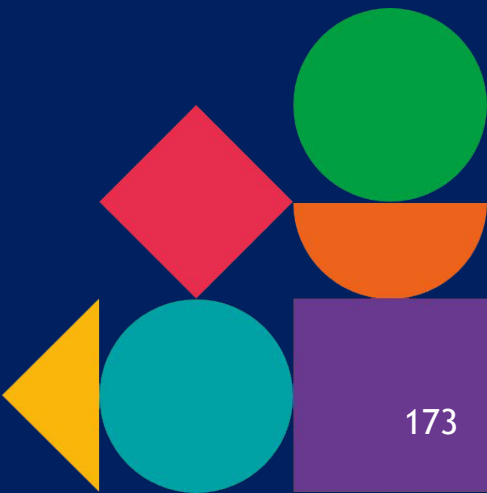
Git file.

https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise17A

1. Open the skill starter template in a text editor
2. Discuss in pairs about the main elements of the description using guiding questions

   [What it does] + [When to use it] + [Trigger keywords] + [What NOT to use it for]
3. Write a description of under 1024 characters in the third person
4. Fill in the other metadata of the frontmatter (between ---)
5. Save the file



How does SKILL.md body look like

```
# My Skill

One-paragraph purpose statement.

## Workflow

1. Step with expected outcome.
2. Step, with the script or reference it uses.
3. Validation step, with failure handling.
```

How does SKILL.md body look like

```
...

## Rules

- Boundaries: what the skill must never do.

## Examples

One realistic input → expected output pair.

## Reference files

- Read [references/DEEP_TOPIC.md](references/DEEP_TOPIC.md)
when [condition].
```

Activity

17B - Create a zipped SKILL

Git file.

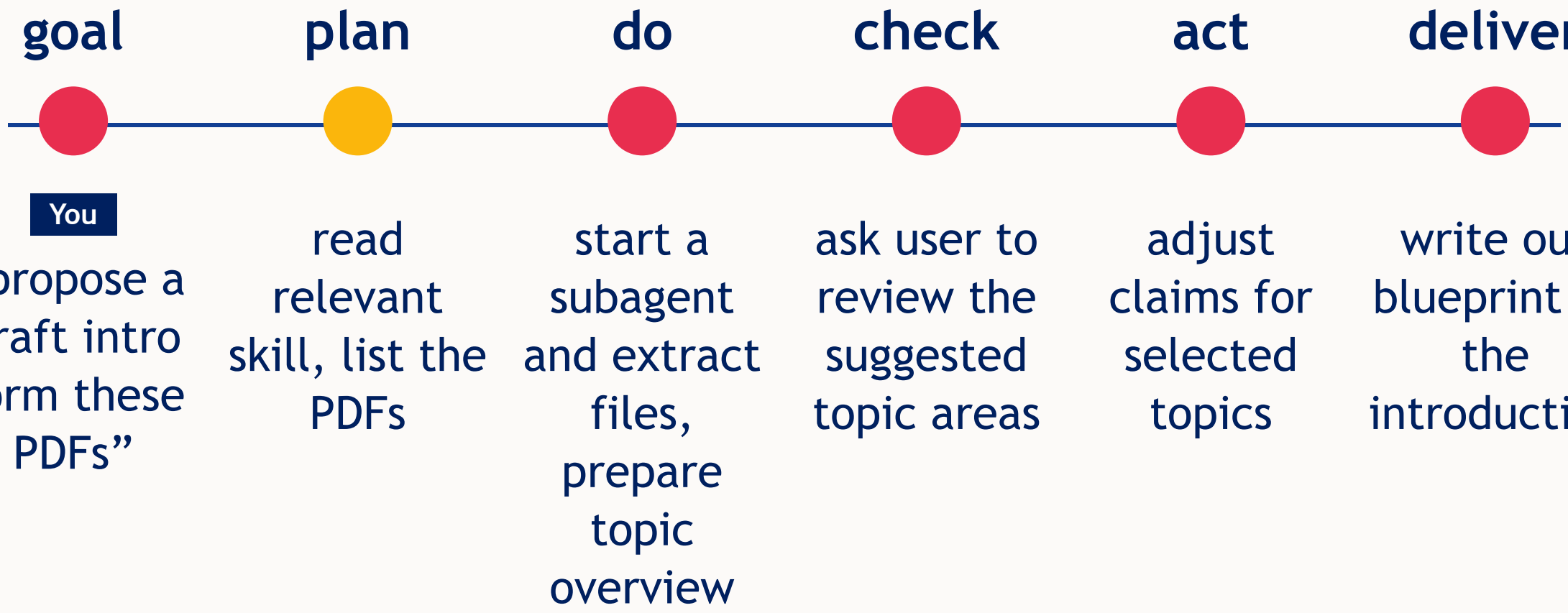


https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise17B

1. Go back to the text editor with the drafted frontmatter of the SKILL.md
2. Discuss in pairs about the main elements of the purpose, and potential workflow
3. Based on the elements from 2, prepare a SKILL.md for an intro planner with a chat-based tool
4. Create a relevant folder structure for your skill and place the SKILL.md in it
5. Prepare a zip file of the folder

What one turn could look like for a writing task



Activity

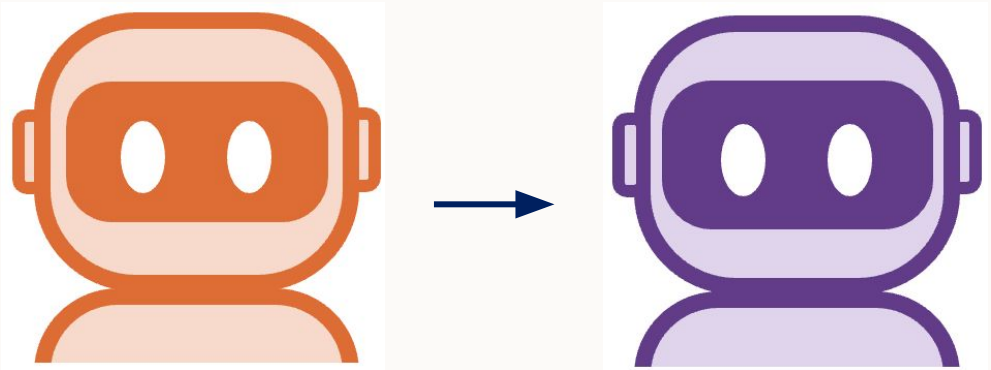
18 - Prepare a blueprint for an introduction

Git file.
https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise18

1. **Create a basic Copilot agent**
2. **Prepare the general part of the 'Intro Planner' agent**
3. **Repeat basic instructions from skill or leave empty.**
4. **Add the zipped skill to the Skills agent setup.**
5. **Add an example prompts to the agent setup.**
6. **Check the knowledge settings to avoid search the web.**
7. **Use preview of the agent to test its functionality.**

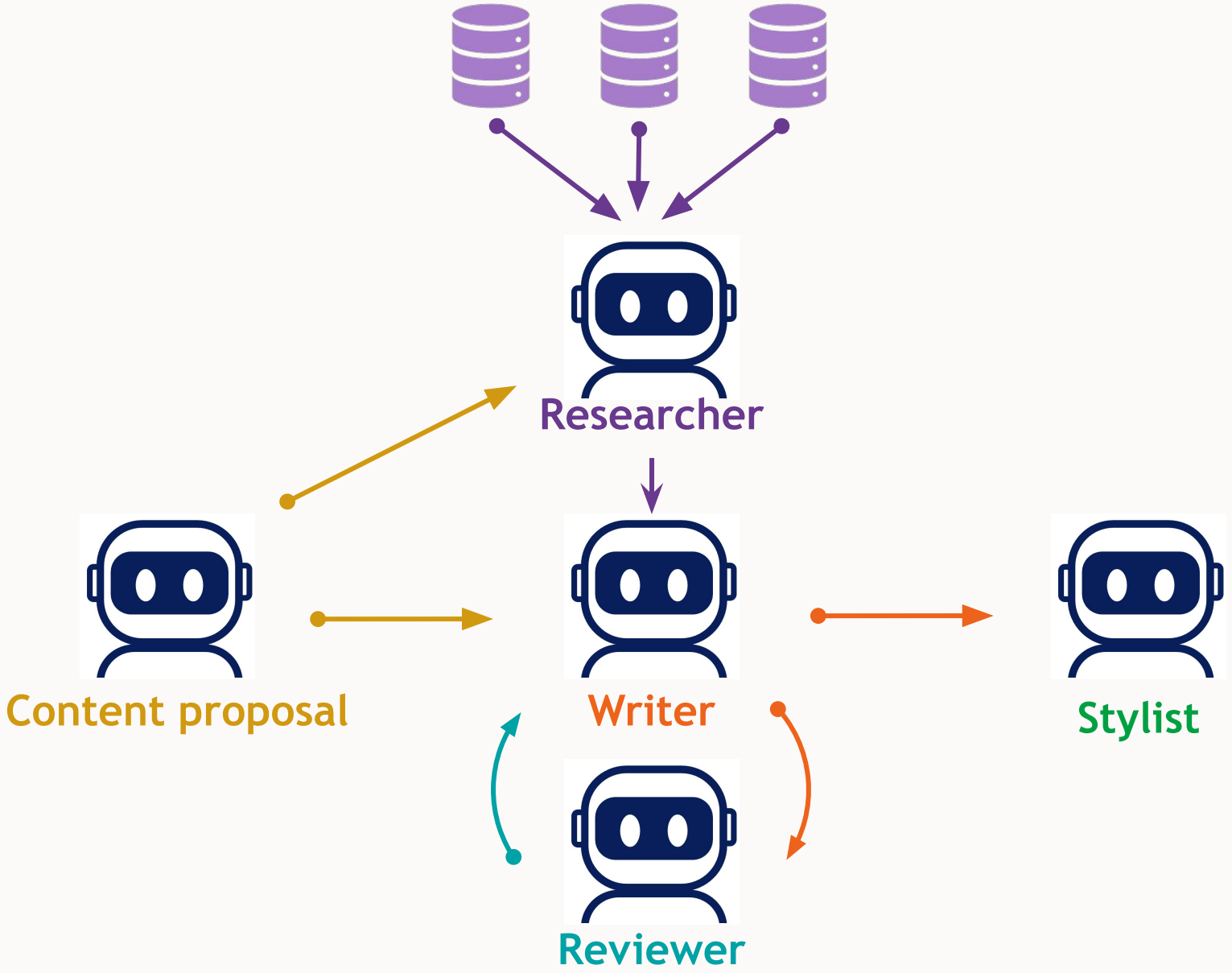
Demo with Claude Code (command line)



Intro planner

Writer academic

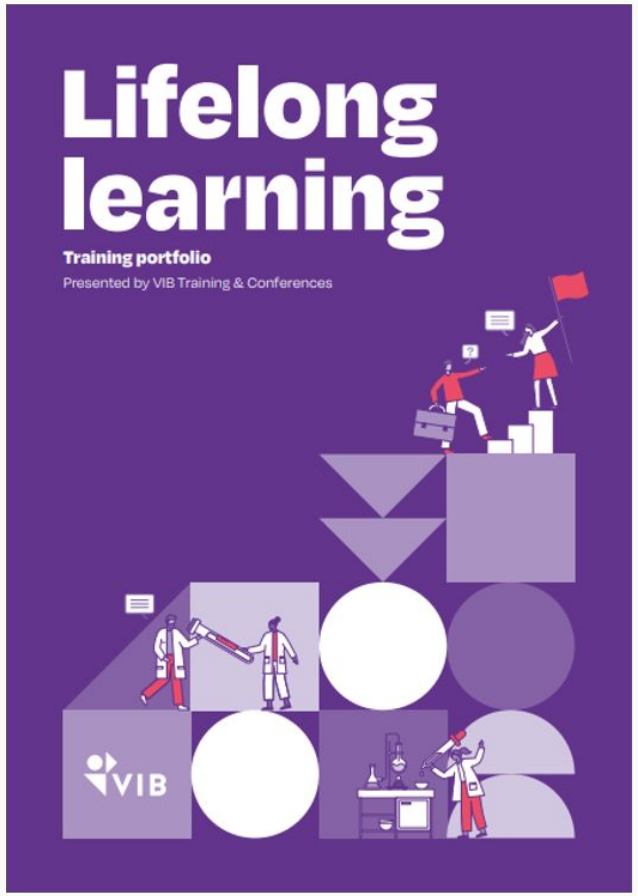
Workflow example



Agents

Agent	= Chef deciding what to cook and when
Model	= Chef's training and talent
Tool	= Stove, oven, knife, mixer
Context window	= Counter space
Goal	= "Serve an excellent dinner"
Harness	= The kitchen's operations: tickets, layout, kitchen manager
Memory	= Chef's notes about this kitchen and these guests
Workflow	= Fixed menu + preparation schedule (chef = line cook)
Skill	= Recipe / cooking technique (general, reusable)

Eager to grow your skills?
Discover our interactive brochure!



vibtrainconf.be

Your feedback helps us make the course better

https://shorturl.at/14tm0

Vocabulary

Component	Definition
Goal	The outcome the user wants. It sets what success means and when the work is done.
Prompt / task	The specific request or instruction that starts a run, usually in plain language.
System prompt	Standing instructions that shape the agent's role, tone, priorities and limits on every task.
Model (LLM)	The trained language model that understands, reasons and generates. It sets the upper limit on quality.
Agent	A model working in a loop. It plans, picks actions, uses tools, checks results and adapts until the goal is met.
Tool	A function or external service the agent can call to act or get information, such as search, code execution or sending email.
Connector (MCP)	A standard protocol for linking tools and data sources to an agent without custom wiring for each one.
Context window	The limited working space the model can see at once: instructions, conversation, files and tool results. When it's full, something has to be cleared or summarized.
Memory	Information saved outside the context window, such as preferences or past decisions, and recalled in later sessions.
Retrieval (RAG)	Searching documents or databases at run time and bringing only the relevant pieces into the context.
Skill	A packaged, reusable set of instructions and resources the agent loads when a task calls for it.
Workflow	A predefined sequence of steps where the code, not the model, decides the path. It's more predictable than an agent and less flexible.
Subagent / multi-agent	Separate agents that each handle part of a larger task in their own context and report results back to a lead agent.
Harness	The software around the model. It runs the loop, routes tools, manages context, memory and permissions, and handles errors.
Guardrails / permissions	Rules and checks that limit what the agent may do, such as blocking risky actions or requiring approval.
Human in the loop	A point where a person reviews, approves or corrects the agent's work before it continues or goes out.
Evaluation	Systematic testing of agent outputs against criteria or test cases to measure and improve quality.



Bruna Piereck and Alexander Botzki
trainingandconferences@vib.be

Agents raise econometric coding success from 74 to 96 percent

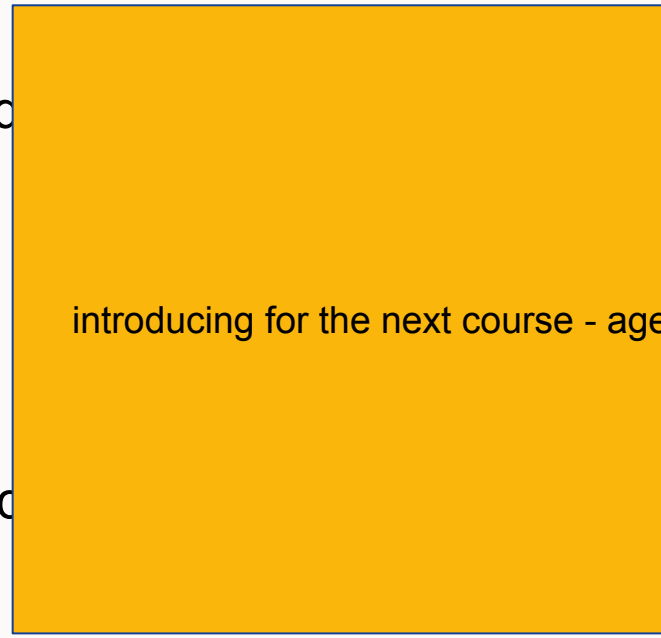
Galiani et al. (2026) benchmark applied econometric tasks in Stata, R and Python. Moving from a chatbot that writes one script to an agent that runs and revises its own code is what raises task success.

Under the chatbot, results differ a lot by software. Under the agent, the c
disappear.

74 → 96% task success, chatbot to constrained agent

8 cents extra cost per run

Galiani et al. (2026), NBER working paper 35588, abstract. “Constrained
runs its code and fixes errors, but within a fixed set of tools.



VIB

https://learn.microsoft.com/en-us/microsoft-365/copilot/extensibility/prerequisites#agent-capabilities-and-licensing-models
https://github.com/seandavi/scriptorium/tree/main

After this module, participants can:

- Explain why Claude is used within VIB Technologies, and assess its strengths and limitations — including the need for a critical eye, because models can hallucinate
- Activate and start your own Claude licence, and determine which data you can or cannot transfer from Copilot or a personal Claude account (or another AI tool via memory import), taking confidentiality into account,…
- Steer Claude effectively with structured prompts (role, task, context, desired format) and tell vague from concrete
- Consciously choose a model (Haiku, Sonnet, Opus or Fable) based on speed, cost and task complexity, with an eye for the link between cost-conscious use and sustainability. Users can keep an eye on their renewal cycle & usage,…
- Set up a Project (a separate workspace within Claude) with its own instructions and a knowledge base of files for recurring work
- Use Artifacts (separate, reusable output windows such as documents, code or dashboards) and refine them in a targeted way, without having everything rewritten
- Use or have a Skill built for a consistently executed, recurring task
- Activate Connectors (links with tools such as Google Drive, Slack, Notion, Jira and VIB SharePoint/MS365), distinguish between merely consulting and letting Claude act (with confirmation), and handle the confidentiality of company data with care
- Set personal preferences and Styles for a consistent writing style
- Use Claude within Word/Excel/PowerPoint/Outlook/Chrome (to be reviewed with IT), and judge when it is better to switch to the desktop app or Claude Code
- Use the memory system and search function to keep building on earlier conversations
- Use Cowork (agentic collaboration with access to several tools at once, aimed at non-developers) for a task involving several tools at once
- (Case 1) Understand how you can gradually build up knowledge about a scientific topic in Claude — for example by bundling earlier conversations and documents in a Project — so that you do not have to start from scratch every time on a new, similar project
- (Case 2) Explain how to turn a collection of loose files (Word, Excel, PDF) into one clear, structured table with Claude, for example with one row per project

How to get the Claude license

email via PI to IT HQ

add tips and tricks for data transfer (chat history)

Claude pricing tier

Free	**$ 0**	Casual chat; no real Claude Code usage
Pro	**$20 / month**	Light daily Claude Code use, projects, research
Max	**$100-200 / month**	High Claude Code use, frontier all day
VIB	**5 * Pro**	(to be checked again with IT)
API	**pay per token**	Automation, batches, custom-built tools

Default model is usually enough; step up for review

Chat	`Haiku`	Conversions, summaries, bulk tasks
Default	`Sonnet`	Daily coding, editing, drafting
Frontier	`Opus`	Hard reasoning, adversarial review
Frontier+	`Fable`	Long-running agents, hardest tasks; costs and waits more

The default model is enough for daily work

check on the intensity
medium

see pareto frontier

add tips and tricks

Grill skills

Pareto frontier

Claude - four good habits

Points at files

`@filename`

Casual chat; no real Claude Code usage

One task, one session

`/clear`

Start a new session for a new task. Thirty turns or a shifted task: restart. Signs of drift.

Convert once

`/pdf-to-markdown`

Long PDFs become markdown once, up front; the markdown is committed and pointed at from then on.

Watch the meter

`/context`

Shows what is filling the window and by how much.

Persistance: what matters goes on disk

3-4 characters of English = 1 token



Effort sets how many tokens the model spends thinking before it answers. It is a setting in the tool, next to the model choice.

15-30k tokens in a short paper

Effort settings

Turn it up for deep analysis and adversarial review. Turn it down for speed on routine tasks.

Projects: keep your assets together

Projects (Claude.ai, ChatGPT)

- Claude and ChatGPT both have projects: upload files, set standing instructions, keep a memory per project. That is the context half of what an agent has, in a chat window.
- Both apps now contain an agent: the Code and Cowork tabs in the Claude app, Codex inside ChatGPT.

Copilot Cowork?

Activity

17 - Write a blueprint for an introduction with Claude Code

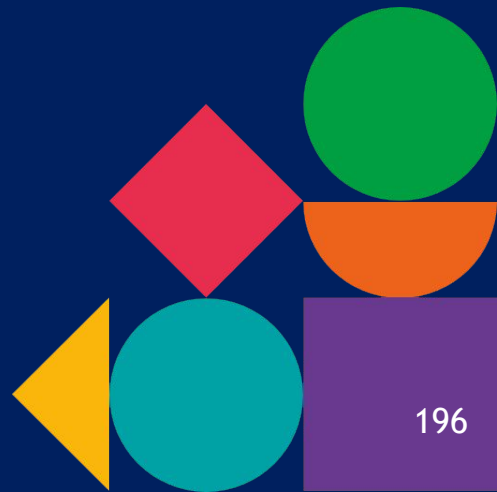
Git file.

https://github.com/vib-training-conferences/gen_AI_for_Scholarly_work

Repo >> exercises >> exercise16

refine procedure
Create one intro agent together

1. **Create a basic Copilot agent**
2. **Prepare the 'Intro Planner' agent**
3. **Review the instructions which serve as knowledge sources to the basic agent**
4. **Add the text instructions which serve as knowledge sources to the basic agent.**
5. **Preview the the basic agent to test its functionality.**



Primary color palette

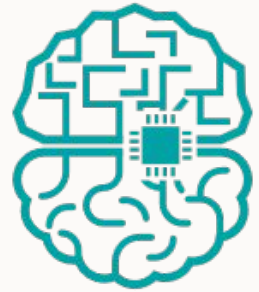
Aqua	Orange	Purple	Navy
HEX #02A2A7 RBG 2, 162, 167	HEX #EC6523 RBG 236, 101, 35	HEX #693A91 RBG 105, 58, 145	HEX #14274C RBG 20, 39, 76
Tint 01 HEX #29B3B8 RBG 41, 179, 184	Tint 01 HEX #EF7739 RBG 239, 119, 57	Tint 01 HEX #784B9A RBG 120, 75, 154	
Tint 02 HEX #99D7D6 RBG 153, 215, 214	Tint 02 HEX #F7C1A3 RBG 247, 193, 163	Tint 02 HEX #C2AFD0 RBG 194, 175, 208	
Tint 03 HEX #CBEAEC RBG 203, 234, 236	Tint 03 HEX #FBE8DF RBG 251, 232, 223	Tint 03 HEX #E3DCEA RBG 227, 220, 234	

Secondary color palette

Blue	Cherry	Green	Yellow
HEX #1F448F RBG 31, 68, 143	HEX #E83050 RBG 232, 48, 80	HEX #00A14A RBG 0, 161, 74	HEX #F9B418 RBG 249, 180, 24
Tint 01 HEX #24539F RBG 36, 83, 159	Tint 01 HEX #EC4D6A RBG 236, 77, 106	Tint 01 HEX #1AAF5E RBG 26, 175, 94	Tint 01 HEX #FCC747 RBG 252, 199, 71
Tint 02 HEX #A1B2D5 RBG 161, 178, 213	Tint 02 HEX #F6AAB9 RBG 246, 170, 185	Tint 02 HEX #9BD3B3 RBG 155, 211, 179	Tint 02 HEX #FCE39B RBG 252, 227, 155
Tint 03 HEX #D5DBEC RBG 213, 219, 236	Tint 03 HEX #FCE3E6 RBG 252, 227, 230	Tint 03 HEX #D1EAD9 RBG 209, 234, 217	Tint 03 HEX #FEF4CF RBG 252, 244, 207

Center icons









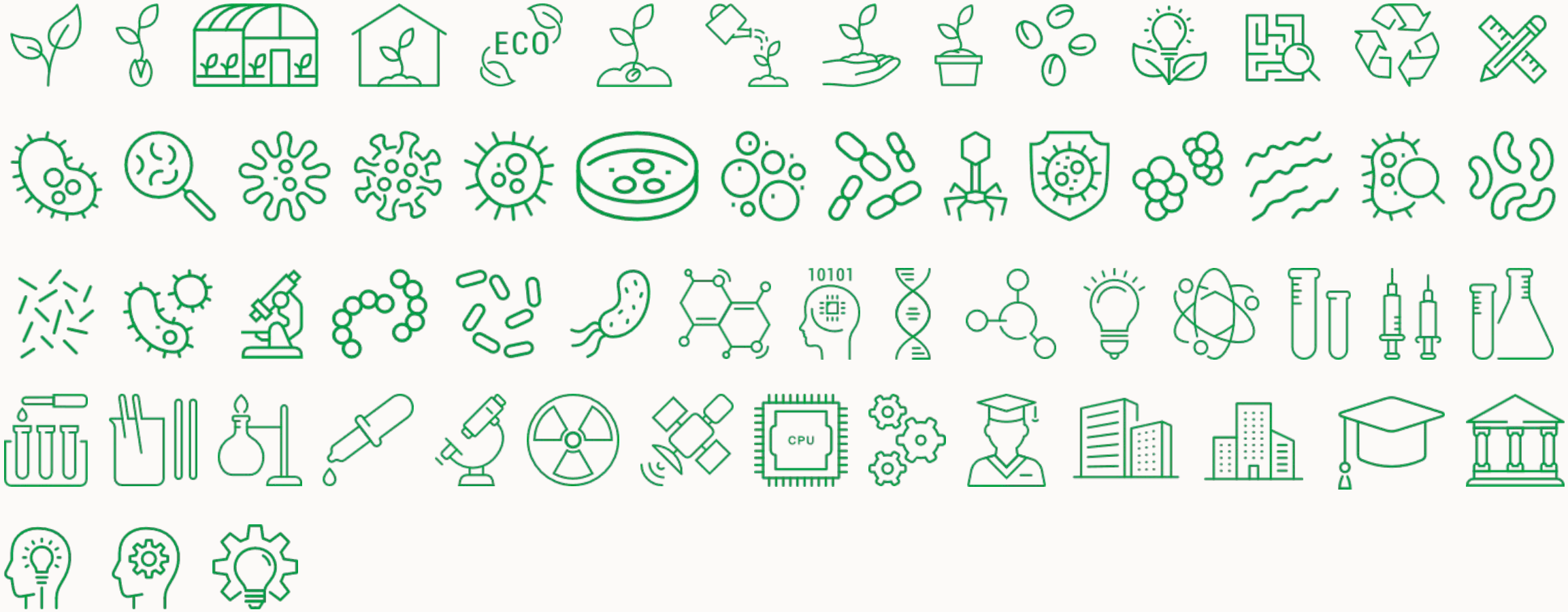
Icon library

Change icon color: click on graphic > graphics format > graphics fill



Icon library

Change icon color: ***click on graphic > graphics format > graphics fill***



Icon library

Change icon color: ***click on graphic > graphics format > graphics fill***



Icon library

Change icon color: ***click on graphic > graphics format > graphics fill***

